Probing Viterbo's Conjecture

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Abstract

Haim–Kislev and Ostrover disproved Viterbo's conjecture with a Lagrangian product of two regular pentagons, the Haim–Kislev–Ostrover (HKO) body. We prove that this body locally maximizes the systolic ratio among polytopes with exactly ten facets. Among bodies sufficiently near HKO, equality in this systolic-ratio bound holds precisely for its translations, positive dilations and linear symplectic images. We also prove an explicit capacity formula and the sharp bound $(3+\sqrt{5})/5$ for products of two independently linearly deformed regular pentagons. Thus neither sufficiently small ten-facet perturbations nor arbitrary deformations within this product family improve the HKO ratio. We also exhibit a distinct ten-facet local maximum: a rational hexagon–quadrilateral product of ratio 1, with a two-parameter local equality family modulo the same symmetries. This second result is a local maximum, not a new counterexample to Viterbo's conjecture.

The placement of product planes supplies another source of variation: every symplectic product of two full-dimensional planar convex bodies admits a capacity-increasing rigid rotation, while the Lagrangian endpoint can have smaller capacity. The proofs and computations use finite capacity formulas, their first-order variation, and an implementation of Haim–Kislev's quadratic program. A flow-graph algorithm using the local passage geometry of Chaidez and Hutchings gives a second method for capacity computation under explicit regularity assumptions.

In broader sampled populations of four-dimensional convex polytopes, symplectic ridge area has a strong inverse association with the computed systolic ratio. Selecting bodies by geometric measurements raises mean ratios in fresh samples, but the ridge association weakens among low-ridge or high-ratio subsets. Area-preserving factor deformations that balance vertex covariances raise ratios in most tested products and weaken their ridge–ratio association, supporting a contribution from affine factor shape to that association in the tested sample. This does not give a universal ordering: exact relative-rotation laws for regular pentagon and equilateral-triangle products order ridge area and systolic ratio in opposite ways. None of the exploratory searches produced a new numerical ratio above one, although retaining previously active capacity branches improved median best ratios over literal gradient ascent in a matched-start comparison.

Disclosure of AI Use

Generative artificial intelligence was used materially throughout the research and production of this thesis, principally through Codex and Claude Code agents. Agents were not limited to language polishing or executing fully specified tasks. They participated in choosing questions and methods; developing mathematical arguments, software, experiments, and prose; interpreting and criticizing results; revising work; and planning, delegating, and reviewing work by other agents. This breadth of AI use was approved by my advisor.

There was no stable division of labor by category. I am not aware of any kind of work I performed on this thesis that I did not also delegate to agents at some point. Agents and I both planned, produced, interpreted, corrected, and reviewed work, often through repeated handoffs. This overlap does not imply equal labor shares or identical contributions to every result. The research and production process, and many retained components, therefore have mixed human–AI provenance. I do not claim sole production or sole intellectual origin of every part of the thesis. Because work often passed between me and agents, I cannot reliably say who first originated every idea, passage, program, or decision.

The analytic endpoint-comparison proof of the rotated-pentagon profile in Section 9 was supplied during an AI-assisted review by ChatGPT Pro. I subsequently checked and accepted that mathematical argument. The same review supplied the restricted-family ridge–capacity identity in equation 35. These contributions are distinct from the earlier computational exploration of the rotation family.

The independent-linear-deformation argument in Section 10 originated in a further ChatGPT Pro return on 18 September 2026. Codex agents checked the argument and its capacity-formula conventions and prepared the repository and thesis versions. This records agent review, not a completed personal verification of the new argument.

As of 13 July 2026, I had proofread the then-current thesis and personally worked to verify its mathematical and factual claims and the overall message it presented. The manuscript was revised substantially afterwards. This statement therefore does not assert that I have completed a final review of the current candidate.

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1 Introduction

How large can a convex body's symplectic capacity be if its volume is fixed? Viterbo's conjecture proposed that a ball gives the largest value [Vit00]. For the Ekeland–Hofer–Zehnder capacity in dimension four, the proposed inequality was

$$\mathrm{sys}(K) := \frac{c_{\mathrm{EHZ}}(K)^2}{2\,\mathrm{vol}_4(K)} \leq 1.$$

The quantity on the left is called the systolic ratio. Capacity scales by the square of a dilation and four-dimensional volume by its fourth power, so this ratio compares shape independently of overall size. Translations and linear symplectic maps also preserve it.

The inequality is false. Haim–Kislev and Ostrover found a counterexample formed from two regular pentagons in complementary Lagrangian planes [HO26]. Writing the coordinates as $(q, p) \in \mathbb{R}^2 \oplus \mathbb{R}^2$, one representative is

$$K_{\mathrm{HKO}} = P_5 \times R(-\pi/2)P_5, \qquad \mathrm{sys}(K_{\mathrm{HKO}}) = \frac{3+\sqrt{5}}{5} > 1.$$

Here P_5 is a centred regular pentagon with a vertex on the positive horizontal axis, and $R(\theta)$ is planar rotation. A product placed in the q- and p-planes is called a Lagrangian product. The relative position of its factors affects capacity even though it does not affect volume.

The counterexample leaves several geometric questions. Is the HKO body a local maximum of the systolic ratio among convex bodies? Are there other families with large ratios? Can the computations reveal geometric relationships involving capacity? This thesis investigates these questions through local analysis, finite algorithms and data science. This is the sense in which we continue to probe Viterbo's conjecture: although its proposed bound is false, the geometric relationship between symplectic capacity and volume remains to be understood.

The mathematical results identify two ways of changing the counterexample that cannot improve its ratio: sufficiently small ten-facet perturbations and arbitrary independent linear deformations of its regular-pentagon factors. They also exhibit a distinct non-HKO local maximum in the same ten-facet space, at ratio one. These are different restrictions and examples: the first is local in the full ten-facet space, the second is global within a product family, and the third is a separate product body with a non-strict local maximum. Finite capacity formulas make these questions accessible to proof. The empirical investigation asks whether geometric measurements can explain capacity in broader populations and guide the search for further examples.

1.1 Capacities, dynamics and convex geometry

Symplectic capacities measure constraints on embeddings that volume alone cannot detect. Two basic examples are the Gromov width c_G, defined by the largest symplectic ball that embeds into a domain, and the cylindrical capacity c_Z, defined by the smallest symplectic cylinder into which the domain embeds. With the usual normalization, every symplectic capacity lies between these two [Edt24]. The EHZ capacity has a different description: on convex bodies it is the least action of a closed generalized characteristic. Thus comparing capacities connects an embedding question to a question about Hamiltonian trajectories.

The conjecture that all normalized capacities agree on convex domains was stronger than Viterbo's volume bound. Indeed, the Gromov width satisfies that bound directly by volume preservation. The HKO example therefore also separates the EHZ capacity from the Gromov width [HO26]. Abbondandolo, Edtmair and Kang prove that EHZ and cylindrical capacity coincide on every convex body in four dimensions [AEK24, Corollary 1]. Thus the capacity computed in this thesis also determines cylindrical capacity. Their proof builds on Edtmair's

earlier work, which characterizes cylindrical capacity on smooth star-shaped domains in $\mathbb{R}^4$ with dynamically convex boundary by the least area of a disk-like global surface of section, and proves coincidence of all normalized capacities for convex domains sufficiently C^3-close to the ball [Edt24]. A global surface of section is a surface whose interior is transverse to the flow and is met in both forward and backward time by every trajectory outside its boundary. These results illustrate how information about the flow can determine an embedding obstruction.

There is also a direct connection to a central problem in convex geometry. For a centrally symmetric convex body $L \subset \mathbb{R}^n$ and its polar L°, Artstein–Avidan, Karasev and Ostrover prove $c_{\mathrm{EHZ}}(L \times L^\circ) = 4$ [AKO14]. Viterbo's proposed inequality on these particular products is consequently equivalent to the lower bound $\mathrm{vol}(L)\,\mathrm{vol}(L^\circ) \geq 4^n/n!$ in the symmetric Mahler conjecture. The general counterexample does not invalidate this equivalence for the restricted family. Capacity–volume questions therefore retain significance beyond the original universal bound.

Local maximality has a dynamical precedent as well. Abbondandolo and Benedetti prove that Zoll contact forms, whose Reeb orbits are all closed with the same minimal period, locally maximize the contact systolic ratio [AB23]. Their results include the volume–capacity inequality near the ball. The HKO body presents a different setting: it has corners and a ratio larger than the ball's. Our local theorem studies how such a nonsmooth example behaves under perturbations of its supporting facets.

Haim–Kislev studies the HKO body through hyperplane cuts. A convex body is a *cuts extremizer* if every cut into two pieces with nonempty interior strictly decreases the EHZ capacity on each piece. Numerical experiments suggest this property for HKO, while also suggesting that its capacity is not always the sum of the capacities of the pieces [Hai25, Definitions 1.1–1.2 and Example 1.12]. For sufficiently shallow cuts in certain directions, including all normals supported in one pentagon factor, Haim–Kislev proves this additivity identity [Hai25, Proposition 1.13].

These cuts give a complementary way to perturb the counterexample. Cutting off a small cap typically adds a facet, whereas our theorem varies the ten existing facets. The systolic ratio also compares capacity with volume: for a retained piece K', the inequality $\mathrm{sys}(K') \leq \mathrm{sys}(K_{\mathrm{HKO}})$ is equivalent to

$$\frac{c_{\mathrm{EHZ}}(K')}{c_{\mathrm{EHZ}}(K_{\mathrm{HKO}})} \leq \left(\frac{\mathrm{vol}_4\, K'}{\mathrm{vol}_4\, K_{\mathrm{HKO}}}\right)^{1/2}.$$

Thus the capacity decrease must be large enough to compensate for the lost volume. Theorem 7.1 establishes this balance throughout a neighborhood in the space of ten-facet polytopes.

1.2 A local theorem around the counterexample

Our main result partially answers the first question: every sufficiently Hausdorff-close ten-facet polytope has ratio at most that of K_{HKO}. In a smaller neighborhood, equality occurs precisely for translations, positive dilations and linear symplectic images of K_{HKO}; see Theorem 7.1. Thus a small change of shape that is not one of these symmetries lowers the ratio. Allowing facets to appear or disappear is a different local problem, which the theorem leaves open.

The fixed-facet space has other local maxima. Section 7 also contains a rational product of a hexagon and a quadrilateral with ten genuine facets and ratio one. An exact certificate, together with an analytic neighborhood argument, proves a local maximum including nonproduct directions, and its equality set contains a two-parameter family after quotienting by translations, positive dilations and linear symplectic maps. Its 24 vertices and ratio one distinguish it from HKO; it is not a new violation of Viterbo's inequality.

The proof uses upper bounds that agree with the systolic ratio at HKO. For every nonzero direction transverse to the symmetries, at least one of these bounds has negative derivative. A uniform estimate then gives strict decrease away from HKO on a sufficiently small transverse slice. The finite task is to construct enough such bounds and check their derivatives. Feasible

weights in Haim–Kislev's capacity formula provide the bounds. An algebraic computation establishes the derivative conditions; the proof explains why those conditions imply the geometric local maximum.

1.3 From boundary dynamics to finite calculations

For a smooth convex body, the EHZ capacity is the least action of a closed characteristic on its boundary. Polytopes have corners, so the relevant curves are generalized characteristics. Their velocities are governed by the normals of the facets they meet. Haim–Kislev's theorem makes the capacity computable by a finite optimization over facet orders and nonnegative weights [Hai19].

We derive the finite formula from the dual action principle, expanding Haim–Kislev's argument for a master's-level reader. We state the imported variational results and explain the existence of a minimizing curve with a finite list of distinct facet directions.

For Lagrangian products, we derive a further reduction from the bilinear structure of the objective: some maximizing word uses at most three facets from each factor. This gives a smaller finite search for the capacity, without classifying every minimizing orbit.

Using the local passage geometry of Chaidez and Hutchings [CH21], we develop a flow-graph algorithm with a finite search justified by the simple-minimizer theorem. Its implementation provides an independent cross-check of the quadratic-program method on selected rational inputs satisfying the flow algorithm's regularity hypotheses.

For regular pentagon products, the finite formula also gives the complete relative-rotation profile. The systolic ratio of $P_5 \times R(\theta)P_5$ is

$$\frac{5+2\sqrt{5}}{10}\sec^2 d(\theta),$$

where $d(\theta)$ is the distance to the nearest multiple of $\pi/5$. A short analytic comparison proves the formula, whose maximum is the HKO value. A separate matrix-containment argument extends the capacity formula and sharp systolic bound to independent linear deformations of both regular pentagons; see Section 10. This family is larger than the rotation family but does not include arbitrary pentagons.

A different extension keeps one regular pentagon fixed and allows its partner to be any centrally symmetric planar convex body, including a smooth one. Section 11 identifies capacity with the largest scale of a prescribed regular decagon that fits inside the partner. The decagon is the unique ratio maximizer up to translation and positive dilation, with its orientation fixed relative to the pentagon. Its sharp ratio is $(10+6\sqrt{5})/25 < 1$. Thus this unrestricted symmetric-partner class cannot reproduce HKO's ratio greater than one: an asymmetric partner is essential when the regular pentagon is fixed.

We also vary the placement of the factor planes themselves. For any two full-dimensional planar convex bodies, a small rigid rotation away from the symplectic-product arrangement increases capacity. The Lagrangian endpoint need not be better: an explicit triangle product has smaller capacity there; see Section 12.

1.4 Looking for mathematical structure in data

Symplectic ridge area gives a strong inverse association with the computed systolic ratio in the sampled populations. Selection based on geometric measurements also raises mean computed ratios in fresh samples. Yet the overall rank association is substantially weaker in subsets selected within each sampling group for either low ridge scores or high computed ratios. Its success as a coarse predictor leaves open how to improve a body already selected for a promising score.

Area-preserving linear maps that make each factor's vertex covariance isotropic raise the computed ratio in 314 of 320 tested products and substantially weaken the ridge–ratio correlation. These paired changes support a contribution from affine factor shape to the broad association in this sample. Their effects cannot be attributed to covariance imbalance alone: the maps change several geometric features together.

Exact family laws further constrain the explanation. An edge–width formula describes the ridge measurement on Lagrangian products. Under relative rotation of regular pentagon factors, it has an inverse-square relation to the systolic ratio; under relative rotation of equilateral-triangle factors, the two quantities have the same strict ordering; see (35) and (36). A general explanation must therefore account for the shapes and relative positions of the factors.

In a comparison of fixed local-search implementations from the same 64 starting bodies, retaining previously active capacity branches attains larger median best ratios than literal gradient ascent under the same evaluation limits. This connects the first-order branch model to practical improvement; the finite searches still do not establish local maximality. None of the exploratory searches produced a new numerical ratio above one.

1.5 How to read the thesis

The foundations in Sections 2–4 explain how nonsmooth boundary dynamics become a finite quadratic program. Section 5 presents the alternative flow-graph method, and Section 6 develops first-order variation.

A reader primarily interested in the HKO theorem can start with its statement and proof strategy in Section 7, returning to the foundations for the capacity formula and to Section 6 for variation. The flow-graph and data-science chapters are not prerequisites for that proof. Section 8 describes the empirical investigation. Sections 9 and 10 give the rotation profile and its extension to independently deformed pentagons. Section 11 instead fixes one regular pentagon and treats every centrally symmetric convex partner through a calibrated decagon. Section 12 then rotates the full product out of symplectic position, rather than rotating one factor within its plane. The later sections document visualizations, numerical methods, code and data, and reflect on the use of AI. The appendices contain computational detail for checking or reproducing the results.

2 Preliminaries

A polytope has edges and vertices where the smooth definition of a characteristic no longer applies. We first extend that definition to convex bodies with nonsmooth boundaries. We then introduce a variational description of capacity in terms of closed curves that need not lie on the boundary. This will allow us to modify curves when deriving the finite capacity formula.

All convex bodies below are compact convex subsets of $\mathbb{R}^4$ with non-empty interior. Unless stated otherwise, the origin lies in the interior.

2.1 Convex and symplectic notation

We write $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2). The standard complex structure is

$$J_0 = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix},$$

and the standard symplectic form is

$$\omega_0(u, v) = \langle J_0 u, v \rangle = \sum_{i=1}^{2} (u_{q_i} v_{p_i} - u_{p_i} v_{q_i}).$$

We use the standard primitive

$$\lambda_0 = \frac{1}{2}\sum_{i=1}^{2}(q_i\,dp_i - p_i\,dq_i).$$

Thus $d\lambda_0 = \omega_0$ and $\lambda_0|_x(v) = \frac{1}{2}\langle -J_0 v, x\rangle$. For an absolutely continuous closed curve $\gamma\colon [0,T] \to \mathbb{R}^4$, its action is

$$A(\gamma) \coloneqq \int_\gamma \lambda_0 = \frac{1}{2}\int_0^T \langle -J_0\dot{\gamma}(t), \gamma(t)\rangle\,dt. \tag{1}$$

This action is invariant under orientation-preserving reparametrization. If γ bounds a smooth map $u\colon D^2 \to \mathbb{R}^4$, then Stokes' theorem gives

$$A(\gamma) = \int_{D^2} u^*\omega_0.$$

Thus action is the signed symplectic area enclosed by the curve. For Reeb orbits this is the quantity that the EHZ capacity minimizes.

Let $K \subset \mathbb{R}^4$ be a smooth convex body with $0 \in \operatorname{int}(K)$. At $x \in \partial K$, the restriction of ω_0 to the three-dimensional space $T_x\partial K$ has a one-dimensional kernel,

$$\ell_x = \{v \in T_x\partial K : \omega_0(v,w) = 0 \text{ for all } w \in T_x\partial K\}.$$

This is the characteristic line. A *closed characteristic* is a closed curve on ∂K whose velocity lies in the positive characteristic direction. We choose the positive direction by requiring $\lambda_0(\dot{\gamma}) > 0$.

The restriction $\alpha = \lambda_0|_{\partial K}$ is a contact form. Its Reeb vector field R is the contact-normalized characteristic direction,

$$\alpha(R_x) = 1, \qquad \omega_0(R_x, v) = 0 \quad \text{for all } v \in T_x\partial K.$$

A Reeb orbit is a periodic integral curve of R. It is the same geometric object as a closed characteristic, with the parametrization fixed by $\lambda_0(\dot{\gamma}) = 1$. Consequently, if γ is a Reeb orbit on $[0,T]$, then

$$A(\gamma) = \int_0^T \lambda_0(\dot{\gamma}(t))\,dt = T.$$

This is the sense in which the action of a contact-normalized Reeb orbit is also its period.

2.2 Convex Hamiltonians

The smooth characteristic line field has a convex-analytic formulation that continues to make sense when ∂K is not smooth. For a convex body K containing the origin, its support and gauge functions are

$$h_K(y) \coloneqq \max_{x\in K}\langle x, y\rangle, \qquad g_K(x) \coloneqq \inf\{r > 0 : x \in rK\}.$$

The boundary is the level set $g_K = 1$. We use the convex, 2-homogeneous Hamiltonian

$$H_K(x) \coloneqq g_K(x)^2.$$

For a convex function H, its subdifferential at x is

$$\partial H(x) = \{y : H(z) \geq H(x) + \langle y, z - x\rangle \text{ for every } z\}.$$

When H is differentiable, this set consists only of $\nabla H(x)$.

Definition 2.1 (Contact-normalized generalized characteristic)**.** *A* contact-normalized generalized characteristic *on* ∂K *is a curve* $\gamma \in W^{1,2}([0,T],\mathbb{R}^4)$*, with* $T>0$ *and* $\gamma(0)=\gamma(T)$*, whose image lies in* ∂K *and which satisfies*

$$-J_0\dot{\gamma}(t) \in \partial H_K(\gamma(t)) \quad \text{for almost every } t \in [0,T]. \tag{2}$$

For smooth K, the subdifferential in (2) is the singleton $\{\nabla H_K\}$. The resulting Hamiltonian vector is tangent to the characteristic line on ∂K. Thus the definition recovers the Reeb orbits described above, including their parametrization.

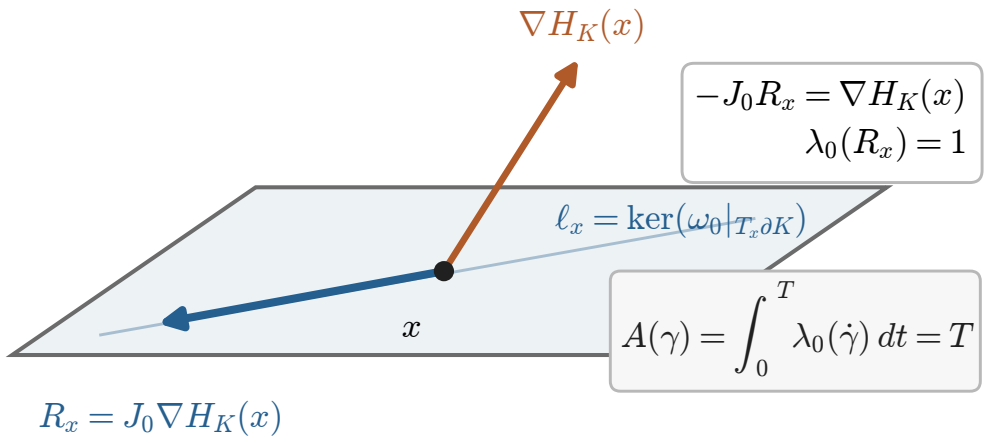


Figure 1: The characteristic and contact normalizations at a smooth boundary point. This is a dimension-reduced tangent-space schematic: it records that $R_x = J_0\nabla H_K(x)$ spans the characteristic line inside $T_x\partial K$, but projected angles and lengths carry no meaning. The contact condition $\lambda_0(R_x) = 1$ turns elapsed time into action.

The normalization also persists without smoothness. Indeed, if $y \in \partial H_K(x)$, then the 2-homogeneity of H_K gives

$$\langle y, x\rangle = 2H_K(x).$$

Along a generalized characteristic, $H_K(\gamma) = 1$, and hence

$$\lambda_0|_{\gamma(t)}(\dot{\gamma}(t)) = \frac{1}{2}\langle -J_0\dot{\gamma}(t), \gamma(t)\rangle = 1 \quad \text{for almost every } t.$$

Consequently every contact-normalized generalized characteristic satisfies

$$A(\gamma) = T. \tag{3}$$

This formulation separates the general nonsmooth dynamical object from its finite description on a polytope. The latter will follow by calculating the subdifferential of H_K from the active facets.

2.3 EHZ capacity

For smooth convex bodies, the Ekeland–Hofer and Hofer–Zehnder capacities coincide. Their common value is the minimum action among closed characteristics on the boundary. We use the resulting EHZ capacity in the form

$$c_{\mathrm{EHZ}}(K) = \min\{A(\gamma) : \gamma \text{ is a closed characteristic on } \partial K\}. \tag{4}$$

This is the smooth least-action characterization of [AO14, Theorem 2.2]. Equivalently, the minimum may be taken over Reeb orbits. The contact normalization only chooses the parametrization

of each characteristic, while the action is a line integral and is invariant under orientation-preserving reparametrization. For a contact-normalized representative, $A(\gamma)$ is also the period. The same least-action theorem gives a positive attained minimum.

For a nonsmooth convex body, we define capacity as the limit of the capacities of smooth convex approximations in the Hausdorff metric. This limit is independent of the approximating sequence. The least-action description remains valid when closed characteristics are replaced by the generalized characteristics of Definition 2.1 [AO14, Proposition 2.7]. Their finite facet description on a polytope is derived in Section 3.

As a symplectic capacity on convex bodies, c_{EHZ} is monotone under inclusion, invariant under symplectomorphisms, and conformal:

$$c_{\mathrm{EHZ}}(\lambda K) = \lambda^2 c_{\mathrm{EHZ}}(K) \quad (\lambda > 0).$$

It is normalized by

$$c_{\mathrm{EHZ}}(B^4(r)) = c_{\mathrm{EHZ}}(Z^4(r)) = \pi r^2,$$

where $B^4(r)$ is the radius-r ball and $Z^4(r) = B^2(r) \times \mathbb{R}^2$ is the symplectic cylinder with disk factor in the (q_1, p_1)-plane and unrestricted coordinates (q_2, p_2).

Throughout the thesis, vol_4 denotes four-dimensional Lebesgue measure in the coordinate order (q_1, q_2, p_1, p_2). In dimension four we define the systolic ratio by

$$\mathrm{sys}(K) \coloneqq \frac{c_{\mathrm{EHZ}}(K)^2}{2\,\mathrm{vol}_4(K)}. \tag{5}$$

With this normalization, Viterbo's proposed inequality for four-dimensional convex bodies was

$$\mathrm{sys}(K) \le 1, \qquad \text{equivalently} \qquad c_{\mathrm{EHZ}}(K)^2 \le 2\,\mathrm{vol}_4(K)$$

[Vit00].

Definition 2.2 (Systolic-ratio symmetries)**.** *Let*

$$G_{\mathrm{sys}} = \{x \mapsto b + \rho M x : b \in \mathbb{R}^4,\ \rho > 0,\ M \in \mathrm{Sp}(4, \mathbb{R})\}.$$

This is the group of affine transformations generated by translations, positive scalings, and linear symplectic transformations.

Lemma 2.3 (Systolic-ratio invariance)**.** *For every convex body $K \subset \mathbb{R}^4$ and every $g \in G_{\mathrm{sys}}$,*

$$\mathrm{sys}(gK) = \mathrm{sys}(K).$$

Proof. Translations and linear symplectic transformations are symplectomorphisms, so they preserve c_{EHZ}. They also preserve four-dimensional volume, because translations do not change volume and every symplectic matrix has determinant 1. A positive scaling by ρ gives

$$c_{\mathrm{EHZ}}(\rho K) = \rho^2 c_{\mathrm{EHZ}}(K), \qquad \mathrm{vol}_4(\rho K) = \rho^4\,\mathrm{vol}_4(K).$$

Substituting these identities in (5) shows that the ratio is unchanged. □

Proposition 2.4 (Continuity)**.** *Let K_t be a family of four-dimensional convex bodies which is continuous in the Hausdorff topology. Then $c_{\mathrm{EHZ}}(K_t)$ is continuous in t. If, moreover, $t \mapsto \mathrm{vol}_4(K_t)$ is positive, finite, and continuous, then the systolic ratio of the family is continuous.*

Proof. For the capacity assertion it suffices to consider $K_n \to K$. Translate all bodies by the same vector so that $rB^4(1) \subset K$ for some $r > 0$. If $\delta_n = d_H(K_n, K)$, their support functions satisfy $|h_{K_n}(u) - h_K(u)| \le \delta_n$ for every unit vector u. Since $h_K(u) \ge r$, for $\delta_n < r$ this implies

$$(1 - \delta_n/r)K \subset K_n \subset (1 + \delta_n/r)K.$$

Monotonicity and conformality give

$$(1 - \delta_n/r)^2 c_{\mathrm{EHZ}}(K) \le c_{\mathrm{EHZ}}(K_n) \le (1 + \delta_n/r)^2 c_{\mathrm{EHZ}}(K),$$

so capacity converges. The assertion for sys is ordinary continuity of the quotient in (5). In the applications below the volume hypothesis is usually automatic, since four-volume is continuous under Hausdorff convergence of convex bodies; for the rotated-polygon family it is even constant. □

2.4 Polytope input language

The finite methods represent polytopes by normalized halfspaces. Let $K \subset \mathbb{R}^4$ be full-dimensional with $0 \in \mathrm{int}(K)$. We write

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x\rangle \le 1,\ i = 1, \dots, F\}. \tag{6}$$

If n_i is the outward unit normal to the corresponding supporting hyperplane and $h_i = h_K(n_i) > 0$ is its height, then

$$a_i = \frac{n_i}{h_i}.$$

For an irredundant description, the vectors a_i are precisely the vertices of the polar body

$$K^\circ = \mathrm{conv}\{a_1, \dots, a_F\}.$$

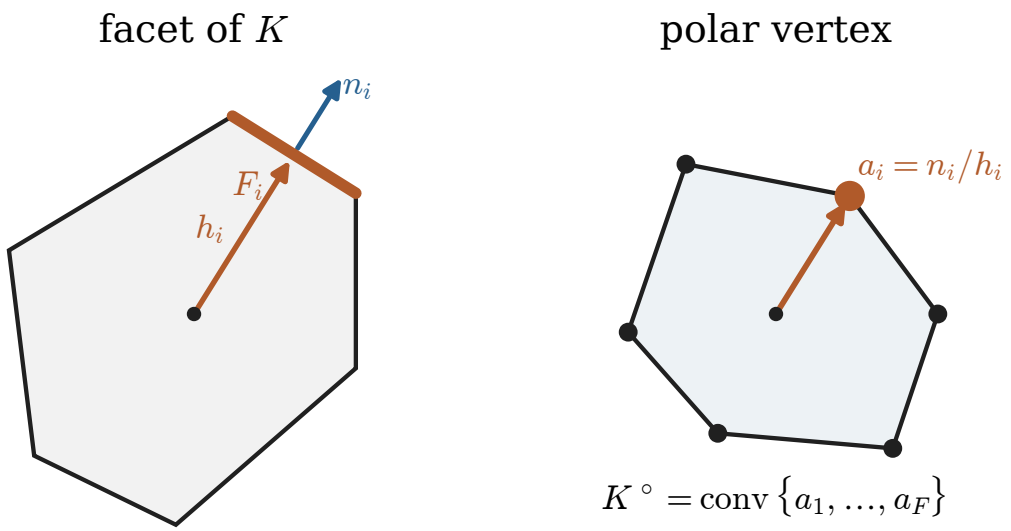


Figure 2: Facet normalization and polarity in a planar model. The supporting line $\langle n_i, x\rangle = h_i$ becomes $\langle a_i, x\rangle = 1$ after setting $a_i = n_i/h_i$, and that row is a vertex of the polar polygon. The same facet–polar-vertex correspondence holds in four dimensions; the planar metric proportions are specific to this example and are not part of the duality statement.

Conversely, the normalized halfspaces define a bounded full-dimensional polytope containing the origin in its interior exactly when $0 \in \mathrm{int}\,\mathrm{conv}\{a_i\}$. The description is irredundant exactly when the listed vectors are pairwise distinct and every a_i is an extreme point of this convex hull.

The ordered list $(a_1, \dots, a_F)$ is a labelled presentation. Relabelling the same inequalities does not change the unlabelled convex body, but labels matter when an algorithm records facet words or when a local calculation uses the a_i as coordinates.

2.5 Lagrangian products

Write

$$\mathbb{R}^4 = \mathbb{R}^2_q \oplus \mathbb{R}^2_p = \{(q_1, q_2, p_1, p_2)\}.$$

Definition 2.5 (Lagrangian product)**.** *For planar convex bodies $K_q, K_p \subset \mathbb{R}^2$, their Lagrangian product is*

$$K_q \times_L K_p = \{(q_1, q_2, p_1, p_2) : (q_1, q_2) \in K_q,\ (p_1, p_2) \in K_p\}. \tag{7}$$

Thus K_q occupies the Lagrangian q-plane and K_p occupies the Lagrangian p-plane. This is different from taking a product of planar domains in the symplectic coordinate planes (q_1, p_1) and (q_2, p_2).

The two decompositions pair the same four coordinates in different ways:

$$\underbrace{(q_1, q_2)}_{K_q} \mid \underbrace{(p_1, p_2)}_{K_p} \qquad \text{versus} \qquad \underbrace{(q_1, p_1)}_{\text{first symplectic plane}} \mid \underbrace{(q_2, p_2)}_{\text{second symplectic plane}}.$$

In block notation $J_0(u, v) = (-v, u)$. A normalized facet row $a_i = (u, 0)$ from the q-factor therefore has $J_0 a_i = (0, u)$ in the p-plane, and a row $a_j = (0, v)$ from the p-factor has $J_0 a_j = (-v, 0)$ in the q-plane. Section 3 will identify the corresponding Reeb velocity as $2J_0 a_i$. Thus the characteristic dynamics exchange the two factors; they do not decouple into independent planar dynamics.

The central examples of the thesis are Lagrangian products of polygons; their lifted facet rows and the resulting restricted word enumeration are introduced with the Haim–Kislev formula, where they are first used.

2.6 Clarke's dual action principle

A curve constrained to lie on ∂K is difficult to modify: changing one velocity can move the rest of it off the boundary. We will instead work with closed curves in $\mathbb{R}^4$ and a functional that depends only on velocity. The dual action principle says that minimizing this functional recovers the capacity, and that a minimizer can be placed back on the boundary. This is the freedom used in the next section to rearrange facet velocities.

Recall $H_K = g_K^2$, and define its convex conjugate by $H_K^*(y) = \sup_x(\langle x, y\rangle - H_K(x))$. For a closed curve $z \in W^{1,2}([0, T], \mathbb{R}^4)$ put

$$I_K(z) = \int_0^T H_K^*(-J_0 \dot{z}(t))\, dt = \frac{1}{4}\int_0^T h_K(-J_0 \dot{z}(t))^2\, dt. \tag{8}$$

The last identity will be proved below. Our normalization is

$$d(K) = \inf\{I_K(z) : T > 0,\ z(0) = z(T),\ A(z) = T\}. \tag{9}$$

The infimum is over Sobolev curves as above. Their positions are unrestricted; the action constraint fixes their size relative to their period.

Theorem 2.6 (Dual action principle)**.** *For a convex body $K \subset \mathbb{R}^4$ with $0 \in \operatorname{int} K$, the infimum in* (9) *is attained and*

$$d(K) = c_{\mathrm{EHZ}}(K) = \min_{\gamma \subset \partial K} A(\gamma),$$

where the last minimum is over contact-normalized generalized characteristics. Every such characteristic satisfies $I_K(\gamma) = A(\gamma)$. Every dual minimizer can be rescaled and translated into a minimizing generalized characteristic, without changing its value of I_K. If a dual minimizer already has $I_K(z) = T$, only a translation is needed.

We use the nonsmooth variational results underlying Clarke's principle [AO14, Proposition 2.5 and Lemmas 5.1–5.2]; for polytopes the correspondence is also stated in [Hai19, Lemma 2.1]. The following argument explains the conjugacy, normalization and reconstruction used here.

The conjugate and its equality condition.

Lemma 2.7. *One has $H_K^*(y) = h_K(y)^2/4$, and*

$$H_K(x) + H_K^*(y) \geq \langle x, y \rangle.$$

Equality holds exactly when $y \in \partial H_K(x)$, equivalently $x \in \partial H_K^*(y)$.

Proof. Write $x = ru$ with $r \geq 0$ and $g_K(u) = 1$. For a fixed r, the largest value of $\langle x, y \rangle$ is $rh_K(y)$. Thus

$$H_K^*(y) = \sup_{r \geq 0}(rh_K(y) - r^2) = \tfrac{1}{4}h_K(y)^2.$$

The inequality follows from the definition of the supremum. Equality says that x maximizes $\langle x, y \rangle - H_K(x)$, which is precisely the subgradient inequality defining $y \in \partial H_K(x)$. Applying conjugacy in the other direction gives the equivalent condition. □

The variational equation. A minimizer satisfies a weak Euler–Lagrange equation. Its useful form is

$$\nu z(t) + b \in \partial H_K^*(-J_0 \dot{z}(t)) \quad \text{for almost every } t, \tag{10}$$

for a constant vector b and a real multiplier ν.

Definition 2.8 (Weak dual critical curve)**.** *A feasible curve satisfying this equation is called* weak dual critical*.*

The constant b reflects the freedom to translate the curve. Conjugacy rewrites the same equation as $-J_0\dot{z} \in \partial H_K(\nu z + b)$, bringing it close to the characteristic equation.

To see the form of the variational equation, choose $p(t) \in \partial H_K^*(-J_0\dot{z}(t))$. The derivative of action in a periodic variation v is $DA(z)[v] = \int \langle -J_0\dot{z}, v \rangle$. The cited nonsmooth multiplier theorem provides a measurable L^2 choice of p such that, for every smooth periodic variation v,

$$\int_0^T \langle p, -J_0\dot{v} \rangle \, dt = \nu \int_0^T \langle -J_0\dot{z}, v \rangle \, dt.$$

The left side pairs the chosen subgradient with the change in velocity; the right side is the variation of the action constraint. Integration by parts gives $-J_0\dot{p} = -\nu J_0\dot{z}$ in distributions, hence $\dot{p} = \nu\dot{z}$ and $p = \nu z + b$. The action constraint is regular because $DA(z)[z] = 2T \neq 0$. These are the variational assertions invoked from the cited nonsmooth theorem; the calculation fixes their signs and constants in our convention.

For any weak critical curve, the multiplier is determined by its value. The two-homogeneity of H_K^* gives $\langle p, y \rangle = 2H_K^*(y)$ when $p \in \partial H_K^*(y)$. Pairing (10) with $-J_0\dot{z}$ and integrating, the constant vector contributes zero because the curve is closed. We obtain

$$2I_K(z) = 2\nu A(z) = 2\nu T, \qquad \nu = \frac{I_K(z)}{T} > 0. \tag{11}$$

Positivity follows since a feasible curve has positive action and is nonconstant, while H_K^* is positive away from zero.

Reconstructing a boundary curve. Let z be weak critical and set $x = \nu z + b$ and $y = -J_0\dot{z}$. Then $y \in \partial H_K(x)$. The convex chain rule gives

$$\frac{d}{dt} H_K(x(t)) = \langle y(t), \dot{x}(t) \rangle = \nu \langle -J_0\dot{z}(t), \dot{z}(t) \rangle = 0.$$

Thus $H_K(x)$ is constant. Since $y \in \partial H_K(x)$, Fenchel equality gives $H_K(x) + H_K^*(y) = \langle x, y \rangle$. Two-homogeneity gives $\langle x, y \rangle = 2H_K(x)$, so $H_K^*(y) = H_K(x)$ almost everywhere. Integrating the constant value yields

$$I_K(z) = \int_0^T H_K^*(y)\, dt = T H_K(x), \qquad H_K(x) = I_K(z)/T = \nu.$$

Put $c = \sqrt{\nu}$ and define

$$\gamma(t) = c\, z(t/c^2) + b/c, \qquad 0 \le t \le c^2 T.$$

Since $\gamma = x(t/c^2)/c$, it satisfies $H_K(\gamma) = 1$. Subgradient homogeneity also gives

$$-J_0\dot{\gamma}(t) = y(t/c^2)/c \in \partial H_K(\gamma(t)).$$

This is the contact-normalized characteristic equation. Its period and action are $c^2 T = I_K(z)$. Direct substitution in (8) gives $I_K(\gamma) = I_K(z)$. In particular, when $I_K(z) = T$ we have $c = 1$, so reconstruction only translates the curve.

Conversely, a contact-normalized characteristic satisfies $H_K(\gamma) = 1$ and $-J_0\dot{\gamma} \in \partial H_K(\gamma)$. Fenchel equality therefore gives $H_K^*(-J_0\dot{\gamma}) = 1$ almost everywhere. Consequently $I_K(\gamma) = T = A(\gamma)$, and it is a feasible dual curve. An attained dual minimum reconstructs to a characteristic with the same value; every characteristic supplies a feasible dual value. This proves equality of the minima and, by the least-action characterization, their equality with capacity.

Period convention and attainment. The cited fixed-period formulation uses $w : [0, 1] \to \mathbb{R}^4$ with $A(w) = 1/2$. If z is feasible in our convention, then

$$w(s) = \frac{z(Ts)}{\sqrt{2T}}, \qquad I_K(w) = \tfrac{1}{2} I_K(z).$$

Conversely, $z(t) = \sqrt{2T}\, w(t/T)$ is feasible for any chosen $T > 0$ and has $I_K(z) = 2I_K(w)$. If the fixed-period formulation imposes mean zero, subtract the mean of w; translation changes neither its action nor I_K. Thus the fixed-period attainment theorem applies to our infimum, with its factor of two accounted for. Choosing $T = 2I_K(w)$ gives an attained representative with $I_K(z) = T$.

The distinction needed below is now concrete. Feasibility allows us to compare functional values, but does not place a curve on the boundary. Once a modified curve is shown to minimize I_K, the variational equation and reconstruction justify placing it there. This is why rearranging velocities will be done in the dual problem first.

3 Generalized Reeb orbits on polytopes

On the interior of a polytope facet, the Reeb velocity is constant. At an intersection of facets, the generalized characteristic equation permits convex combinations of their velocities. Haim–Kislev's theorem nevertheless guarantees a capacity minimizer that uses only the individual facet velocities, each at most once. Its action is therefore determined by a finite ordered list of facets and the time spent in each direction.

We first derive this finite description and distinguish closure of the curve from its placement on the boundary. We then prove the existence theorem by rearranging curves in the dual variational problem.

3.1 Facet velocities

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x\rangle \le 1,\ i = 1, \dots, F\}$$

be a full-dimensional polytope with $0 \in \mathrm{int}(K)$. We assume that the halfspace description is irredundant, so the facet corresponding to a_i is

$$F_i = \{x \in K : \langle a_i, x\rangle = 1\}.$$

The normalized halfspace description gives

$$g_K(x) = \max_{1\le i\le F} \langle a_i, x\rangle.$$

If $x \in \partial K$, let $I(x) = \{i : x \in F_i\}$ be its active facets. The subdifferential rules for a finite maximum and for the square on the level set $g_K = 1$ give

$$\partial H_K(x) = 2\,\mathrm{conv}\{a_i : i \in I(x)\}. \tag{12}$$

Thus Definition 2.1 becomes a finite differential inclusion. In this normalization the pure contact-normalized Reeb direction on F_i is

$$R_i = 2J_0 a_i. \tag{13}$$

Indeed R_i is tangent to F_i, because $\langle a_i, J_0 a_i\rangle = 0$, and for $x \in F_i$,

$$\lambda_0|_x(R_i) = \frac{1}{2}\langle -J_0 R_i, x\rangle = \langle a_i, x\rangle = 1.$$

For a Lagrangian product, this formula makes the exchange between the two factors explicit. A row $a_i = (u, 0)$ from a facet of the q-factor and a row $a_j = (0, v)$ from a facet of the p-factor give

$$R_i = 2J_0(u, 0) = (0, 2u), \qquad R_j = 2J_0(0, v) = (-2v, 0).$$

Thus a segment on a q-facet moves only in the p-coordinates, while a segment on a p-facet moves only in the q-coordinates.

A *generalized Reeb orbit* on K is a contact-normalized generalized characteristic in the sense of Definition 2.1. By (12), this means precisely that $\gamma \in W^{1,2}([0, T], \mathbb{R}^4)$ is closed, has image in ∂K, and satisfies

$$\dot\gamma(t) \in \mathrm{conv}\{R_i : \gamma(t) \in F_i\} \quad \text{for a.e. } t \in [0, T]. \tag{14}$$

The general normalization already shows that such an orbit has

$$A(\gamma) = T.$$

We call such an orbit *simple* if, after choosing its starting point at a velocity change, it consists of finitely many segments with velocities R_i, and each R_i occurs in at most one segment. Here simplicity refers to the use of facet directions. In particular, it includes the requirement that the velocities be individual R_i, rather than nontrivial convex combinations of them.

3.2 Words, dwell times, and closure

A simple orbit is described by the ordered facet labels it uses, the time spent in each direction, and a starting point. The *active word* contains only directions used for positive time. A cyclic word

$$\sigma = (\sigma_1, \dots, \sigma_m)$$

with distinct entries, together with dwell times $\tau_k > 0$ and a base point $b = \gamma(0)$, describes the velocity data

$$\dot\gamma(t) = R_{\sigma_k} \quad \text{for a time interval of length } \tau_k.$$

Such data can represent a closed orbit only if the closing equation holds:

$$\sum_{k=1}^{m} \tau_k R_{\sigma_k} = 0. \tag{15}$$

This equation closes the broken path. A further condition is needed to place each segment on its prescribed facet: there must be a suitable starting point b. Lemma 3.2 expresses that condition by linear equalities and inequalities.

For example, consider the planar square $[-1,1]^2$. The right and left facet rows give opposite directions $R_r = (0,2)$ and $R_\ell = (0,-2)$. Equal dwell times satisfy $\tau R_r + \tau R_\ell = 0$, but the breakpoint between the two segments would have to lie in the disjoint facets F_r and F_ℓ. No translation can realize this closed two-segment path on the prescribed facets.

The period and action of a closed pure-word orbit are

$$T = \sum_{k=1}^{m} \tau_k, \qquad A(\gamma) = T.$$

Cyclically shifting the facet–duration pairs and moving the starting point to the corresponding breakpoint describes the same orbit. We therefore regard the word as cyclic. A zero duration, when allowed in a limiting argument, is simply deleted from the active word.

The action of a finite word can be calculated directly from its velocities and dwell times.

Lemma 3.1 (Symplectic shoelace formula)**.** *Let $z\colon [0,T] \to \mathbb{R}^4$ be a closed piecewise affine curve. Suppose that z has constant velocity w_k on consecutive intervals of length ℓ_k, for $k = 1, \dots, m$. Then*

$$2A(z) = \sum_{1 \le j < k \le m} \ell_j \ell_k \, \omega_0(w_j, w_k). \tag{16}$$

In particular, the action of a closed pure-word Reeb orbit depends only on the cyclic velocity word and the dwell times, not on the base point.

Proof. Write $t_k = \sum_{j<k} \ell_j$. On the k-th interval,

$$z(t) = z(0) + \sum_{j<k} \ell_j w_j + (t - t_k) w_k.$$

The contribution of $z(0)$ to $\int_0^T \langle -J_0 \dot z, z\rangle \, dt$ vanishes because z is closed, so $\sum_k \ell_k w_k = 0$. The self-term $\langle -J_0 w_k, w_k\rangle$ vanishes by skew-symmetry. Hence

$$2A(z) = \sum_k \ell_k \left\langle -J_0 w_k, \sum_{j<k} \ell_j w_j \right\rangle = \sum_{j<k} \ell_j \ell_k \, \omega_0(w_j, w_k),$$

using $\omega_0(u,v) = \langle J_0 u, v\rangle$. □

Breakpoints of $\dot\gamma$ are velocity changes, not necessarily changes in the minimal face containing $\gamma(t)$: an orbit may move inside a nonsmooth face while switching from one active facet direction to another.

Lemma 3.2 (Base-point recovery)**.** *Fix an active word $\sigma = (\sigma_1, \dots, \sigma_m)$ and dwell times $\tau_k > 0$ satisfying the closure equation* (15)*. Put*

$$v_1 = 0, \qquad v_k = \sum_{j=1}^{k-1} \tau_j R_{\sigma_j} \quad (k = 2, \dots, m+1).$$

Then $v_{m+1} = 0$. A point $b \in \mathbb{R}^4$ recovers a simple generalized Reeb orbit with these velocity data, by

$$\gamma(t) = b + v_k + sR_{\sigma_k}, \qquad 0 \le s \le \tau_k$$

on the k-th segment, if and only if

$$\langle a_{\sigma_k}, b + v_k \rangle = 1 \quad \text{for } k = 1, \dots, m, \tag{17}$$

and

$$\langle a_i, b + v_k \rangle \le 1 \quad \text{for } i = 1, \dots, F \text{ and } k = 1, \dots, m. \tag{18}$$

Equivalently, the equalities have the form

$$N_\sigma b = r_\sigma, \qquad (N_\sigma)_{k\bullet} = a_{\sigma_k}^{\mathsf{T}}, \qquad (r_\sigma)_k = 1 - \langle a_{\sigma_k}, v_k \rangle.$$

When this system is nonempty, its solution space is an affine space of dimension $4 - \operatorname{rank} N_\sigma$. The inequalities cut out the valid base points as a convex polytope inside this affine space. In particular, the set of valid base points is bounded, because $v_1 = 0$ and (18) imply $b \in K$.

Proof. On the k-th segment, the prescribed facet equation is

$$\langle a_{\sigma_k}, b + v_k + sR_{\sigma_k} \rangle = 1.$$

The s-term vanishes because $\langle a_{\sigma_k}, R_{\sigma_k} \rangle = 2\langle a_{\sigma_k}, J_0 a_{\sigma_k} \rangle = 0$. Hence the whole segment lies in the prescribed supporting hyperplane exactly when (17) holds.

The segment lies in K exactly when

$$\langle a_i, b + v_k + sR_{\sigma_k} \rangle \le 1 \quad \text{for all } i \text{ and all } s \in [0, \tau_k].$$

For fixed i, k this expression is affine in s, so it is enough to check the endpoints. The endpoint at $s = 0$ is $b + v_k$, and the endpoint at $s = \tau_k$ is $b + v_{k+1}$. As k runs cyclically and $v_{m+1} = v_1 = 0$, these endpoint checks are exactly (18). With these equalities and inequalities, the resulting closed broken path lies on ∂K and has velocity R_{σ_k} on the k-th segment, hence is a simple Reeb orbit with the prescribed data. □

Remark 3.3 (Non-uniqueness and computation)**.** *The base point is not part of the finite action data. If $\operatorname{rank} N_\sigma = 4$, the facet equations determine it uniquely when they are consistent. If $\operatorname{rank} N_\sigma < 4$, the inequalities may leave a positive-dimensional set of base points. This non-uniqueness does not affect the action, which is determined by the closed velocity polygon via Lemma 3.1. Computationally, base-point recovery is therefore a finite linear feasibility check after the word and dwell times have been chosen.*

3.3 Existence of simple minimizers

The next theorem is the reason a finite search can compute capacity. It is Haim–Kislev's simple-minimizer theorem [Hai19, Theorem 1.5], written with the contact-normalized velocities $R_i = 2J_0 a_i$.

Theorem 3.4 (Simple minimizer)**.** *Let $K \subset \mathbb{R}^4$ be a full-dimensional convex polytope with $0 \in \operatorname{int}(K)$. Among all generalized Reeb orbits on ∂K, there exists a minimum-action orbit that is simple in the contact-normalized convention above. Hence*

$$c_{\mathrm{EHZ}}(K) = \min\{A(\gamma) : \gamma \text{ is a simple Reeb orbit on } \partial K\}.$$

The difficulty is that splitting or rearranging velocities need not leave a curve on ∂K. We make those changes to closed curves in the dual problem instead. The changes preserve the total time and can be chosen to increase action. Rescaling all durations restores the action–period constraint. Dual minimality controls the possible action change, and a convergent subsequence then supplies a minimizer with the desired finite form.

During the proof, a *simple pure-velocity dual loop* means a closed piecewise affine curve using each R_i on at most one cyclic interval. It need not lie on the boundary. The next five lemmas give the approximation, rearrangement and compactness steps.

Lemma 3.5 (Piecewise affine approximation)**.** *Let $z \in W^{1,2}([0,T],\mathbb{R}^4)$ be closed and satisfy*

$$\dot z(t) \in \operatorname{conv}\{R_1,\dots,R_F\} \quad \text{for a.e. } t.$$

For every $\varepsilon > 0$, there is a closed piecewise affine curve z' with

$$\dot z'(t) \in \operatorname{conv}\{R_1,\dots,R_F\}$$

on each affine piece and $\|z' - z\|_{W^{1,2}} < \varepsilon$. Consequently $A(z') \to A(z)$ and $I_K(z') \to I_K(z)$ as $\varepsilon \to 0$.

Proof. Let $C = \operatorname{conv}\{R_1,\dots,R_F\}$. Averages over intervals of refining uniform partitions converge to $\dot z$ in L^2. Choose a partition $0 = t_0 < \cdots < t_m = T$ so that the conditional averages of $\dot z$ on the partition intervals are L^2-close to $\dot z$. On $[t_k, t_{k+1}]$, put

$$w_k = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \dot z(s)\, ds.$$

Because C is convex and closed, each w_k lies in C. Define z' by $z'(0) = z(0)$ and $\dot z' = w_k$ on $[t_k, t_{k+1}]$. Then

$$z'(T) - z'(0) = \int_0^T \dot z'(t)\, dt = \int_0^T \dot z(t)\, dt = 0,$$

so z' is closed. Since $\dot z' \to \dot z$ in L^2, integration gives $z' \to z$ uniformly, hence in $W^{1,2}$. The convergence of A follows from bilinearity in $(z, \dot z)$. Both $-J_0\dot z'$ and $-J_0\dot z$ take values in the compact set $-J_0 C$, on which h_K^2 is Lipschitz; hence the L^2-convergence also gives $I_K(z') \to I_K(z)$. □

Lemma 3.6 (Splitting mixed velocities)**.** *Let z be a closed piecewise affine curve of period T whose velocities lie in $\operatorname{conv}\{R_1,\dots,R_F\}$. Then there is a closed piecewise affine curve z', with the same period, whose velocity on each segment is one pure Reeb vector R_i, such that $A(z') \geq A(z)$ and*

$$I_K(z') = T.$$

Proof. On a segment with velocity $v = \sum_i \theta_i R_i$, $\theta_i \geq 0$ and $\sum_i \theta_i = 1$, replace the segment by subsegments with velocities R_i and lengths θ_i times the original length. The total displacement and period are unchanged. The only change in the action is the sum of the new internal cross-terms from Lemma 3.1. Reversing the order of the inserted subsegments changes the sign of this internal contribution, so one of the two orders does not decrease the action.

For a pure Reeb velocity $R_i = 2J_0 a_i$,

$$-J_0 R_i = 2a_i, \qquad h_K(-J_0 R_i) = 2h_K(a_i) = 2,$$

because $F_i \neq \varnothing$ and $\max_{x\in K}\langle a_i, x\rangle = 1$. Thus $h_K^2(-J_0 R_i) = 4$ on every pure segment, and

$$I_K(z') = \frac{1}{4}\int_0^T 4\, dt = T.$$

□

Lemma 3.7 (Merging repeated velocities)**.** *Let z be a closed piecewise affine curve whose velocity on each segment is one of the R_i. If some R_i occurs in two non-adjacent segments, then there is a rearrangement z' that merges those two occurrences into one contiguous segment and satisfies*

$$A(z') \geq A(z), \qquad I_K(z') = I_K(z).$$

Repeating this operation, and merging adjacent equal-velocity intervals, gives a closed piecewise affine curve in which each pure Reeb vector occurs in at most one interval. Here adjacency is understood cyclically, since the curve is closed.

Proof. Suppose a velocity v occurs in two segments I_r and I_s, with middle segments $M = \{r+1, \dots, s-1\}$. Here I_i is a time interval, $|I_i|$ is its duration, and w_i is the velocity on that interval. Merging I_s forward into I_r changes only the action terms involving v and the middle velocities, and the change is

$$\Delta A_{\mathrm{fwd}} = |I_s| \sum_{i \in M} |I_i|\, \omega_0(v, w_i).$$

Merging I_r backward into I_s gives

$$\Delta A_{\mathrm{bwd}} = -|I_r| \sum_{i \in M} |I_i|\, \omega_0(v, w_i).$$

The two changes have opposite signs, or both vanish, so one merge has nonnegative action change. Closure is preserved because the total duration assigned to each velocity is unchanged, and I_K is unchanged because it depends only on the velocities and their durations. □

Lemma 3.8 (Rescaling)**.** *Let z be a closed piecewise affine curve with pure Reeb velocities and period T. If all durations are multiplied by $\beta > 0$, the resulting curve z' satisfies*

$$A(z') = \beta^2 A(z), \qquad I_K(z') = \beta I_K(z).$$

Proof. In Lemma 3.1, every action term contains a product of two durations, so A scales by β^2. The integrand in I_K is unchanged and the time intervals are all multiplied by β, so I_K scales by β. □

Lemma 3.9 (Compactness of simple pure-velocity dual loops)**.** *Let z_n be simple pure-velocity dual loops whose periods satisfy $T_n \to T > 0$. After translating the curves and passing to a subsequence, z_n converges to a simple pure-velocity dual loop z_* with period T, and*

$$A(z_*) = \lim_n A(z_n), \qquad I_K(z_*) = T.$$

Proof. Choose the initial point of each periodic curve at a breakpoint, and insert zero dwell times for unused facet directions. Each curve is then determined up to translation by a permutation of the finitely many facet directions and a nonnegative dwell-time vector of total mass T_n. Pass first to a subsequence with a fixed permutation. Since $T_n \to T$, the dwell-time vectors lie in a common compact set and have a convergent subsequence whose entries sum to T. Translating the starting points gives uniform convergence after identifying each domain with $[0, 1]$ by linear time rescaling. Closure passes to the limit because $\sum_i \tau_i R_{\sigma(i)} = 0$ for every curve. Some limiting dwell times may be zero; deleting those zero-duration entries recovers the active word of the limiting simple curve. The action is continuous in the dwell-time data by Lemma 3.1. Since the remaining positive-duration segments still have pure Reeb velocities, the same computation as in Lemma 3.6 gives $I_K(z_*) = T$. □

Proof of Theorem 3.4. Choose a capacity-realizing generalized Reeb orbit γ. By Theorem 2.6, γ is also a dual minimizer in the thesis convention, with

$$I_K(\gamma) = A(\gamma) = T = c_{\mathrm{EHZ}}(K).$$

Moreover $\dot\gamma(t) \in \operatorname{conv}\{R_1, \dots, R_F\}$ for almost every t.

Choose, by Lemma 3.5, a sequence of closed piecewise affine curves $z_{1,n} \to \gamma$ in $W^{1,2}$, with velocities in the convex hull of the R_i. Apply Lemma 3.6 to obtain closed pure-velocity curves $z_{2,n}$, and then Lemma 3.7 to obtain closed simple pure-velocity curves $z_{3,n}$, such that

$$A(z_{3,n}) \geq A(z_{2,n}) \geq A(z_{1,n}), \qquad I_K(z_{3,n}) = T.$$

Since $A(z_{1,n}) \to A(\gamma) = T$, discard finitely many terms so that $A(z_{3,n}) > 0$. Rescale all durations of $z_{3,n}$ by

$$\beta_n = \frac{T}{A(z_{3,n})},$$

and denote the resulting curve by $z_{4,n}$. It satisfies

$$A(z_{4,n}) = \frac{T^2}{A(z_{3,n})}, \qquad I_K(z_{4,n}) = \frac{T^2}{A(z_{3,n})}.$$

Thus $z_{4,n}$ is feasible for the dual problem, with period $T^2/A(z_{3,n})$. Since γ is a dual minimizer, $I_K(z_{4,n}) \geq T$, and therefore $A(z_{3,n}) \leq T$. Together with $A(z_{3,n}) \geq A(z_{1,n}) \to T$, this gives

$$A(z_{3,n}) \to T, \qquad \beta_n \to 1, \qquad \operatorname{period}(z_{4,n}) \to T.$$

By Lemma 3.9, a subsequence of the $z_{4,n}$ converges up to translation to a simple pure-velocity dual loop z_* with $A(z_*) = T$ and $I_K(z_*) = T$. This curve is a dual minimizer. Applying Theorem 2.6 again gives a contact-normalized generalized Reeb orbit on ∂K in the same translate/rescale family. In the reconstruction part of that theorem, the corresponding characteristic has the form

$$t \longmapsto c\, z_*(t/c^2) + b/c$$

for some $c > 0$, with

$$c^2 = \nu = \frac{I_K(z_*)}{T} = 1$$

by (11). Thus the reconstruction only translates z_*. Its velocities remain the pure facet directions R_i, each used in one connected interval, so the resulting orbit is simple and has minimum action. The translation also recovers the prescribed facet labels. Indeed, on an interval with $\dot\gamma = R_i$, the reconstructed inclusion gives

$$2a_i = -J_0 R_i \in 2\operatorname{conv}\{a_j : \gamma(t) \in F_j\}.$$

Because the irredundant row a_i is an extreme point of K°, this forces F_i to be active along that interval. The reconstructed curve lies in K, so its starting point $\gamma(0)$ is a valid base point in Lemma 3.2; closure supplies the remaining displacement equation. □

The theorem supplies one simple capacity minimizer, which is enough to compute the capacity by a finite optimization. It does not require every minimizer to be simple. The next section expresses the action in terms of the word and normalized dwell times, obtaining Haim–Kislev's quadratic program.

4 The Haim–Kislev Quadratic Program

The preceding section showed that some capacity-minimizing generalized Reeb orbit uses a finite list of distinct facet directions. We now express its action in terms of that list and its dwell times. Haim–Kislev's formula turns the result into a finite quadratic program [Hai19].

The chapter has two tasks. First, it explains the variables and how to compute the global optimum by solving linear stationarity systems and comparing their values. Second, it proves reductions for Lagrangian products of planar polygons. Those reductions make the later family calculations practical.

4.1 The finite quadratic program

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x\rangle \le 1,\ i = 1, \dots, F\}$$

be a full-dimensional polytope with $0 \in \mathrm{int}(K)$, written as in Section 3. Thus $a_i = n_i/h_i$ if n_i is the outward unit normal to the i-th facet and $h_i = h_K(n_i) > 0$ is its height. Haim–Kislev state their formula in the (n_i, h_i)-variables. We use the equivalent dual-vertex variables

$$\beta_i^{\mathrm{here}} = \beta_i^{\mathrm{HK}} h_i.$$

We write β_i for β_i^{here} below. For a contact-normalized orbit represented by a pure facet word, β_i is the fraction of its period spent with velocity $2J_0a_i$. The weights therefore sum to one. Closure of the curve says that the weighted sum of its velocities vanishes; since J_0 is invertible, this is equivalent to $\sum_i \beta_i a_i = 0$. Thus the constraints are

$$\beta_i \ge 0, \qquad \sum_i \beta_i = 1, \qquad \sum_i \beta_i a_i = 0.$$

For an ordered facet word $\sigma = (\sigma_1, \dots, \sigma_m)$ with distinct entries, define

$$M_\sigma = \left\{ \beta \in \mathbb{R}^m_{\ge 0} : \sum_{k=1}^m \beta_k = 1,\ \sum_{k=1}^m \beta_k a_{\sigma_k} = 0 \right\}.$$

The quadratic objective for this active word is

$$Q_\sigma(\beta) = \sum_{1 \le j < k \le m} \beta_j \beta_k\, \omega_0(a_{\sigma_j}, a_{\sigma_k}). \tag{19}$$

Here σ is written in the Reeb traversal order used in Section 3. Haim–Kislev's displayed formula uses the reversed ordered word after translating their symplectic convention to ω_0. Since word reversal is a bijection on the ordered candidate family, the maximum is unchanged; the convention matters only when a single fixed word is interpreted.

Let

$$Q_{\max} = \max_\sigma \max_{\beta \in M_\sigma} Q_\sigma(\beta),$$

where σ ranges over ordered words with distinct facets and $M_\sigma \ne \varnothing$. Equivalently, one may range over all permutations of all facets and allow zero weights; the zero-weight entries are then deleted to recover the active word. This is only an enumeration convention: HK's theorem is the full-permutation statement, and the active-word form is obtained by deleting unused zero-weight facets and reindexing the remaining entries. This formulation intentionally permits duplicates of the same geometric orbit, for example from changing the starting segment.

Theorem 4.1 (Haim–Kislev quadratic program)**.** *For a full-dimensional convex polytope $K \subset \mathbb{R}^4$ with $0 \in \mathrm{int}(K)$, written in the facet normalization above, $Q_{\max} > 0$ and*

$$c_{\mathrm{EHZ}}(K) = \frac{1}{2Q_{\max}}.$$

Proof. Write $c = c_{\mathrm{EHZ}}(K)$. We derive the formula from the dual principle and the simple-minimizer theorem proved above.

First let $\beta \in M_\sigma$ have $q = Q_\sigma(\beta) > 0$, and set $T = 1/(2q)$. Traverse the velocities $R_{\sigma_k} = 2J_0 a_{\sigma_k}$ for times $T\beta_k$. The resulting curve z is closed because $\sum_k \beta_k a_{\sigma_k} = 0$. By Lemma 3.1,

$$2A(z) = 4T^2 Q_\sigma(\beta) = 2T.$$

Moreover, $h_K(a_i) = 1$ for each facet row, so $H_K^*(-J_0 R_i) = h_K(2a_i)^2/4 = 1$. Hence $I_K(z) = T = A(z)$: this curve is feasible for the dual problem (9). The dual action principle, Theorem 2.6, gives

$$c \le I_K(z) = \frac{1}{2q}, \qquad q \le \frac{1}{2c}.$$

Nonpositive feasible values also satisfy this last inequality, since $c > 0$. Thus $Q_{\max} \le 1/(2c)$.

For the reverse inequality, choose the simple minimum-action orbit supplied by Theorem 3.4. Its period is c. If its distinct facet word is σ and its dwell times are $\tau_k > 0$, put $\beta_k = \tau_k/c$. Closure and the period identity show that $\beta \in M_\sigma$. The same shoelace formula now gives

$$2c = 4c^2 Q_\sigma(\beta), \qquad Q_\sigma(\beta) = \frac{1}{2c} > 0.$$

Consequently $Q_{\max} \ge 1/(2c)$, proving both the formula and positivity. The maximum exists: there are finitely many words, each M_σ is compact, and the simple minimizer supplies a nonempty feasible set. □

This is Haim–Kislev's formula [Hai19, Theorem 1.1], with the change of weights and word convention described above. The derivation uses the dual action principle and Haim–Kislev's simple-minimizer theorem as its foundations. The next proposition identifies when the dual curve constructed in the proof can be placed on the boundary.

Proposition 4.2 (From finite weights to a dual curve)**.** *Let $\beta \in M_\sigma$ and put $q = Q_\sigma(\beta)$. If $q > 0$, define*

$$T = \frac{1}{2q}, \qquad \tau_k = T\beta_k.$$

Traversing the velocities R_{σ_k} for the durations τ_k, and deleting zero-duration pieces, gives a closed simple dual curve z with

$$A(z) = I_K(z) = T.$$

If (σ, β) attains $Q_{\max}$ across all words, then $T = c_{\mathrm{EHZ}}(K)$, and a translation of z is a minimum-action simple generalized Reeb orbit on ∂K, with the active word obtained from σ by deleting its zero-weight entries.

Conversely, if a simple generalized Reeb orbit has period T, active word σ, and positive dwell times τ_k, then $\beta_k = \tau_k/T$ lies in M_σ and $Q_\sigma(\beta) = 1/(2T)$.

Proof. The closure constraint gives

$$\sum_k \tau_k R_{\sigma_k} = 2TJ_0 \sum_k \beta_k a_{\sigma_k} = 0,$$

while $\sum_k \tau_k = T$. The shoelace formula gives

$$2A(z) = 4T^2 Q_\sigma(\beta) = 2T.$$

Every positive-duration piece has velocity R_i, for which

$$H_K^*(-J_0 R_i) = \frac{1}{4} h_K^2(2a_i) = 1.$$

Hence $I_K(z) = T$, and z is feasible for the free-period dual problem.

If (σ, β) attains $Q_{\max}$, the Haim–Kislev formula gives $T = 1/(2Q_{\max}) = c_{\mathrm{EHZ}}(K)$. Thus z is a dual minimizer. Its multiplier in (11) is $\nu = I_K(z)/T = 1$. The reconstruction in Theorem 2.6 therefore has scale one: it only translates z onto ∂K. Translation preserves the ordered pure-velocity pieces. On a piece with velocity R_i, the reconstructed inclusion is

$$2a_i = -J_0 R_i \in \partial H_K(\gamma) = 2\,\mathrm{conv}\{a_j : F_j \text{ is active at } \gamma\}.$$

The irredundant rows a_i are extreme points of K°, so this convex combination forces F_i to be active there. Thus, after zero durations are deleted, the resulting minimum-action orbit is simple and has the reduced active word.

For the converse, closure of the orbit gives $\sum_k \beta_k a_{\sigma_k} = 0$, and positivity and $\sum_k \beta_k = 1$ are immediate. Since $A(\gamma) = T$, the shoelace formula yields

$$2T = 4T^2 \sum_{1 \le r < s \le m} \frac{\tau_r}{T}\frac{\tau_s}{T}\,\omega_0(a_{\sigma_r}, a_{\sigma_s}) = 4T^2 Q_\sigma(\beta), \qquad \text{so} \qquad Q_\sigma(\beta) = \frac{1}{2T}.$$

□

4.2 Solving one word and comparing the results

For a fixed word, the constraints are linear and the objective is quadratic. Write

$$C_\sigma = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \qquad e = (0,0,0,0,1)^T, \qquad Q_\sigma(\beta) = \tfrac{1}{2}\beta^T H_\sigma \beta.$$

Here H_σ is symmetric with zero diagonal and, for $j < k$, $(H_\sigma)_{jk} = (H_\sigma)_{kj} = \omega_0(a_{\sigma_j}, a_{\sigma_k})$. Thus the problem is to maximize this quadratic form subject to $C_\sigma \beta = e$ and $\beta \ge 0$.

First suppose that the maximizing weights are all positive. Small perturbations in $\ker C_\sigma$ remain feasible, so the first derivative must vanish on that kernel. Equivalently, there is a vector of multipliers λ such that

$$H_\sigma \beta = C_\sigma^T \lambda, \qquad C_\sigma \beta = e.$$

These are the equality-constrained Karush–Kuhn–Tucker (KKT) equations:

$$\begin{pmatrix} H_\sigma & -C_\sigma^T \\ C_\sigma & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ e \end{pmatrix}. \tag{20}$$

They are a linear system even though the objective is quadratic. After solving it, one checks whether the weights are positive and evaluates Q. A stationary point can be a minimum or a saddle, so this calculation does not yet identify the maximum.

Example 4.3 (A four-facet calculation)**.** *Take $K = [-1,1]^4$ and the ordered facet rows $(e_1, e_3, -e_1, -e_3)$, where the coordinate order is (q_1, q_2, p_1, p_2). Closure forces the weights to have the form (x, y, x, y), and normalization gives $x + y = 1/2$. The ordered symplectic pairings give*

$$Q = 2xy = 2x(1/2 - x), \qquad 0 \le x \le 1/2.$$

Thus stationarity reduces to $1 - 4x = 0$. The second derivative is -4, so $x = y = 1/4$ maximizes this word, with $Q = 1/8$; both boundary choices have value zero. The corresponding capacity upper bound is 4. Identifying the capacity still requires a global comparison or another argument giving the matching lower bound.

If a maximum has zero weights, delete those entries from the word. The remaining weights are positive, and deleting zeros changes neither closure nor the objective. It follows that solving all active words covers maxima on every face of the nonnegative feasible set. Conversely, a shorter word can be padded with zero weights to recover a full-facet permutation. Comparing the feasible stationary values over this complete family therefore gives the global maximum in Haim–Kislev's formula.

A singular system requires care, but not a search over infinitely many objective values. Suppose (β, λ) and $(\beta + z, \lambda + \mu)$ solve (20). Then $C_\sigma z = 0$ and $H_\sigma z = C_\sigma^T \mu$, whence

$$Q_\sigma(\beta + z) - Q_\sigma(\beta) = z^T H_\sigma \beta + \tfrac{1}{2} z^T H_\sigma z = \lambda^T C_\sigma z + \tfrac{1}{2} \mu^T C_\sigma z = 0.$$

Thus all solutions of the same stationary system have the same quadratic value. One must still decide whether that affine solution set contains positive weights. A singular matrix by itself means neither infeasibility nor failure of the capacity formula. This observation will also be used in the pentagon classification.

A feasible positive value q gives a dual curve with value $1/(2q)$. Capacity is the infimum of the dual values, so $c_{\mathrm{EHZ}}(K) \le 1/(2q)$. Once q is proved to be the largest value across all words, that curve is a dual minimizer and can be translated onto the boundary. For a nonglobal candidate, boundary placement instead requires the conditions in Lemma 3.2.

The full enumeration can be reduced when a geometric argument proves that a smaller family still contains a capacity minimizer. For example, an actual transition from facet F_i to F_j requires an intersection of those facets and the directed sign $\omega_0(a_i, a_j) \ge 0$. These necessary conditions can exclude words before solving their systems. They need not hold for every feasible dual curve; what makes them valid for computing capacity is the existence of a boundary-realized global minimizer. A stronger reduction for Lagrangian products is proved next.

4.3 Finite enumeration for Lagrangian products

Let $K_q, K_p \subset \mathbb{R}^2$ be nondegenerate convex polygons containing the origin in their interiors, and let

$$K = K_q \times_L K_p \subset \mathbb{R}^2_q \oplus \mathbb{R}^2_p$$

be their Lagrangian product. A facet of K is either a q-facet, coming from an edge of K_q, or a p-facet, coming from an edge of K_p. We call a single facet a one-facet block, and an ordered adjacent pair of facets of the same type a two-facet block.

Theorem 4.4 (Finite enumeration for Lagrangian products)**.** *For a Lagrangian product $K = K_q \times_L K_p$ of nondegenerate planar convex polygons containing the origin in their interiors, $c_{\mathrm{EHZ}}(K)$ is obtained by maximizing Q_σ over the Haim–Kislev quadratic programs associated to cyclic words formed by k alternating q-blocks and p-blocks, where $k \in \{2, 3\}$. Each block contains one or two facets from its factor, distinct blocks use disjoint facets, and a two-facet block must use adjacent polygon edges. Thus every word in the family uses at most twelve facets. This is an exhaustive candidate family for the capacity value; it may contain overcounting or infeasible words, and it is not a classification of all minimum-action simple Reeb orbits.*

Proof. We first specify the billiard data used in the proof. For a convex body C and $x \in \partial C$, write

$$N_C(x) = \{u : \langle u, y - x \rangle \le 0 \text{ for every } y \in C\}$$

for the outward normal cone. A closed *strong Minkowski billiard* in K_q is a polygonal curve with bounce points $q_1, \dots, q_m \in \partial K_q$, $m \ge 2$, admitting a dual polygon with vertices $p_1, \dots, p_m \in \partial K_p$ such that

$$q_{j+1} - q_j \in N_{K_p}(p_j), \qquad p_{j+1} - p_j \in -N_{K_q}(q_{j+1}).$$

Indices are cyclic; consecutive q_j are distinct, and no bounce point lies between its two neighbors on a straight segment. Its length is

$$\ell_{K_p}(q) = \sum_j h_{K_p}(q_{j+1} - q_j) = \sum_j \langle p_j, q_{j+1} - q_j \rangle.$$

Here the support function uses the Euclidean pairing between the p-plane and the q-plane: $h_{K_p}(u) = \max_{p \in K_p} \langle p, u \rangle$. The last equality follows from the first normal-cone relation and $p_j \in \partial K_p$.

Rudolf's theorem applies to arbitrary convex bodies in $\mathbb{R}^2$, including polygons: the least strong-billiard length equals $T = c_{\mathrm{EHZ}}(K_q \times_L K_p)$, and a minimizing pair (q, p) exists with $m \leq 3$ [Rud24, Definition 2 and Theorem 1]. No smoothness or strict convexity is required. Choose such a pair and consider the alternating broken line through

$$(q_j, p_j),\ (q_{j+1}, p_j),\ (q_{j+1}, p_{j+1}).$$

On a q-leg, the fixed coordinate p_j lies on ∂K_p, and convexity keeps the moving coordinate inside K_q; the corresponding statement holds on a p-leg. Thus the entire line lies on the product boundary. Its two leg velocities have the forms $(u, 0) = -J_0(0, u)$ and $(0, -v) = -J_0(v, 0)$, with $u \in N_{K_p}(p_j)$ and $v \in N_{K_q}(q_{j+1})$. It is therefore a generalized characteristic for Rudolf's complex-structure convention. That convention is opposite to ours, so reverse the loop. Since λ_0 differs from $-p\,dq$ by an exact form, its action in our convention is

$$-\oint_{\mathrm{rev}} p\,dq = \oint_{\mathrm{orig}} p\,dq = \sum_j \langle p_j, q_{j+1} - q_j \rangle = \ell_{K_p}(q) = T.$$

Thus the reversed lift is a capacity-realizing generalized characteristic on $\partial(K_q \times_L K_p)$ whose legs alternately move only one factor. Parametrize it in the contact normalization. Its period is then T, and, as a dual curve z, it satisfies

$$A(z) = I_K(z) = T.$$

The following surgery is performed on z as a competitor in the dual variational problem, not as an orbit already constrained to ∂K. In particular, splitting a normal-cone direction into pure facet directions does not assert that the corresponding broken path, drawn directly in K_q or K_p, would remain inside that polygon. Boundary placement is recovered only after the surgery has been shown to preserve dual minimality.

At the level of a cyclic velocity word, we split each mixed leg into pure facet directions, transfer the duration of a repeated direction to one of its occurrences and delete the other, then rescale all durations together if needed. Apply this splitting–merging–rescaling part of the proof of Theorem 3.4 to z. Approximation and compactness are unnecessary here because the billiard lift is already a finite broken line. The only extra point to retain from the billiard is its block bound. On each leg, the polygonal normal cone is generated by one facet normal or by the two adjacent normals at a vertex, so splitting replaces that leg by one or two pure directions inside the same block. When a repeated direction is merged, it is removed from one same-type block and absorbed into another block that already contains it. Thus merging cannot increase the number of blocks of either factor, although it may decrease them. If the source block is a singleton, deleting it coalesces its two neighboring opposite-type blocks and reduces both block counts by one. This possible loss is harmless: only the inherited upper bound of three blocks of each type is needed below.

The rescaling argument in the cited proof uses dual minimality to show that the resulting distinct-facet curve is again a dual minimizer. Indeed, write $A' \geq T$ for its action before

rescaling; its dual value is still T. Multiplying every duration by T/A' gives a dual-feasible curve with

$$A = I_K = \frac{T^2}{A'}.$$

The dual minimum is T, so $T^2/A' \geq T$, hence $A' \leq T$. Therefore $A' = T$: no rescaling is needed, and the distinct-facet curve z' is a dual minimizer with $A(z') = I_K(z') = T$. By the preceding observations, its cyclic word still has at most three blocks of each type.

Only now do we return to the primal problem. Normalize the positive dwell times of z' by their sum. The resulting finite weights have QP value $1/(2T) = Q_{\max}$, so Proposition 4.2 translates z' to a simple generalized Reeb orbit with the same active word on $\partial(K_q \times_L K_p)$. Its factor coordinates therefore remain in K_q and K_p; this containment comes from dual-minimizer reconstruction, not from the splitting operation. A simple orbit cannot contain three consecutive facets of the same type. Indeed, along three consecutive q-facet segments the q-coordinate is fixed; the two breakpoints would place it at both endpoints of the middle edge of K_q, a contradiction. The same holds for p-facets. Hence every maximal block in the final word has length one or two, and a two-facet block uses adjacent edges.

If only one factor type occurred, the whole word would be a single block of length at most two, whose positive closure equation would require one outward normal, or a positive combination of two adjacent outward normals, to vanish. This is impossible. Thus both types occur. Their maximal blocks alternate cyclically, so there is a common number $k \leq 3$ of q- and p-blocks. One q-block and one p-block cannot close, by the same argument in each factor, so $k \neq 1$. Hence $k \in \{2, 3\}$. Since each block contains at most two facets, the resulting simple word uses at most $4k \leq 12$ distinct facets. This proves that the restricted family contains a capacity maximizer. It does not exclude other minimum-action orbits with more blocks. □

Algorithm 4.5 (Lagrangian-product capacity enumeration)**.** *For each factor, enumerate all one-facet blocks and all ordered adjacent two-facet blocks. For $k = 2, 3$, choose k non-overlapping q-blocks and k non-overlapping p-blocks, order them cyclically in the alternating pattern q-block, p-block, q-block, p-block, and so on. Read the facet labels in that order, discard infeasible words, and solve the corresponding Haim–Kislev quadratic program. The capacity is $1/(2Q_{\max})$, where $Q_{\max}$ is the largest objective value among the feasible words in this candidate family.*

4.4 A six-facet reduction for the product capacity

Algorithms for polygonal Lagrangian products have also been developed through Minkowski billiards. Krupp and Rudolf enumerate face choices for two- and three-bounce trajectories and solve the resulting geometric problems using linear programming [KR22, Section 7.3]. The reduction below works instead with the balanced facet weights in the Haim–Kislev formula: it replaces the continuous weight search by pairs of vertices of two closure polytopes.

The product structure gives a reduction that is unavailable for a general polytope. The q- and p-coordinate planes are Lagrangian, so pairings between two q-facets or two p-facets vanish. After separating the total weight in each factor from its normalized distribution, the objective is therefore linear in either distribution when the other is fixed. Maximizing these two linear functions at vertices is the mechanism behind both the six-facet bound and the following algorithm.

Let $u_1, \ldots, u_r$ be the planar dual vertices of K_q, and let $v_1, \ldots, v_s$ be those of K_p. Define

the normalized closure polytopes

$$C_q = \left\{ \alpha \in \mathbb{R}^r_{\geq 0} : \sum_i \alpha_i = 1,\ \sum_i \alpha_i u_i = 0 \right\},$$
$$C_p = \left\{ \gamma \in \mathbb{R}^s_{\geq 0} : \sum_j \gamma_j = 1,\ \sum_j \gamma_j v_j = 0 \right\}.$$

Theorem 4.6 (Six-facet product maximizer)**.** *For a Lagrangian product $K_q \times_L K_p$ of nondegenerate planar convex polygons containing the origin in their interiors, the complete Haim–Kislev family has a global maximizer using at most three q-facets and at most three p-facets. The total weight in each factor is $1/2$. Consequently the capacity is represented by a word of length at most six.*

Algorithm 4.7 (Product capacity from closure vertices)**.** *Enumerate the vertices of C_q and C_p. For each pair, enumerate the cyclic orders of the union of their supports, give each factor total weight $1/2$, and evaluate Q_σ. The largest value is $Q_{\max}$. Each support union has at most six facets, so fixing one cyclic starting point leaves at most $5! = 120$ orders.*

an arbitrary global maximizer (σ, β)
↓ fix γ, maximize the linear function in α
α is a vertex of C_q, hence $|\operatorname{supp}\alpha| \leq 3$
↓ fix α, maximize the linear function in γ
γ is a vertex of C_p, hence $|\operatorname{supp}\gamma| \leq 3$
↓ maximize $t(1-t)$ and delete zero weights
a global maximizer with $t = \frac{1}{2}$ and at most six facets

Figure 3: The proof mechanism for the six-facet reduction. Linearity in each factor distribution is the product-specific step.

Proof. Choose a global maximizer and let I_q and I_p be its q- and p-positions. Its objective is positive by Theorem 4.1. A weight vector supported in only one factor has objective zero, because every same-factor symplectic pairing vanishes. Thus both factor weights are nonzero. Put

$$t = \sum_{k \in I_q} \beta_k, \qquad \alpha_k = \frac{\beta_k}{t}\ (k \in I_q), \qquad \gamma_k = \frac{\beta_k}{1-t}\ (k \in I_p).$$

Closure splits between the factors, so the zero-extended vectors α and γ lie in the faces

$$C_q(\sigma) = \{\alpha \in C_q : \alpha_i = 0 \text{ if } q\text{-facet } i \text{ is omitted from } \sigma\},$$

and likewise $C_p(\sigma) \subset C_p$. Since the symplectic form vanishes on each factor, there is a bilinear function

$$B_\sigma : C_q(\sigma) \times C_p(\sigma) \longrightarrow \mathbb{R}$$

such that

$$Q_\sigma(\beta) = t(1-t) B_\sigma(\alpha, \gamma). \tag{21}$$

Its coefficients are the signed pairings $\langle u_i, v_j \rangle$, with signs determined by the order in σ.

With γ fixed, the bilinear factor is linear in α, so it has a maximizer at a vertex of $C_q(\sigma)$. Replacing α by that vertex, while keeping t, γ and σ fixed, preserves feasibility and cannot decrease the objective. It cannot increase it either, since the original value was the global maximum over all words and weights. The replacement therefore retains the positive value $Q_{\max}$.

Fixing the new α and repeating the same argument in $C_p(\sigma)$ gives another global maximizer for which both normalized factor weights are vertices of the support-restricted closure polytopes. Coordinate restrictions define faces of C_q and C_p, and a vertex of a face is a vertex of the ambient polytope.

A vertex of

$$\left\{ x \geq 0 : \sum_i x_i = 1,\ \sum_i x_i w_i = 0 \right\}, \qquad w_i \in \mathbb{R}^2,$$

has at most three positive coordinates. Indeed, more than three positive columns in the three scalar equality constraints have a nontrivial linear dependence. Extend its coefficients by zero to a nonzero vector d supported on the positive coordinates of x. Both equality constraints are preserved by $x \pm \varepsilon d$, and choosing $0 < \varepsilon < \min_{d_i \neq 0} x_i/|d_i|$ keeps both vectors nonnegative. They are distinct feasible points with midpoint x, contradicting extremality. Thus each factor uses at most three facets.

For the selected vertices the bilinear value is positive. Varying only the total factor weight gives

$$\tau(1-\tau)B_\sigma(\alpha,\gamma), \qquad 0 \leq \tau \leq 1,$$

which is uniquely maximal at $\tau = 1/2$. Every such mass replacement is feasible, so global maximality forces $t = 1/2$. Delete the remaining zero-weight positions. The relative cyclic order and objective are unchanged, and the complete Haim–Kislev family contains the resulting word of length at most six.

The same argument proves Algorithm 4.7: the selected vertices occur in the enumeration, while every enumerated candidate is feasible for the complete Haim–Kislev family and therefore cannot exceed $Q_{\max}$. □

Remark 4.8 (Why the reduction is product-specific)**.** *The argument does not extend directly to a general polytope. It uses that each factor is Lagrangian: same-factor pairings vanish, leaving a bilinear function which is linear in one normalized closure distribution after the other is fixed. Without such a splitting, the objective remains quadratic in the full weight vector, and moving to a vertex of the closure polytope need not preserve or improve it.*

The theorem asserts the existence of one sparse capacity maximizer. It neither classifies all minimizing orbits nor says that every maximizer has support at most six. The preceding twelve-facet block family remains useful when an argument must compare a specified family of geometric branches. No genericity assumption is needed: the closure polytopes are closed, and degeneracy can only lower the rank used in the support bound.

4.5 Use in later certificates

The later proofs use these results in two ways. The HKO local-maximum proof uses selected active words to build smooth upper bounds. The rotated-pentagon proof compares fixed feasible choices at different angles using the quadratic formula directly.

The numerical implementation follows the same distinction between general polytopes and products. In the general case it examines the candidate words and their stationary points, using proved exclusion criteria to reduce the search. It returns an interval containing the capacity. For a recognized Lagrangian product it instead enumerates the closure-polytope vertices from Algorithm 4.7 and compares the resulting objective values. This route returns the exact capacity of the supplied input, whose floating-point coordinates are treated as exact rational numbers. Both routes use interval bounds to control numerical error and exact arithmetic when a comparison cannot otherwise be decided. Section 14 describes those checks. The HKO proof uses a separate SageMath program over its number field.

The QP is therefore the appropriate language when the global objective value, its competing branches, or inequalities between them matter. The next section keeps the same simple facet

words but asks a different geometric question: it propagates actual breakpoints through facet-pair sections and detects a closed orbit as a fixed point of an affine return map. This second route is conditional on explicit regularity assumptions, but it exposes the trajectory geometry and provides an independent implementation comparison.

5 Computing closed trajectories with a flow graph

The quadratic program computes capacity by comparing possible facet orders and weights. A second approach follows trajectories through the facets themselves. In the relative interior of a facet, the contact-normalized velocity is constant. Following a chosen facet direction carries a point on an incoming two-dimensional section to an outgoing section by an affine map. Composing these maps around a facet word produces a return map; a fixed point gives a closed trajectory.

We use the local passage geometry of Chaidez and Hutchings [CH21]. The exhaustive search below is justified by the simple-minimizer theorem: some capacity-minimizing trajectory uses each of its facet directions once. It is therefore enough to search finitely many cyclic words, retaining the conditions that make each passage possible. We state the regularity hypotheses before the correctness theorem and discuss the rational implementation after its proof.

5.1 Affine tubes

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x\rangle \le 1,\ i = 1, \dots, F\}$$

be a bounded full-dimensional polytope with $0 \in \operatorname{int} K$, and assume that all listed inequalities define genuine facets F_i. Put

$$R_i = 2J_0 a_i.$$

As in Section 3, a trajectory on F_i with velocity R_i has action equal to its elapsed time.

For each ordered pair (i, j) for which a_i, a_j are linearly independent, choose affine coordinates

$$\psi_{ij} : \mathbb{R}^2 \longrightarrow H_i \cap H_j, \qquad H_i = \{x : \langle a_i, x\rangle = 1\},$$

and let

$$P_{ij} = \{u \in \mathbb{R}^2 : \psi_{ij}(u) \in K\}.$$

The set P_{ij} is allowed to be empty or lower-dimensional. Thus the construction also permits a breakpoint to lie on additional facets; it does not assume that every breakpoint lies in the relative interior of a two-face.

Consider a passage from facet F_p along F_i to F_n, with three distinct facet labels. We consider the case where the incoming direction approaches H_i and the direction along F_i approaches H_n:

$$\langle R_p, a_i\rangle > 0, \qquad \langle R_i, a_n\rangle > 0.$$

For $x \in F_p \cap F_i$, the time at which the line $x + tR_i$ reaches H_n is

$$\tau_{pin}(x) = \frac{1 - \langle a_n, x\rangle}{\langle a_n, R_i\rangle}. \tag{22}$$

The denominator is positive. Restrict the starting section to those points for which $x \in K$, $x + \tau_{pin}(x)R_i \in K$, and $\tau_{pin}(x) \ge 0$. On this closed polygonal domain, the endpoint map

$$\varphi_{pin}(x) = x + \tau_{pin}(x)R_i$$

and the action contribution τ_{pin} are affine. Convexity of K ensures that the whole segment lies in $K \cap H_i$.

We describe a tube by a closed starting polygon P, an affine endpoint map Φ, and an affine total-action function A. If (P_1, Φ_1, A_1) and (P_2, Φ_2, A_2) are consecutive tubes in matching coordinates, their concatenation is

$$(P_1 \cap \Phi_1^{-1}(P_2),\ \Phi_2 \circ \Phi_1,\ A_1 + A_2 \circ \Phi_1). \tag{23}$$

Thus an empty intermediate intersection excludes every longer word containing that partial tube.

A *simple cyclic facet word* is a cyclic word $w = [s_1, \ldots, s_k]$ with distinct entries. To close it, concatenate the primitive tubes for

$$(s_1, \ldots, s_k, s_1, s_2).$$

Each primitive consumes three consecutive labels: the first passage is (s_1, s_2, s_3), from $H_{s_1} \cap H_{s_2}$ to $H_{s_2} \cap H_{s_3}$. The appended labels make the last passage (s_k, s_1, s_2), returning to the initial section. Write the resulting data as (P_w, Φ_w, A_w). A point $u \in P_w$ is a closed search candidate precisely when $\Phi_w(u) = u$. It represents the displayed word as a strict generalized Reeb orbit precisely when every reconstructed segment time is positive. In that case its action is $A_w(u)$.

Lemma 5.1 (Tube semantics)**.** *For every simple cyclic word whose primitive denominators are positive, the fixed points of Φ_w in P_w are exactly its closed nonnegative-time tube realizations. The fixed points with all segment times positive are exactly the strict closed generalized Reeb orbits represented by w, and their action is A_w.*

Proof. Equation (22) gives the unique next breakpoint and elapsed time for one passage. The endpoint and the elapsed time are affine, and the domain inequalities state exactly that the endpoints and the passage belong to K. Formula (23) says that two passages can be concatenated exactly when their common breakpoint agrees; it also composes their endpoint maps and adds their elapsed times. Induction over the word proves the assertion for the closed tube. Its starting and ending sections have the same coordinates, so closure is equivalent to $\Phi_w(u) = u$. Finally, the closed tube was defined with nonnegative times, whereas the active word omits zero-time passages and records only the positive-duration pieces. Thus the entire displayed word is active exactly when every passage time is positive. Since $\lambda_0(R_i) = 1$ on F_i, the sum of the elapsed times is the Reeb action. □

Example 5.2 (A return map on a two-dimensional section)**.** *Figure 4 follows the word $w = [1, 2, 4, 5, 3]$ on a six-facet polytope. For this example we number the facets from 0 to 5, as in the figure. The normalized facet rows, rounded here to four decimal places, are*

$$\begin{pmatrix} -0.2091 & -0.8088 & 1.6079 & -0.0729 \\ 0.2405 & 0.5222 & 0.6397 & -0.2308 \\ -0.8794 & -0.3786 & 0.2356 & 0.0382 \\ 0.2108 & -0.2660 & 0.1605 & -0.3711 \\ -0.3527 & 0.1313 & -0.4414 & 0.0589 \\ -0.1226 & 0.1707 & -0.1480 & 0.4331 \end{pmatrix}.$$

The exact dyadic rows and all rational calculations for this example are stored in `experiments/dev-flow-graph/visualize-tube/flow-graph-f6-tube.json` *(the six-facet instance with seed 20260605 and attempt 3); see also Section 15. The rounded numbers below illustrate the calculation; the exact values decide membership and closure.*

Start on $F_1 \cap F_2$, immediately before the passage along F_2. A convenient affine chart uses the last two ambient coordinates:

$$\psi_{12}(u, v) \simeq \begin{pmatrix} -2.446722 \\ 3.041760 \\ 0 \\ 0 \end{pmatrix} + u \begin{pmatrix} 0.991996 \\ -1.681765 \\ 1 \\ 0 \end{pmatrix} + v \begin{pmatrix} -0.183156 \\ 0.526320 \\ 0 \\ 1 \end{pmatrix}.$$

The five passages take this section successively to $F_2 \cap F_4$, $F_4 \cap F_5$, $F_5 \cap F_3$, $F_3 \cap F_1$, and back to $F_1 \cap F_2$. Composing their affine maps gives

$$\Phi_w(u,v) \simeq \begin{pmatrix} -1.601002 & 1.364891 \\ 4.482887 & -4.446374 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} 1.626615 \\ -1.821973 \end{pmatrix},$$

$$A_w(u,v) \simeq 36.063319 + 6.721729u - 6.397030v.$$

The fixed point is $(u,v) \simeq (0.791856, 0.317243)$. It belongs to the closed tube, and its five passage times are approximately

$$0.538394, \quad 5.134160, \quad 14.681452, \quad 16.367140, \quad 2.635397.$$

They are all positive and sum to the orbit's action, approximately 39.356542. *Thus the fixed-point calculation produces a closed orbit. Finding the capacity still requires comparison with the other words.*

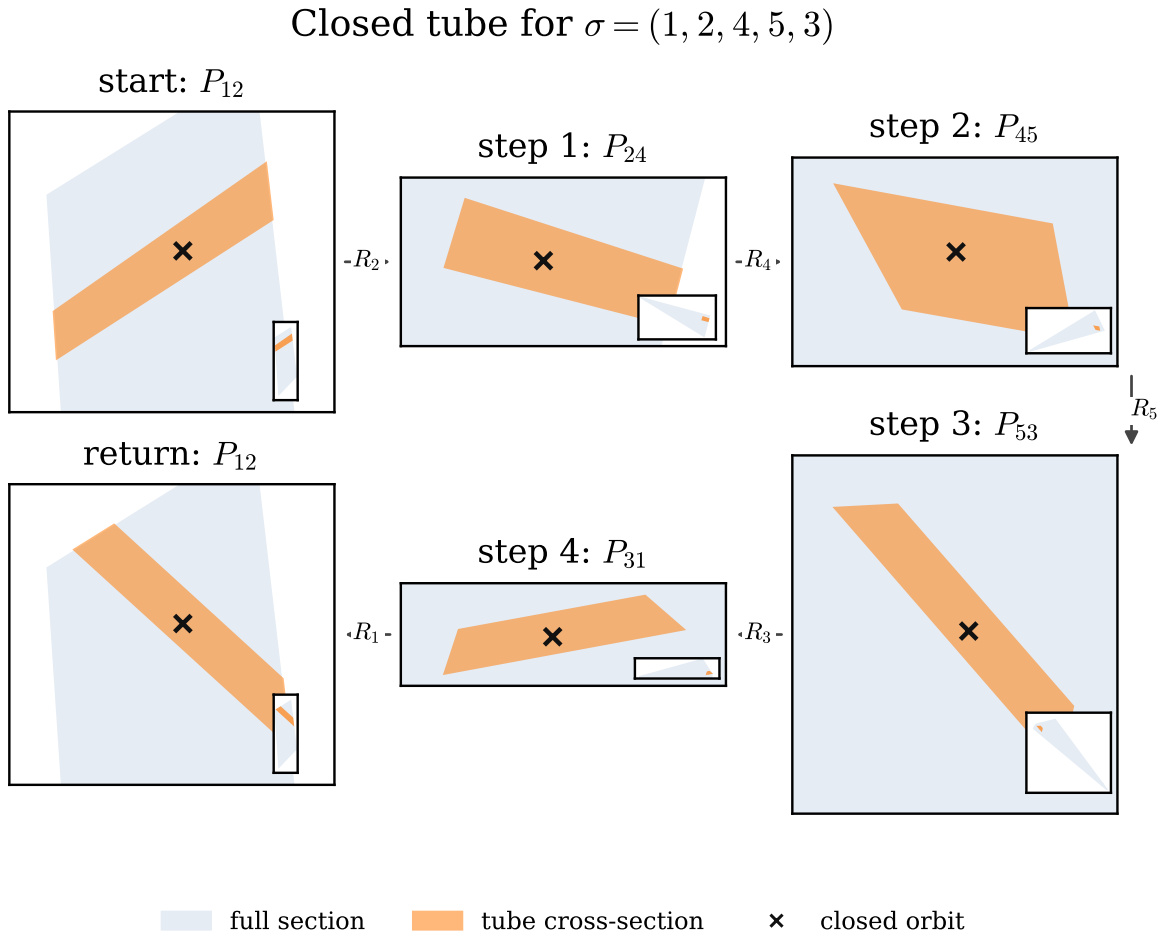


Figure 4: The five passages of Example 5.2. The colored polygons are the successive sections of the tube; each inset locates the displayed region in the full facet-pair section. The black cross follows the closed orbit. Each panel uses a Euclidean-orthonormal frame, so its coordinates differ from the affine chart used in the text. The start and return panels share a frame and scale: closure returns the cross to the same position.

5.2 Exhaustive search and correctness

The fixed-point calculation is simplest when every surviving return map has a unique fixed point. The following hypotheses make this precise and also justify the two pruning rules used in the search.

Definition 5.3 (Flow-graph regular presentation)**.** *The facet presentation above is* flow-graph regular *if:*

1. *for distinct i, j, $F_i \cap F_j \neq \varnothing$ implies $\omega_0(a_i, a_j) \neq 0$;*
2. *every sublist of at most four of the vectors $a_1, \dots, a_F$ is linearly independent; and*
3. *for every simple cyclic word of length at least five which survives the facet-intersection and positive-transition tests and has nonempty closed tube domain, the linear part M_w of Φ_w satisfies $\det(I - M_w) \neq 0$.*

The last condition is called finite-orbit regularity*.*

Algorithm 5.4 (Idealized exhaustive flow-graph search)**.** *Enumerate every simple cyclic facet word once. For each word:*

1. *discard it if an adjacent facet intersection is empty or an adjacent transition sign is not positive;*
2. *discard it if its length is at most four: under the regularity hypothesis, these facet directions are linearly independent and cannot form a closed positive-time path;*
3. *construct its closed tube (P_w, Φ_w, A_w); discard the word if $P_w = \varnothing$, and otherwise solve $\Phi_w(u) = u$;*
4. *reject a singular fixed-point equation as outside the regular case, and otherwise record the unique solution only if it lies in P_w and every reconstructed segment time is positive.*

Return the least positive recorded action.

We justify both pruning steps. If a strict trajectory changes from R_i to R_j at $x \in F_i \cap F_j$, then the inequalities for K immediately before the breakpoint give

$$\langle a_j, R_i \rangle \geq 0.$$

The first condition in Definition 5.3 excludes equality. Thus every trajectory transition has the positive sign used by the search, even when x lies on additional facets. Second, a strict closed orbit with active word $(s_1, \dots, s_k)$ satisfies

$$\sum_{r=1}^{k} \tau_r R_{s_r} = 0, \qquad \tau_r > 0.$$

If $k \leq 4$, this contradicts the linear independence in condition 2.

Theorem 5.5 (Idealized real-arithmetic flow-graph correctness)**.** *Let $K \subset \mathbb{R}^4$ be a bounded full-dimensional convex polytope with $0 \in \operatorname{int} K$, equipped with a flow-graph regular ordered normalized irredundant facet presentation. Then the finite procedure in Algorithm 5.4, with its algebraic and order operations performed exactly over $\mathbb{R}$, returns*

$$c_{\mathrm{EHZ}}(K).$$

Proof. There are finitely many simple cyclic words. By Lemma 5.1, every recorded point is a strict closed generalized Reeb orbit with the recorded action. The strict transition-sign argument and the short-word linear-independence argument above show that neither pruning rule can remove such an orbit. Empty tubes contain no orbit and are discarded before solving a fixed-point equation. Finite-orbit regularity makes every remaining fixed-point equation nonsingular, so no long word reaches the rejection branch and each word has at most one fixed point.

By Theorem 3.4, a capacity-minimizing orbit can be chosen with an active cyclic word in which every used facet direction occurs in one positive-duration interval. Its breakpoints lie in the corresponding facet-pair sections, its transitions have the required positive signs, and its passages satisfy the primitive tube equations. Lemma 5.1 therefore represents it by a strict fixed point with action $c_{\mathrm{EHZ}}(K)$. The returned minimum is no larger than this action, while soundness of every recorded orbit makes it no smaller. □

Corollary 5.6 (Exact rational-arithmetic flow-graph correctness)**.** *Under the hypotheses of Theorem 5.5, assume in addition that $a_i \in \mathbb{Q}^4$ for every facet row. Then rational affine coordinates can be chosen for every facet-pair plane, and Algorithm 5.4 is a finite exact rational-arithmetic algorithm returning $c_{\mathrm{EHZ}}(K)$.*

Proof. Rational Gaussian elimination gives a rational base point and basis for every facet-pair plane. In these coordinates, the section inequalities, primitive times and maps, tube intersections and compositions, and nonsingular fixed-point systems all have rational coefficients. Their feasibility, signs, solutions, and action comparisons can therefore be decided in exact rational arithmetic. The conclusion follows from Theorem 5.5. □

5.3 Why the regular class is generic

The regularity assumptions are explicit algebraic nonvanishing conditions. For fixed facet count there are only finitely many of them. Pairwise symplectic degeneracy and short-word dependence are ordinary polynomial conditions. The only non-immediate point is that the long-word determinant is not the zero rational function.

When all cyclic primitive denominators of a word are nonzero, let M_w^{alg} denote the composition of the primitive linear maps, regardless of their signs or whether the geometric tube domain is nonempty. For a searched tube, this is the linear part M_w of its affine return map.

Proposition 5.7 (Presentation-chamber genericity)**.** *Fix F, and let $\mathcal{C} \subset (\mathbb{R}^4)^F$ be a nonempty Euclidean-open set of ordered normalized dual-row lists defining bounded full-dimensional polytopes with 0 in the interior and all F inequalities irredundant. An open dense subset of $\mathcal{C}$ satisfies the stronger conditions*

1. *$\omega_0(a_i, a_j) \neq 0$ for every $i \neq j$;*
2. *every sublist of at most four rows is linearly independent; and*
3. *$\det(I - M_w^{\mathrm{alg}}) \neq 0$ for every simple cyclic word w of length at least five.*

Consequently every presentation in this subset satisfies Theorem 5.5, and every rational presentation in it satisfies Corollary 5.6.

Proof. The first two families are finite collections of nonzero polynomial conditions. Fix a simple word $w = (s_1, \dots, s_k)$, and write $b_r = a_{s_r}$, with cyclic indices. For its nonvanishing argument the k rows vary independently in the affine space $X_k = (\mathbb{R}^4)^k$. Let $D_k \subset X_k$ be the set where all cyclic pairings $\langle b_{r+2}, 2J_0 b_{r+1} \rangle$ are nonzero. They imply that every adjacent pair is linearly

independent, so $V_{pi} = \ker b_p \cap \ker b_i$ is a two-dimensional section direction. The primitive linear map is

$$D_{pin} : V_{pi} \longrightarrow V_{in}, \qquad D_{pin}\xi = \xi - \frac{\langle b_n, \xi\rangle}{\langle b_n, 2J_0 b_i\rangle} 2J_0 b_i.$$

Choosing a nonzero coordinate minor for each adjacent pair gives bases for these planes that depend rationally on the rows. Thus the return-map determinant is rational on each such chart; its values agree on overlaps because the determinant is independent of the starting basis.

We first record an identity that permits witnesses of arbitrary length. Here x, A, B, y denote row labels, and $R_A = 2J_0 b_A$. Whenever the displayed primitive denominators are nonzero,

$$D_{BAy} D_{ABA} D_{xAB} = D_{xAy} \quad \text{on } V_{xA}.$$

Indeed, for $\xi \in V_{xA}$, put $t = \langle b_B, \xi\rangle / \langle b_B, R_A\rangle$ and $\eta = \xi - tR_A$. Then $\eta \in V_{AB}$, so $D_{ABA}\eta = \eta$, since $\langle b_A, \eta\rangle = 0$. The last passage gives

$$D_{BAy}\eta = \xi - tR_A - \frac{\langle b_y, \xi\rangle - t\langle b_y, R_A\rangle}{\langle b_y, R_A\rangle} R_A = D_{xAy}\xi.$$

Both sides have domain V_{xA} and target V_{Ay}. Thus replacing (x, A, B, A, y) by (x, A, y) preserves the entire return map when the cyclic cut is taken at this unchanged incoming section.

Now specialize the row variables to

$$A = (2, 3, -4, -3), \quad B = (0, 1, 2, -1), \qquad C = (-4, -3, 0, -1),$$
$$D = (4, -1, 1, 1), \qquad E = (-4, 2, -4, -4).$$

The required pairings are all nonzero, as the following complete table shows (reversing a pair reverses its sign):

pq	AB	AC	AD	AE	BC	BD	BE	CD	CE	DE
$\langle q, 2J_0 p\rangle$	8	−56	36	−60	8	−16	12	−16	60	−20

Direct substitution into the primitive formula gives

$$\det(I - M^{\mathrm{alg}}_{ABCDE}) = -\frac{10643}{600}, \qquad \det(I - M^{\mathrm{alg}}_{ACDE}) = \frac{1}{168}.$$

For odd length $k = 2m + 3$, $m \geq 1$, specialize to $(A, B)^m, C, D, E$. Repeatedly contracting (E, A, B, A, B) reduces this tuple to (A, B, C, D, E). For even length $k = 2m + 4$, $m \geq 1$, use $(A, B)^m, A, C, D, E$. The same contractions reduce it to (A, B, A, C, D, E), and contracting (E, A, B, A, C) leaves (A, C, D, E). Taking the cyclic cut at V_{EA} makes each contraction preserve the return map. Changing the cut does not change its determinant: each primitive is invertible, with inverse obtained by projecting back along the same R_i, so cyclic return maps are conjugate.

These are algebraic specializations of independent row variables; the repeated vectors do not describe simple searched words or irredundant presentations. They are legitimate evaluations of the rational function on D_k, and suffice to show that it is not identically zero for any $k \geq 5$. They also show that D_k is nonempty. As a nonempty Zariski-open subset of the irreducible affine space X_k, it meets every nonempty coordinate-minor chart in a dense set. A nonzero rational function cannot vanish identically on such a chart. Its nonzero locus is therefore open dense in D_k, and hence in X_k, since the excluded denominators are nonzero polynomials.

For fixed F, selecting the rows of a simple word pulls this open dense condition back to $(\mathbb{R}^4)^F$: the remaining rows are free variables. There are only finitely many simple words and sublists. Intersecting their open dense nonvanishing loci, and then the Euclidean-open chamber $\mathcal{C}$, preserves density. The all-pairs condition is stronger than the nonempty-pair condition in Definition 5.3, and the remaining conditions imply finite-orbit regularity. □

This is chamber-relative genericity for ordered normalized irredundant presentations. It is not the conjectural CH2021 claim that generic polytopes have no Type 2 orbit below an action bound.

5.4 Relation to CH2021

Chaidez–Hutchings study a finite model that compares Reeb dynamics on a four-dimensional polytope with Reeb dynamics on its smoothings [CH21]. Their setting and proof route differ from the general Haim–Kislev formula. They assume that the polytope is *symplectic*; in particular, its two-faces are not Lagrangian, so the relevant combinatorial Reeb direction on each two-face is one-dimensional.

A Chaidez–Hutchings combinatorial Reeb orbit is a cyclic sequence of oriented line segments in the boundary. Their symplectic flow graph records the orbits which avoid the one-skeleton: its vertices are open two-faces, an edge is a Reeb passage through a three-face, and periodic graph orbits correspond to these Type 1 combinatorial Reeb orbits with the same action [CH21, Proposition 2.14]. Their smoothing theory relates this model to Reeb orbits on smooth convex approximations and yields a capacity formula involving Type 1 and Type 2 combinatorial orbits [CH21, Theorems 1.11 and 1.12, Corollary 1.15].

The rational-arithmetic algorithm developed here uses part of the same local geometry but a different global argument. Its nodes are closed facet-pair sections, and it searches simple facet words whose affine return maps have fixed points. The simple-minimizer theorem supplies completeness on the stated regular class. Consequently this algorithm neither implements the CH2021 rotation pruning nor imports the Type 1/Type 2/Type 3 classification as its capacity proof. In particular, an action cutoff alone would not control arbitrarily long helical combinatorial words in the CH2021 model; the finite search here instead comes from simple words with no repeated facet direction.

5.5 Scope and rational implementation

The regularity hypotheses exclude some important structured examples. In a Lagrangian polygon product, all facet normals from either factor lie in a two-dimensional subspace. Any three of them are linearly dependent, violating condition 2 of Definition 5.3. The genericity proposition therefore does not make this algorithm applicable to the products studied later in the thesis. The product quadratic program of Section 4 provides a separate method for those bodies.

The rational implementation follows the construction above: it enumerates simple directed cycles with positive transitions, intersects and composes affine tubes, and reconstructs the segment times at a return-map fixed point. It assumes a bounded, irredundant rational facet presentation in dimension four, with correct facet-incidence and symplectic-sign matrices supplied by the caller. These are input requirements, not conclusions of the tube search. It checks short-word independence and rejects a nonempty facet pair with zero symplectic pairing. A singular fixed-point equation is harmless if it is inconsistent or if its fixed set misses the tube; otherwise the implementation reports an unsupported case.

On selected deterministic five-, six-, and seven-facet examples, this search agrees with the certified quadratic-program capacity computation. The tests also exercise tube composition, positive-time reconstruction, and unsupported inputs. These comparisons check the correspondence between the rational program and the theorem on particular inputs. They do not establish a performance advantage over the quadratic program; no common-input benchmark is claimed here.[1]

[1]The exact search and its regression tests are in `crates/symplectic/src/algorithms/flow_graph/exact_search.rs`; the worked example and its input are in `experiments/dev-flow-graph/visualize-tube/`.

6 First-Order Variation in Facet Coordinates

We now let the supporting hyperplanes move and ask how the systolic ratio changes. The volume has a direct derivative in a neighborhood with fixed simple face structure. Capacity is more delicate: the Haim–Kislev formula expresses it through a maximum over words, and the maximizing word can change.

There are two useful calculations. A smooth feasible choice of weights gives a differentiable upper bound for the ratio. If the constrained optimizer is nondegenerate, its weights also vary smoothly, and we can differentiate its optimized value. The first construction needs less: it will supply the upper bounds in the HKO proof even at singular stationarity systems. For local search, the second construction gives branch gradients that can be compared before proposing a step.

Leipold studies a related optimization problem for volume-preserving linear images of a simplex [Lei26, Sections 9.2, 11.3, and 12.3]. There the balance weights are fixed, capacity evaluation reduces to a linear-ordering integer program, and active permutation branches provide first-order information for numerical search. Here the facets move independently, so both the feasibility constraints and the optimizing weights can change. We derive the corresponding derivatives in facet coordinates.

6.1 Row coordinates and the volume variation

Fix a labelled presentation

$$K(a) = \{x \in \mathbb{R}^4 : \langle a_i, x\rangle \le 1,\ i = 1, \dots, F\}, \qquad a = (a_1, \dots, a_F). \tag{24}$$

The vectors a_i are the dual vertices used in Section 4. A row perturbation is a tuple $h = (h_1, \dots, h_F) \in (\mathbb{R}^4)^F$, and we write $a(t) = a + th$. Throughout this section we work in a neighborhood in which $K(a)$ remains a bounded full-dimensional polytope, the listed rows remain irredundant, and the same simple face combinatorics persists. In particular, $V(a) > 0$, and the Haim–Kislev formula applies. These are open conditions at a simple labelled polytope.

For the facet F_i, let S_i be its three-dimensional Euclidean volume and let $\bar{x}_i$ be its centroid. The volume derivative has a particularly simple expression in row coordinates.

Proposition 6.1 (Volume first variation)**.** *In the stable simple chart above, with* $V(a) = \mathrm{vol}_4(K(a))$*,*

$$DV(a)[h] = -\sum_{i=1}^{F} \frac{S_i}{\|a_i\|}\langle \bar{x}_i, h_i\rangle. \tag{25}$$

Proof. At a point $x \in F_i$, differentiating the moving hyperplane equation $\langle a_i(t), x(t)\rangle = 1$ gives

$$\langle a_i, \dot{x}\rangle = -\langle h_i, x\rangle.$$

Thus the outward normal velocity of this facet is $-\langle h_i, x\rangle/\|a_i\|$. The first variation of volume is the integral of the outward normal velocity over the boundary. Integrating on each facet and using $\int_{F_i} x\, dS = S_i\bar{x}_i$ gives (25). Intersections of facets have three-dimensional measure zero and do not contribute to the boundary integral. □

The formula displays both parts of a row perturbation at once: changing $\|a_i\|$ moves the supporting hyperplane, while changing its direction tilts that hyperplane. It is also convenient computationally, because it needs only facet volumes and centroids once the incidence data are known.

6.2 Feasible sections and smooth maxima

Fix an ordered word $\sigma = (\sigma_1, \dots, \sigma_m)$ with distinct facets. Write

$$C_\sigma(a)\beta = \left(\sum_{k=1}^{m} \beta_k a_{\sigma_k}, \sum_{k=1}^{m} \beta_k \right), \qquad e = (0,0,0,0,1),$$

so that the Haim–Kislev feasibility equations are $C_\sigma(a)\beta = e$ and $\beta \geq 0$. We use the quadratic objective from (19):

$$Q_\sigma(a,\beta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \, \omega_0(a_{\sigma_j}, a_{\sigma_k}). \tag{26}$$

For later differentiation, its explicit row variation at fixed β is

$$D_a Q_\sigma(a,\beta)[h] = \sum_{1 \leq j < k \leq m} \beta_j \beta_k (\omega_0(h_{\sigma_j}, a_{\sigma_k}) + \omega_0(a_{\sigma_j}, h_{\sigma_k})). \tag{27}$$

First consider any smooth feasible choice, without imposing stationarity.

Lemma 6.2 (Feasible-section upper function)**.** *Suppose that $\beta(a)$ is smooth near a_0,*

$$C_\sigma(a)\beta(a) = e, \qquad \beta(a) > 0, \qquad Q_\sigma(a,\beta(a)) > 0.$$

Define

$$A_\sigma(a) = \frac{1}{2Q_\sigma(a,\beta(a))}, \qquad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}.$$

Then

$$\operatorname{sys}(K(a)) \leq U_\sigma(a) \tag{28}$$

throughout this neighborhood. If the chosen weights attain the global Haim–Kislev maximum at a_0, then equality holds in (28) *at a_0.*

Proof. The chosen $\beta(a)$ is feasible for the global maximum defining $Q_{\max}(a)$. Hence $Q_{\max}(a) \geq Q_\sigma(a,\beta(a))$. The Haim–Kislev formula gives

$$c_{\mathrm{EHZ}}(K(a)) = \frac{1}{2Q_{\max}(a)} \leq A_\sigma(a).$$

Squaring and dividing by the positive number $2V(a)$ proves the upper bound. A globally maximizing choice gives equality at the base point. □

The lemma requires no KKT system. For example, if $C_\sigma(a_0)$ has rank five and $\beta_0 > 0$ is feasible, choose an invertible five-column minor and hold the other $m-5$ coordinates of β fixed. Solving the five feasibility equations gives a smooth local section through β_0; positivity persists after shrinking the neighborhood. This elementary construction is the one used for the singular seven-facet data in the HKO certificate.

At a nondegenerate constrained maximum, the KKT equations instead choose a canonical optimizing branch and remove the derivative of β from the value derivative. Let $H_\sigma(a)$ denote the Hessian of $Q_\sigma(a,\cdot)$ with respect to β.

Definition 6.3 (Nondegenerate Haim–Kislev branch)**.** *A feasible pair (σ, β_0) at a_0 is a* nondegenerate branch *if*

1. *$C_\sigma(a_0)$ has rank five;*
2. *$\beta_0 > 0$ and $Q_\sigma(a_0,\beta_0) > 0$;*
3. *β_0 is a constrained critical point of $Q_\sigma(a_0,\cdot)$; and*

4. $H_\sigma(a_0)$ is negative definite on $\ker C_\sigma(a_0)$.

For $m = 5$, the kernel is zero and the last condition is vacuous.

Proposition 6.4 (Derivative of a nondegenerate branch)**.** *Let (σ, β_0) be a nondegenerate branch at a_0. Then, after shrinking the row chart, there is a unique smooth constrained maximizer $\beta_\sigma(a) > 0$ near β_0. Put*

$$q_\sigma(a) = Q_\sigma(a, \beta_\sigma(a)), \qquad U_\sigma(a) = \frac{1}{2V(a)} \left(\frac{1}{2q_\sigma(a)} \right)^2 .$$

If $\lambda_\sigma \in \mathbb{R}^5$ is chosen so that the KKT equation is

$$D_\beta Q_\sigma(a_0, \beta_0) = C_\sigma(a_0)^{\mathsf{T}} \lambda_\sigma,$$

then

$$Dq_\sigma(a_0)[h] = D_a Q_\sigma(a_0, \beta_0)[h] - \langle \lambda_\sigma, D_a C_\sigma(a_0)[h]\, \beta_0 \rangle . \tag{29}$$

Consequently,

$$\frac{DU_\sigma(a_0)[h]}{U_\sigma(a_0)} = -2\frac{Dq_\sigma(a_0)[h]}{q_\sigma(a_0)} - \frac{DV(a_0)[h]}{V(a_0)}. \tag{30}$$

Proof. Apply the implicit function theorem to the stationarity and feasibility map

$$(\beta, \lambda) \longmapsto \begin{pmatrix} D_\beta Q_\sigma(a, \beta) - C_\sigma(a)^{\mathsf{T}} \lambda \\ C_\sigma(a)\beta - e \end{pmatrix}.$$

Its derivative in these variables at the base point is the KKT block matrix

$$\mathcal{K} = \begin{pmatrix} H_\sigma(a_0) & -C_\sigma(a_0)^{\mathsf{T}} \\ C_\sigma(a_0) & 0_{5\times 5} \end{pmatrix}.$$

If $\mathcal{K}(u, v)^T = 0$, then $C_\sigma u = 0$ and $u^T H_\sigma u = 0$. Negative definiteness on $\ker C_\sigma$ forces $u = 0$, and full row rank then forces $v = 0$. Thus $\mathcal{K}$ is nonsingular. The implicit function theorem gives a unique smooth critical point near β_0. Negative definiteness and positivity are open conditions, so this point remains the positive constrained maximum.

Differentiate $q_\sigma(a) = Q_\sigma(a, \beta_\sigma(a))$. The term containing $D\beta_\sigma$ is

$$D_\beta Q_\sigma \, D\beta_\sigma = \lambda_\sigma^{\mathsf{T}} C_\sigma D\beta_\sigma.$$

Differentiating $C_\sigma(a)\beta_\sigma(a) = e$ gives $C_\sigma D\beta_\sigma = -(D_a C_\sigma)\beta_\sigma$, which proves (29). The logarithmic derivative of $U_\sigma = (2V)^{-1}(2q_\sigma)^{-2}$ is (30). □

Equations (27), (25), and (29) are the practical first-order calculation. They use the current rows, the word, its positive KKT weights and multipliers, and facet volumes and centroids. They do not require finite differences of the full capacity computation.

6.3 From branches to the systolic ratio

A nondegenerate branch always gives the upper function U_σ, but it gives the actual systolic ratio only while it realizes the global Haim–Kislev maximum.

Suppose that finitely many nondegenerate branches have fixed-word maxima $q_1, \dots, q_r$ on a neighborhood of a_0, and that they cover the global finite problem there in the explicit sense

$$Q_{\max}(a) = \max_{1 \le j \le r} q_j(a).$$

For example, this identity follows from a complete retained word list and a persistent value gap for every discarded word. Then

$$\operatorname{sys}(K(a)) = \min_{1\le j\le r} U_j(a). \tag{31}$$

If $I(a_0) = \{j : U_j(a_0) = \operatorname{sys}(K(a_0))\}$, the one-sided directional derivative is

$$D^+ \operatorname{sys}(a_0; h) = \min_{j\in I(a_0)} DU_j(a_0)[h]. \tag{32}$$

Indeed, this is the first-order expansion of a finite minimum of smooth functions. In particular, a unique active nondegenerate branch gives an ordinary gradient. At a tie of nondegenerate branches, the directional derivative is generally only the lower envelope of the tied branch slopes.

The hypothesis preceding (31) contains real mathematical work. A list returned by one numerical solve, or a tolerance window around its smallest actions, is not by itself a certificate of the displayed identity.

6.4 What the first-order model contributes to search

On a nondegenerate branch patch with active branch rows $r_j = DU_j(a_0)$, a common first-order ascent direction in a chosen linear slice S can be sought through the maximin problem

$$\max_{h\in S,\ \|h\|\le 1} \min_{j\in I(a_0)} r_j(h). \tag{33}$$

The slice removes directions generated by the systolic-ratio symmetries from Definition 2.2. The true ratio is constant on the corresponding group orbits, so its derivative vanishes on their tangent space; S is chosen as a complementary linear slice. A concrete instance appears in Lemma 7.5: at the ten-facet HKO body the ambient row space has dimension forty, the symmetry tangent space T has dimension fifteen, and S has dimension twenty-five. A positive value in (33) predicts improvement of every active nondegenerate branch. The prediction still has to be checked by recomputing the capacity after a finite step.

At finite distance, currently inactive branches can matter before they are active. A useful semi-local model therefore retains their base gaps:

$$\min_{j\in W}\bigl(U_j(a_0) - \operatorname{sys}(K(a_0)) + DU_j(a_0)[h]\bigr), \tag{34}$$

where W is a chosen branch window. Formula (34) explains why a single active gradient can miss a nearby branch switch, but it is a prediction model rather than an infinitesimal theorem. Its reliability depends on the window containing the branches that become relevant, on the branches remaining feasible, and on the linearization error at the proposed step.

One retained finite experiment illustrates this distinction. On a fixed panel of twelve random ten-facet starts, the tested multi-direction method accepted a positive move at each of eight configured trace iterations, but all runs stopped at their iteration cap and every endpoint scan still found a further above-threshold move. Thus the evidence supports finite-step progress and a measured computation cost on that panel, not endpoint local maximality. A separate comparison on sixty-four matched starts found larger median terminal best-so-far values for methods using several competing branches than for either single-branch gradient implementation. All 448 terminal values remained below one. Section 8.10 gives the results and the limits of this comparison on a historical numerical objective; its starts are separate from the twelve-start panel above.

A retained three-case diagnostic also explains a failure of the affine branch-window prediction. At the tested finite step, two named KKT branches change too rapidly for their base-point

derivatives to predict even the sign of the action change. Smaller-step differences approach the analytical slopes, while the volume calculation remains consistent. Thus a small Euclidean step alone need not make the branch model accurate; its useful scale also depends on the conditioning of the KKT system. The decomposition and its three cases are retained in `experiments/dev-gradient-ascent/endpoint-model-audit/`.

6.5 The non-generic boundary

The smooth branch calculation can break down in four ways.

- If a maximizing weight is zero, deleting that coordinate is valid for the capacity value at the base point. It need not preserve the nearby slope: the longer word may acquire a positive weight on one side of the perturbation and define a different branch there.
- If the constraint rank drops or the KKT matrix is singular, the implicit function theorem does not provide a unique smooth optimizing branch. A continuum of optimizers may nevertheless have a well-defined value.
- A tie is described by (32) only when all relevant tied branches extend smoothly and the stated coverage and gap hypotheses hold. Base-point branch enumeration alone does not prove this.
- When facet labels become redundant or the face combinatorics changes, both the branch family and the smooth volume formula require a new local chart. Hausdorff continuity of capacity and volume across that boundary does not imply differentiability there.

All constraints and objectives are semialgebraic. A possible route to a fully general one-sided theory would organize germs of feasible optimizing branches by a finite decomposition of direction space. Such a construction would have to record support appearances, singular optimizer sets, and dominance of every discarded germ. Obtaining a complete first-order description at an arbitrary non-generic polytope remains open here.

6.6 Application at HKO

The HKO argument uses the feasible-section calculation. It constructs twenty-six upper bounds that agree with the true ratio at the base body. Together their derivatives force at least one bound to decrease in every direction transverse to the symmetries. Section 7 gives the construction and proves that this directional statement yields a strict local maximum modulo these symmetries among ten-facet bodies. No continuation of an optimizing branch is required.

6.7 Implementation and validation status

The implementation evaluates the branch and volume formulas in dual-vertex coordinates. Regression tests compare the volume derivative with central finite differences and compare a fixed-word capacity derivative with central finite differences on known fixtures. An exact rational implementation of the capacity-gradient formula is also compared with the floating-point formula on a rational simplex fixture. These checks test the correspondence between the formulas and their implementations; they do not certify branch coverage, finite-step prediction, or local maximality.

For the HKO theorem, the evidence standard is stronger and different. The SageMath verifier checks the selected feasible sections and their derivative rows exactly in the relevant algebraic number field. The mathematical use of those rows is the upper-bound and transverse-slice argument in Section 7.

7 Local maximality of the HKO pentagon product

Let P be the regular pentagon of circumradius one, centred at the origin and with vertex $(1, 0)$. In coordinates (q_1, q_2, p_1, p_2) put $K_0 = P \times R(-\pi/2)P$, with symplectic form $\omega_0 = \sum_j dq_j \wedge dp_j$. The Haim–Kislev–Ostrover capacity formula [HO26] for this body gives

$$c_{\mathrm{EHZ}}(K_0) = 2\cos(\pi/10)(1 + \cos(\pi/5)), \qquad \mathrm{sys}(K_0) = \frac{3+\sqrt{5}}{5} > 1.$$

Here $\mathrm{sys}(K) = c_{\mathrm{EHZ}}(K)^2/(2\,\mathrm{vol}_4\,K)$. The counterexample itself is due to Haim–Kislev and Ostrover; the result proved here concerns its local maximality when the number of facets stays ten.

Theorem 7.1. *There is a Hausdorff neighborhood $\mathcal{U}$ of K_0 among convex four-dimensional polytopes with exactly ten facets such that $\mathrm{sys}(K) \le \mathrm{sys}(K_0)$ for $K \in \mathcal{U}$. Equality holds precisely for those bodies in $\mathcal{U}$ obtained from K_0 by translation, positive dilation and a linear symplectic map.*

7.1 The proof strategy

The ten supporting hyperplanes may move in all four coordinate directions, so the nearby bodies include polytopes that are no longer products. Write

$$K(a) = \{x \in \mathbb{R}^4 : a_i \cdot x \le 1,\ i = 1, \dots, 10\}, \qquad a = (a_1, \dots, a_{10}),$$

and let a_0 be the row list of K_0.

We construct twenty-six smooth upper bounds U_j for the systolic ratio, all equal to it at a_0. If one of these bounds decreases, the ratio decreases with it. Different bounds handle different directions. The proof first identifies their derivatives and the symmetry directions. A linear-algebra lemma then gives a uniform negative slope transverse to symmetry. Only at the end do we pass from these derivatives to a finite neighborhood and recover the equality cases.

7.2 Local coordinates and volume

Lemma 7.2 (A simple row neighborhood)**.** *There is a neighborhood $\mathcal{N}$ of a_0 in $(\mathbb{R}^4)^{10}$ on which $K(a)$ is bounded and simple, with the same vertex–facet incidences as K_0. Every sufficiently Hausdorff-close ten-facet polytope has a labelled presentation with $a \in \mathcal{N}$, and its rows tend to a_0 as the body tends to K_0.*

Proof. The origin stays in the interior under sufficiently small Hausdorff perturbations of $K_0 = K(a_0)$. One way to see this is through support functions: $h_K(u) = \max_{x \in K} u \cdot x$. Write B for the Euclidean unit ball. If $rB \subset K_0$ and the Hausdorff distance from K to K_0 is less than $\epsilon < r$, then $h_K(u) \ge h_{K_0}(u) - \epsilon \ge r - \epsilon$ for every unit vector u. Thus $(r - \epsilon)B \subset K$. On such a neighborhood polarity is continuous: the radial function, the distance from zero to the boundary along a unit direction, of K° is $1/h_K$ on the unit sphere, and the common positive lower bound on h_K makes these reciprocals converge uniformly. The polars are uniformly bounded convex bodies, so this also gives Hausdorff convergence.

The polar K_0° has the ten vertices $a_{0,i}$. Choose disjoint small balls about them. Each vertex is uniquely exposed by some linear functional ℓ_i. For polars sufficiently close to K_0°, every maximizer of ℓ_i lies in the corresponding ball. Otherwise a convergent sequence of maximizers outside that ball would limit to a maximizer on K_0° different from $a_{0,i}$. A maximizing face contains a vertex. A polytope K with exactly ten facets has a polar with exactly ten vertices; hence there is exactly one polar vertex in each ball and none elsewhere. Labelling these vertices a_i gives

$$K = \{x : a_i \cdot x \le 1,\ i = 1, \dots, 10\}, \qquad a_i \longrightarrow a_{0,i} \quad \text{as } K \longrightarrow K_0.$$

For completeness, the face structure is stable in this neighbourhood. The base body is a product of two polygons and is simple: precisely four facets meet at each vertex, with independent normals. Their four equations therefore determine a smoothly varying intersection under a small change of a. All the remaining inequalities have strict slack at that intersection, so it remains a vertex with exactly those incident facets.

No new vertex can appear. The convex hull of the base normals contains a ball about zero; the convex hull of sufficiently close normals contains a smaller common ball. Consequently the bodies $K(a)$ are uniformly bounded. If arbitrarily close parameters had a new vertex, choose four independent active constraints there. There are finitely many four-element subsets, so along a subsequence the same subset is active and the vertices converge to a point of K_0 where those four constraints are equalities. Simplicity forces them to be the incident facets of one of the original vertices. Their intersection is the continuation already constructed, contradicting newness. This argument also excludes a new vertex arising from a subset whose normals are dependent at the base point; no invertibility assumption for such a subset is needed.

The vertex–facet incidences therefore remain unchanged. In particular each inequality still defines a facet. Conversely, small changes of the ten normals give Hausdorff-small changes of $K(a)$: their convex hulls converge, contain a common ball about zero, and polarity is continuous there.

□

Lemma 7.3 (Smooth volume and its derivative)**.** *On a sufficiently small neighborhood $\mathcal{N}$ as above, $V(a) = \operatorname{vol}_4 K(a)$ is smooth and positive. If $F_i(a)$ has centroid $\bar{x}_i(a)$, then*

$$DV(a)[h] = -\sum_{i=1}^{10} \frac{\operatorname{vol}_3(F_i(a))}{\|a_i\|} \langle h_i, \bar{x}_i(a) \rangle.$$

Proof. The fixed incidences express each vertex smoothly in the rows. A fixed boundary triangulation then expresses volume smoothly in those vertices. On the moving facet, differentiating $\langle a_i + \epsilon h_i, x(\epsilon) \rangle = 1$ gives outward normal velocity $-\langle h_i, x \rangle / \|a_i\|$. Integrating this velocity over the facets and using their centroids gives the formula. □

The concrete base point and its volume

Put $t = \tan(\pi/5)$, $s = (5 - t^2)/2 = \sqrt{5}$, and

$$t^4 - 10t^2 + 5 = 0, \quad 0 < t < 1, \qquad \alpha = (3 - s)/2, \quad b = t(1 + s)/2, \quad d = s - 1.$$

The selected root is unique in $(0, 1)$: the polynomial is strictly decreasing there and changes sign. For the circumradius-one pentagon with vertex $(1, 0)$, the edge normals have arguments $(2k + 1)\pi/5$ and lengths $\sec(\pi/5) = d$. Rotating the second factor by $-\pi/2$ gives the following ten rows $a_{0,i}$, in (q_1, q_2, p_1, p_2) order:

$$\begin{pmatrix} 1 & t & 0 & 0 \\ -\alpha & b & 0 & 0 \\ -d & 0 & 0 & 0 \\ -\alpha & -b & 0 & 0 \\ 1 & -t & 0 & 0 \\ 0 & 0 & t & -1 \\ 0 & 0 & b & \alpha \\ 0 & 0 & 0 & d \\ 0 & 0 & -b & \alpha \\ 0 & 0 & -t & -1 \end{pmatrix}.$$

These rows give the normalized facet presentation of K_0. Writing $\theta = \pi/5$, the area of one pentagon is the sum of five triangles with central angle 2θ, hence $B = \frac{5}{2}\sin(2\theta) = 5\sin\theta\cos\theta$. Fubini therefore gives

$$V_0 = B^2 = 25\sin^2\theta\cos^2\theta = \frac{25(5+s)}{32} = \frac{25(15-t^2)}{64}.$$

To specialize Lemma 7.3, observe that F_i is an edge times the other centered pentagon. Its centroid is $\bar{x}_i = a_{0,i}/\|a_{0,i}\|^2$, its three-volume is $2\sin\theta\, B$, and $\|a_{0,i}\| = \sec\theta$. Consequently

$$DV_0[h] = -\mu\sum_{i=1}^{10} a_{0,i}\cdot h_i, \qquad \mu = 10\sin^2\theta\cos^4\theta = \frac{25}{32} + \frac{5s}{16}.$$

This gives the volume derivative for every row perturbation at the base point.

7.3 The touching upper bounds

For an ordered facet word $\sigma = (\sigma_1, \dots, \sigma_m)$, write

$$C_\sigma(a) = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \qquad e = (0,0,0,0,1)^T.$$

The equations $C_\sigma(a)\beta = e$ impose closure and normalization. For positive feasible weights let

$$Q_\sigma(a,\beta) = \sum_{j<i} \beta_i\beta_j\omega_0(a_{\sigma_j}, a_{\sigma_i}), \qquad A_\sigma(a) = \frac{1}{2Q_\sigma(a,\beta(a))}, \qquad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}.$$

When $Q_\sigma > 0$, the variational capacity formula gives $\mathrm{sys}(K(a)) \le U_\sigma(a)$.

Let T be the tangent space at a_0 to the orbit generated by translations, positive dilations and linear symplectic maps. The following lemma records the finite calculation that will be used in the rest of the proof.

Lemma 7.4 (Exact upper bounds and symmetry directions)**.** *The space T has dimension fifteen. On a neighborhood of a_0 there are twenty-six smooth functions U_j such that*

$$\mathrm{sys}(K(a)) \le U_j(a), \qquad U_j(a_0) = \mathrm{sys}(K_0).$$

Their derivative rows $r_j = DU_j(a_0)$ vanish on T, have rank twenty-five, and satisfy

$$\sum_{j=1}^{26} \lambda_j r_j = 0, \qquad \lambda_j > 0, \qquad \sum_{j=1}^{26} \lambda_j = 1.$$

The construction and computer-algebra verification of this lemma occupy the following calculations. The number twenty-six reflects the twenty-five-dimensional space transverse to symmetry: covectors spanning that space and admitting a strictly positive relation need at least twenty-six members. Indeed, with at most twenty-five members a nonzero linear relation would prevent full rank. Here the twenty-six restricted rows form the vertices of a simplex whose interior contains zero; the positive coefficients in the lemma are its barycentric coordinates. This is the finite geometry that makes one upper bound decrease in every nonzero transverse direction.

Holding selected weights fixed and solving the five closure and normalization equations for the others produces the smooth sections: an invertible five-column minor gives smoothness, and strict positivity at the base point persists nearby.

One of the twenty-six upper bounds

We now work through one of the twenty-six upper bounds to illustrate the construction and differentiation. The other twenty-five are constructed in the same way. The exact computations for all twenty-six were verified using a computer algebra system. Appendix B.1 gives the complete program with explanations of its checks.

Consider the seven-facet word and position sets

$$\sigma = (2,3,8,9,5,4,6), \quad I = (1,2,3,4,5), \quad J = (6,7).$$

We solve closure and normalization for the weights at positions I and hold the weights at positions J fixed. These are positions in the word, not facet labels. Set

$$r = \frac{6909830056250513}{10^{17}}, \quad c = \frac{t^2+5}{40}, \quad z = \frac{5-t^2}{20} = \frac{s}{10}, \quad k = \frac{3-t^2}{4}.$$

Fix $\beta_6 = r$ and $\beta_7 = z$; the decimal-looking rational is an exact rational choice, not a rounding assertion. The complete base vector is

$$\beta_0 = (c+kr,\ c-r,\ c,\ c,\ z-kr,\ r,\ z)^T.$$

At this word the closure matrix's selected minor is

$$M = C_\sigma(a_0)_I = \begin{pmatrix} -\alpha & -d & 0 & 0 & 1 \\ b & 0 & 0 & 0 & -t \\ 0 & 0 & 0 & -b & 0 \\ 0 & 0 & d & \alpha & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \qquad \det M = 10(1-t^2) > 0.$$

The full matrix is obtained by adjoining columns $(a_4, 1)^T$ and $(a_6, 1)^T$. Writing $e = (0,0,0,0,1)^T$, define the nearby section by

$$\beta_J(a) = (r,z)^T, \qquad \beta_I(a) = C_\sigma(a)_I^{-1}(e - C_\sigma(a)_J(r,z)^T).$$

Direct substitution gives $C_\sigma(a_0)\beta_0 = e$ and $\beta_0 > 0$. For example $0.726 < t < 0.727$ and $0.069 < r < 0.070$ suffice to check all seven signs from the displayed formulas. Invertibility and positivity persist in a neighborhood.

Use $\omega(x,y) = x_{q_1}y_{p_1} + x_{q_2}y_{p_2} - x_{p_1}y_{q_1} - x_{p_2}y_{q_2}$ and

$$Q_\sigma(a,\beta) = \sum_{j<i} \beta_i\beta_j\omega(a_{\sigma_j}, a_{\sigma_i}).$$

At the fixed base geometry, solving closure with arbitrary $(\beta_6,\beta_7) = (u,v)$ gives the particularly simple identity

$$Q_\sigma = -4tv^2 + (2t - 2t^3/5)v.$$

Indeed,

$$\frac{\partial Q_\sigma}{\partial u} = 0, \qquad \frac{\partial Q_\sigma}{\partial v} = -8tv + 2t - \frac{2t^3}{5} = -8t(v-z).$$

Since $t > 0$, the maximum occurs at $v = z$, and substitution gives $Q_0 = t/5$ using $t^4 - 10t^2 + 5 = 0$. Thus its Hessian on the two-dimensional closure affine space is singular (in the (u,v) coordinates it is $\mathrm{diag}(0,-8t)$). The five-by-five closure minor is nevertheless invertible, so the displayed feasible section continues smoothly and supplies an upper bound nearby.

$$A_0 = \frac{1}{2Q_0} = \frac{5}{2t} = 5t - \frac{t^3}{2}, \qquad U_0 = \frac{A_0^2}{2V_0} = \frac{11-t^2}{10} > 1.$$

The HKO capacity formula gives $c_{\mathrm{EHZ}}(K(a_0)) = 2\cos(\pi/10)(1+\cos(\pi/5)) = A_0$. Thus the constructed upper bound agrees with the systolic ratio at the base point.

A worked coordinate outside product-preserving perturbations

We now choose a perturbation that varies the first momentum component of the second q-facet's normal:

$$h_2 = (0, 0, 1, 0), \qquad h_i = 0 \quad (i \neq 2).$$

Put $w = c + kr = \beta_{0,1}$. Differentiated closure gives

$$D\beta_J[h] = 0, \qquad D\beta_I[h] = -wM^{-1}(0, 0, 1, 0, 0)^T = w \begin{pmatrix} t^3/20 - t/4 \\ t^3/20 - t/4 \\ t^3/40 - 3t/8 \\ -3t^3/20 + 5t/4 \\ t^3/40 - 3t/8 \end{pmatrix}.$$

Inserting these values into the product rule for Q gives the exact scalar

$$DQ[h] = \frac{3 - t^2}{80} + \left(\frac{3t^2}{20} - \frac{1}{4}\right) r - \frac{t^2 + 5}{4} r^2 =: q_h.$$

Since a_2 has zero momentum coordinates, $DV_0[h] = 0$, while

$$DA[h] = -\frac{25}{2t^2} q_h, \qquad DU[h] = -\frac{125}{4t^3 V_0} q_h.$$

This coefficient of the upper-bound derivative is approximately -0.1801707325. Changing only the two polygon factors keeps every q-facet normal in the q-plane. This perturbation adds a p_1-component to one of those normals, so the derivative describes a change of the upper bound outside that product family.

7.4 Computing the twenty-six derivatives

Table 1 gives the exact fixed values for every section, including the worked entry. Except at entry 22, the minor positions are $I = (1, 2, 3, 4, 5)$ and $J = (6, \ldots, m)$. Entry 22 instead has $I = (1, 2, 3, 4, 6)$ and $J = (5, 7)$. The table is ordered by the witness's zero-based certificate index. For each table row set $\beta_J = u$ and compute

$$\beta_I = C_I^{-1}(e - C_J u), \qquad (D\beta)_J[h] = 0, \qquad (D\beta)_I[h] = -C_I^{-1}(DC[h])\beta.$$

For each of the forty coordinate directions, compute

$$\begin{aligned} DQ[h] &= \sum_{j<i} \big((D\beta_i[h])\beta_j + \beta_i(D\beta_j[h])\big)\omega(a_{\sigma_j}, a_{\sigma_i}) \\ &\quad + \sum_{j<i} \beta_i\beta_j\big(\omega(h_{\sigma_j}, a_{\sigma_i}) + \omega(a_{\sigma_j}, h_{\sigma_i})\big), \\ R[h] &= -\frac{A_0}{2V_0 Q_0^2} DQ[h] - \frac{A_0^2}{2V_0^2} DV_0[h]. \end{aligned}$$

Together with the explicit table these formulas specify the entire 26×40 matrix R exactly.

The weight derivatives are checked against the differentiated closure equations. The remaining finite predicates have four roles in Lemma 7.4:

1. *Touching upper bounds.* Every word has distinct valid labels; $\det C_I \neq 0$, $C\beta = e$, all $\beta_i > 0$, and $Q = t/5$. These checks give local positive feasible sections, whose base values touch the known HKO capacity.

2. *Independence and rank.* The fifteen symmetry columns have rank fifteen, and $\operatorname{rank} R = 25$.

3. *Symmetry annihilation.* Every row of R vanishes on every symmetry column. Together with the ranks, this identifies the common kernel of the rows as exactly T.

4. *Positive relation.* The one-dimensional kernel of R^T has a vector with all entries of one strict sign. Change its sign if needed and normalize its sum to obtain $\lambda > 0$ and $\lambda^T R = 0$.

The symmetry columns used in these checks are explicitly $h_i = -a_{i,j}a_i$ for $j = 1, \dots, 4$, $h_i = -a_i$, and $h_i = -X^T a_i$ for the ten generators

$$X = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}, \qquad B = B^T,\ C = C^T,$$

using the four matrix units for A and the three symmetric matrix units for each of B, C. Their column rank is required to be fifteen. Appendix B.1 maps these four groups to the executable checks. The arithmetic establishes the asserted independence, rank, annihilation and positive relation. The constructions above give smooth feasible sections on a common neighborhood, and the HKO capacity formula gives the touching identities. This proves Lemma 7.4.

7.5 The transverse slope estimate

The exact calculation is now summarized by Lemma 7.4. We use only its rank and positivity statements in the remaining argument.

Lemma 7.5 (Uniformly negative transverse slopes)**.** *Choose a linear complement S to the symmetry tangent space T, so that $(\mathbb{R}^4)^{10} = T \oplus S$ and $\dim S = 25$. There is $\eta > 0$ such that*

$$\min_{1 \le j \le 26} r_j(v) \le -\eta \|v\| \qquad \text{for every } v \in S.$$

Proof. The rows have rank twenty-five and vanish on the fifteen-dimensional space T. Their common kernel is therefore exactly T, and their restrictions span S^*. For $v \in S \setminus \{0\}$, suppose that every $r_j(v)$ were nonnegative. Their positively weighted sum is zero, so each would have to vanish. Then $v \in T \cap S = \{0\}$, a contradiction. Thus $\min_j r_j(v) < 0$ for every nonzero transverse direction.

The function $v \mapsto \min_j r_j(v)$ is continuous on the unit sphere of S. Its maximum there is strictly negative by compactness. Write that maximum as $-\eta$ and use homogeneity to obtain the assertion. □

Figure 5 illustrates the slope condition in one and two transverse dimensions.

7.6 From slopes to a neighborhood

We now pass from the derivative estimate to nearby bodies. The uniform constant η lets us choose one neighborhood for all transverse directions.

Lemma 7.6 (Strict decrease on the slice)**.** *For all sufficiently small nonzero $v \in S$,*

$$\operatorname{sys}(K(a_0 + v)) \le \operatorname{sys}(K_0) - \frac{\eta}{2}\|v\| < \operatorname{sys}(K_0).$$

Proof. There are only twenty-six smooth upper bounds. Their differentiability at a_0 therefore supplies a common neighborhood in which

$$|U_j(a_0 + v) - U_j(a_0) - r_j(v)| \le \frac{\eta}{2}\|v\| \qquad (j = 1, \dots, 26).$$

Shrink it so that all bounds are defined and dominate the ratio there. For each $v \in S$, Lemma 7.5 supplies an index with $r_j(v) \le -\eta\|v\|$. Since $U_j(a_0) = \operatorname{sys}(K_0)$,

$$\operatorname{sys}(K(a_0 + v)) \le U_j(a_0 + v) \le \operatorname{sys}(K_0) - \frac{\eta}{2}\|v\|.$$ □

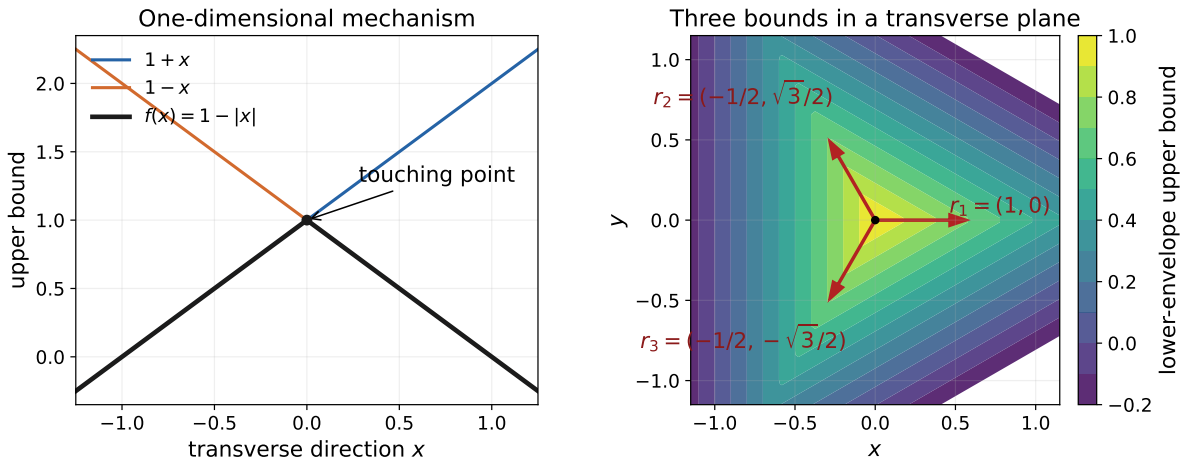


Figure 5: Toy models of touching upper bounds. Left: the lower envelope of $1 + x$ and $1 - x$ is $1 - |x|$. Right: level sets of the lower envelope of three affine functions with gradients $(1, 0)$, $(-1/2, \sqrt{3}/2)$ and $(-1/2, -\sqrt{3}/2)$. The gradients span the plane and sum to zero, and the envelope decreases in every nonzero direction. In the proof the systolic ratio lies below the envelope and agrees with it at the base point.

Lemma 7.7 (Local symmetry decomposition)**.** *Every row list sufficiently close to a_0 is the image of some $a_0 + v$, with $v \in S$ small, under a symmetry close to the identity. The symmetry and v can be chosen smoothly in local group coordinates.*

Proof. Translation by y, dilation by $\rho > 0$ and a linear symplectic map M act on the rows, respectively, by

$$a_i \mapsto \frac{a_i}{1 + a_i \cdot y}, \qquad a_i \mapsto \rho^{-1} a_i, \qquad a_i \mapsto M^{-T} a_i.$$

These actions are smooth near the identity. Choose fifteen local coordinates θ for this symmetry group and write its action as $g(\theta)$. The map

$$\Phi(\theta, v) = g(\theta) \cdot (a_0 + v), \qquad v \in S,$$

has derivative at $(0, 0)$ with image T in the group variables and image S in the slice variables. The fifteen symmetry derivatives are independent by Lemma 7.4. Thus $D\Phi(0, 0)$ is an isomorphism, and the inverse function theorem gives the decomposition. □

The toy model in Figure 6 shows how a strict transverse maximum becomes a family of equal maxima along symmetry.

Proof of Theorem 7.1. By Lemma 7.2, a sufficiently Hausdorff-close ten-facet body has a row list in the neighborhood of Lemma 7.7. It is therefore a symmetry image of $K(a_0 + v)$ for some small $v \in S$. The ratio is unchanged by that symmetry. Lemma 7.6 gives a strict decrease if $v \neq 0$, and equality if $v = 0$. Equality consequently forces the body to be a symmetry image of K_0. Conversely, every such image has the same ratio as K_0. □

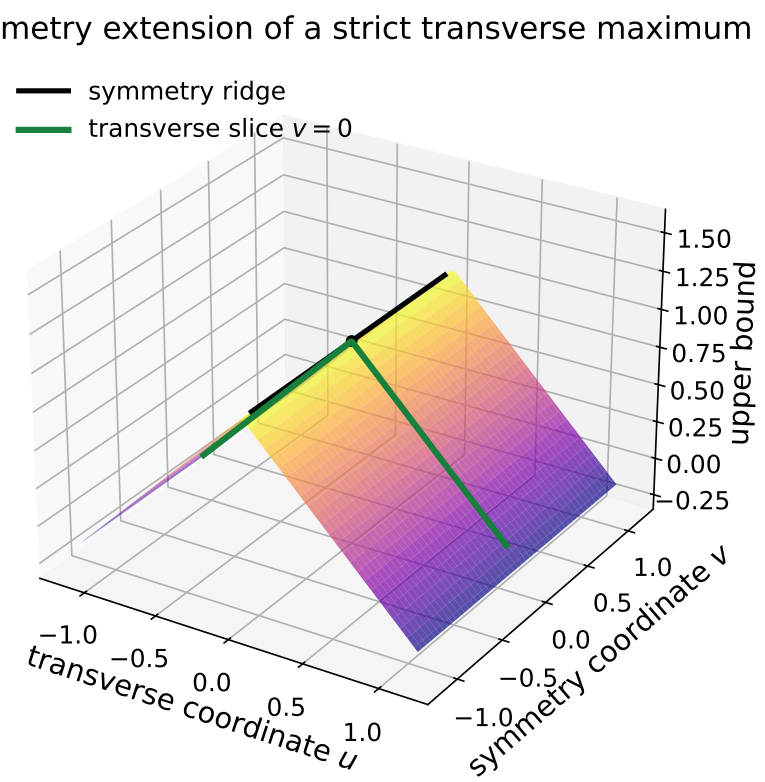


Figure 6: The toy function $f(u, v) = 1 - |u|$ has a strict maximum on the transverse slice $v = 0$ and is constant along the symmetry coordinate v. Its maxima form the ridge $u = 0$. In the HKO proof the symmetry tangent space T has dimension fifteen and the transverse space S has dimension twenty-five. The picture illustrates their roles in the local decomposition.

The twenty-six exact choices

The constants $c = (t^2 + 5)/40$ and $z = (5 - t^2)/20$ are as above. Facet labels and positions start at one; entry numbers start at zero to identify the rows in the accompanying computational witness. Every finite decimal in the table denotes its exact rational value.

Table 1: Exact choices for the twenty-six feasible sections. The solved positions are the complement of J in each word.

Entry	Word σ	Fixed positions J	Exact fixed values
0	$(1, 2, 8, 4, 10, 6)$	(6)	(c)
1	$(1, 2, 8, 7, 4, 10)$	(6)	(z)
2	$(2, 3, 8, 9, 5, 4, 6)$	$(6, 7)$	$(0.06909830056250513, z)$
3	$(1, 8, 7, 3, 4, 10)$	(6)	(z)
4	$(2, 9, 8, 5, 4, 6)$	(6)	(z)
5	$(2, 8, 4, 5, 6, 10)$	(6)	(c)
6	$(1, 6, 2, 8, 5, 4, 10)$	$(6, 7)$	$(0.180901699437495, c)$
7	$(1, 7, 8, 4, 3, 9, 10)$	$(6, 7)$	$(0.020225424859373682 5t^2 + 0.0584220259843842225, 0.18090169943749473)$
8	$(1, 5, 7, 6, 3, 10, 9)$	$(6, 7)$	$(0.02022542485937365t^2 + 0.058422025984384415, 0.18090169943749462)$
9	$(2, 8, 4, 3, 9, 5, 6)$	$(6, 7)$	$(0.1809016994374947, z)$
10	$(2, 3, 9, 8, 4, 5, 6)$	$(6, 7)$	$(0.18090169943749512, z)$
11	$(2, 8, 9, 5, 7, 6, 3)$	$(6, 7)$	$(0.18090169943749473, c)$
12	$(1, 7, 8, 4, 10, 2)$	(6)	(c)
13	$(1, 7, 4, 3, 9, 10)$	(6)	(c)
14	$(2, 9, 5, 6, 7, 3)$	(6)	(c)
15	$(2, 8, 3, 9, 5, 6)$	(6)	(z)
16	$(2, 8, 4, 5, 10, 6)$	(6)	(c)
17	$(1, 7, 3, 2, 8, 4, 10)$	$(6, 7)$	$(0.18090169943749507, z)$
18	$(1, 5, 6, 7, 3, 2, 9)$	$(6, 7)$	$(0.06909830056250547, z)$
19	$(1, 7, 3, 9, 5, 6, 2)$	$(6, 7)$	$(c, 0.06909830056250552)$
20	$(1, 8, 5, 4, 6, 10, 2)$	$(6, 7)$	$(c, 0.14926352961534042)$
21	$(1, 6, 2, 8, 7, 4, 10)$	$(6, 7)$	$(z, 0.1809016994374947)$
22	$(1, 5, 7, 4, 3, 9, 10)$	$(5, 7)$	$(0.18090169943749482, c)$
23	$(1, 7, 3, 4, 10, 9, 5)$	$(6, 7)$	$(c, 0.06909830056250518)$
24	$(1, 7, 6, 3, 10, 9, 5)$	$(6, 7)$	$(0.1809016994374948, c)$
25	$(2, 8, 9, 5, 4, 6, 3)$	$(6, 7)$	$(z, 0.12028993467659169)$

7.7 A second ten-facet local maximum

The HKO body is not the only local maximum in the ten-facet space. Consider the row-normal presentation

$$A_* = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 12/7 & 4/7 & 0 & 0 \\ 12/7 & -4/7 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ -12/7 & -4/7 & 0 & 0 \\ -12/7 & 4/7 & 0 & 0 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & -4 & 2 \\ 0 & 0 & -4 & -2 \\ 0 & 0 & 4/3 & -2 \end{pmatrix}, \qquad P_* := \{x \in \mathbb{R}^4 : A_* x \le \mathbf{1}\}.$$

The first six rows define a hexagon and the last four a quadrilateral in complementary Lagrangian planes. An exact finite certificate, together with the neighborhood argument below, proves the following.

Theorem 7.8 (A non-HKO ten-facet local maximum)**.** *The polytope P_* is simple, has ten genuine facets and 24 vertices, and*

$$\mathrm{vol}_4(P_*) = \frac{1}{2}, \qquad c_{\mathrm{EHZ}}(P_*) = 1, \qquad \mathrm{sys}(P_*) = 1.$$

There is a Hausdorff neighborhood $\mathcal{V}$ of P_* among convex four-dimensional polytopes with exactly ten genuine facets such that

$$\mathrm{sys}(P) \le 1 \qquad (P \in \mathcal{V}).$$

The maximum is non-strict. Locally, equality holds precisely on the images of the two-parameter product family described below under translations, positive dilations and linear symplectic maps. The two parameters remain independent after quotienting by these symmetries.

For completeness, write

$$u = 1 - v + v^2, \quad a = \frac{1}{2} - s, \quad b = \frac{1}{2} + s, \quad t = v - \frac{1}{2},$$

and let $E(v,s) = H(v,s) \times T(v)$, where

$$\begin{aligned} H(v,s) &= \mathrm{conv}\{(-a, \tfrac{1}{2}), (a, \tfrac{1}{2}), (b,t), (a, -\tfrac{1}{2}), (-a, -\tfrac{1}{2}), (-b,-t)\}, \\ T(v) &= \mathrm{conv}\{(u,0), (0,v), (u-1,0), (0,v-1)\}. \end{aligned}$$

At $(v,s) = (1/2, 1/12)$ this is P_*. The areas are identically 1 and $1/2$, and exact continuation of all product closure cases gives $c_{\mathrm{EHZ}}(E(v,s)) = 1$ and $\mathrm{sys}(E(v,s)) = 1$ for parameters sufficiently close to the base point.

Proof. For a word with positive feasible HK weights and objective $Q > 0$, the Haim–Kislev formula gives the upper bound

$$\mathrm{sys}(P(A)) \le U(A) := \frac{1}{8\ \mathrm{vol}_4(P(A))\, Q(A)^2}.$$

Thus the selected feasible sections are upper bounds for the actual ratio; they are not lower bounds and need not remain optimizing away from the base. On the product family, the Lagrangian splitting reduces the complete capacity calculation to the vertices of two planar

closure polytopes. Enumerating all 624 cyclic orders gives $Q_{\max} = 1/2$, 12 tight cases, and a strict gap of $1/14$ for every other case. Exact rational-function continuation of the closure supports and the tight cases proves the capacity and equality claims on $E(v, s)$. In this continuation every zero weight in a consistent base closure system is identically zero, while positive and negative weights retain their signs. The relevant support-system ranks persist, and inconsistent systems remain inconsistent. Thus no new closure vertex appears arbitrarily near the base; the twelve tight values remain $1/2$ identically, and the other values retain a strict gap on a common parameter neighborhood.

For arbitrary ten-facet perturbations, 24 positive feasible sections are continued through the full normal chart. Their gradient bank has rank 22 and has a positive normalized relation whose smallest coefficient is $3/36617$. The symmetry tangent space has rank 15; the two parameters of $E(v, s)$ add two directions, and one remaining transverse direction W is checked by an exact rational-function identity. At the base its only nonzero rows, in the same one-based labelling as A_*, are

$$W_7 = (1/2, 1/3, 0, 0), \qquad W_8 = (3/2, -1, 0, 0), \qquad W_9 = (1, 0, 0, 0).$$

Along this direction, with the feasible weights continued as in the certificate,

$$\operatorname{vol}_4(P(A_* + zW)) = \frac{1}{2} + \frac{13}{1728}z^2, \qquad Q_j(z) = \frac{1}{2}, \qquad U_j''(0) = -\frac{13}{432}$$

for every selected section. The last displayed number is the curvature of the feasible upper sections, not an asserted second derivative of the actual capacity.

To pass from these base-point calculations to a neighborhood, write $A(v, s)$ for the normalized facet rows of $E(v, s)$. The certificate continues the twenty-four contacts and W rationally along this family. Their ambient derivative matrix $G(v, s)$ annihilates the fifteen symmetry directions, the two family directions and $W(v, s)$ identically. Those eighteen directions remain independent near the base, while a rank-22 minor of G remains nonzero. Consequently G has constant rank 22 there. This constant-rank conclusion uses identities along the family; base-point rank alone would not suffice.

Choose a fixed 40×22 matrix Y whose columns complete these eighteen directions to a basis at the base. The exact chart-rank check verifies that this is possible. The matrix $H = GY$ has full column rank. If λ_* is the positive base relation, then

$$\widetilde{\lambda}(v, s) = [I_{24} - H(H^T H)^{-1} H^T]\lambda_*, \qquad \lambda = \frac{\widetilde{\lambda}}{\mathbf{1}^T \widetilde{\lambda}}$$

is a positive normalized relation after shrinking the parameter neighborhood. Indeed it equals λ_* at the base, and H and G have the same column space. Use local transverse coordinates

$$A(v, s, z, y) = A(v, s) + zW(v, s) + Yy, \qquad y \in \mathbb{R}^{22}.$$

The feasible sections give smooth upper bounds $U_j(v, s, z, y)$ with $U_j(v, s, 0, 0) = 1$. Their free-weight derivatives in z are chosen by the exact certificate to give the displayed curvature at the base.

Here is the uniform estimate that controls mixed perturbations. On a small compact parameter patch the positive relation and rank, by the same unit-sphere argument as in Lemma 7.5, give $\min_j (G_j Y)y \le -\alpha\|y\|$ for some $\alpha > 0$. Since $G_j W = 0$, Taylor's theorem gives a common $C > 0$ such that

$$\min_j (U_j - 1) \le -\alpha\|y\| + C(\|y\|^2 + z^2).$$

The weighted upper bound $F = \sum_j \lambda_j(v, s) U_j$ satisfies $F(v, s, 0, 0) = 1$ and $D_{(z,y)} F(v, s, 0, 0) = 0$. Its negative zz derivative persists near the base. Hence, for some $b, C' > 0$,

$$F - 1 \le -bz^2 + C'(\|y\|^2 + |z|\|y\| + (|z| + \|y\|)^3).$$

All constants may be chosen uniformly because these functions are rational and regular on a neighborhood of the compact patch. Choose $M > 2C/\alpha$. If $\|y\| \geq Mz^2$ and $(z, y) \neq 0$, the first estimate is strictly negative for sufficiently small $\|y\|$. If $\|y\| < Mz^2$, then $z \neq 0$ and the positive terms in the second estimate are $O(|z|^3)$, so it too is strictly negative for sufficiently small $|z|$. Each estimate bounds the actual ratio from above. At $z = y = 0$ that ratio equals one by the complete product calculation.

Finally include the fifteen symmetry parameters. The differential of their action on this chart has rank forty at the base, so the inverse function theorem gives coordinates for every nearby row matrix. The simple-polytope and polarity argument of Lemma 7.2 applies to P_* as well and turns this into a fixed-ten-facet Hausdorff neighborhood. The strict decrease for $(z, y) \neq 0$ gives precisely the asserted equality family and its symmetry images. □

The finite witness, exact verifier and independent direct differentiation check are retained at `topics/variation-and-local-maxima/hko-local-maximum/non-hko/`. The packet's `PROOF.md`, Sections 4–7, gives the support continuation, contact formulas and chart choices used above; its Section 8 records the uniform estimate. Exact arithmetic checks the finite predicates and rational-function identities, while the analytic argument supplies their geometric implication. No numerical neighborhood radius is claimed.

This result is deliberately narrower than a global theorem. It does not prove local maximality when facets are added, uniqueness among all ten-facet polytopes, or any new violation of Viterbo's inequality. Since HKO has 25 vertices and ratio $(3 + \sqrt{5})/5$, while P_* has 24 vertices and ratio one, the two bodies are not related by any invertible affine map, in particular not by a symplectic or anti-symplectic symmetry.

8 Geometric signals for large systolic ratios

We sample random polytopes and compute a broad range of readily available geometric quantities, including symplectic invariants and measurements that can change under symplectic transformations. We look for statistical patterns relating these quantities to the systolic ratio, then ask what geometry might explain them. Such patterns can suggest a conjecture, but an explanation must also account for the bodies on which the pattern fails.

A strong association concerns the total symplectic area of the two-dimensional faces, normalized by volume. In the broad samples, bodies with smaller normalized area tend to have larger systolic ratios. This ordering weakens substantially when we restrict attention to small values of this normalized sum or large ratios. Changes to the shapes of polygon factors support a geometric contribution to the broad association, without establishing its cause. Exact rotation formulas sharpen the distinction: regular pentagon products obey an inverse-square law, whereas equilateral triangle products have the opposite ordering.

These findings also distinguish two tasks in the search for large systolic ratios. A geometric measurement can select a population with larger ratios on average without indicating how to improve a body already in that population. The final part of the chapter examines search procedures that use capacity evaluations to propose further improvements, and the limits of numerical tests for local optimality.

8.1 Symplectic ridge area

A two-dimensional face of a four-dimensional polytope is called a ridge. For a ridge F with cyclically ordered vertices $v_0, \ldots, v_{r-1}$, define its absolute symplectic area by

$$A_\omega(F) = \frac{1}{2} \left| \sum_{i=0}^{r-1} \omega_0(v_i, v_{i+1}) \right|, \qquad v_r = v_0, \qquad \omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

This is the polygonal area formula with the symplectic pairing in place of the planar determinant. It measures the area seen by ω_0; a ridge can have positive Euclidean area and zero symplectic area. We normalize the sum over all ridges by

$$R(K) = \frac{\sum_F A_\omega(F)}{\sqrt{\mathrm{vol}_4(K)}}.$$

The absolute value is taken separately on each face: R records symplectic area without cancellation between faces, rather than their Euclidean areas. R is unchanged by translation, common dilation and linear symplectic maps, as is the systolic ratio. Translation invariance follows because the terms involving the translation vector cancel around the closed polygon. A linear symplectic map preserves each $A_\omega(F)$ and the four-dimensional volume. Thus any relation between R and the systolic ratio respects these symmetries.

8.2 A strong inverse association weakens under selection

For numerical evaluations we write $\widehat{\mathrm{sys}} = c_{\mathrm{num}}^2/(2V_{\mathrm{num}})$. Its normalization agrees with sys; its volume is computed in floating-point arithmetic. The statistical analysis retains the original numerical ratios. A separate reevaluation now supplies certified capacity intervals for 14,335 of the 14,336 original bodies, with one input still outside the numerical-size policy. These capacity bounds do not certify the floating-point volume or the numerical ratio (Appendix A.3).

The exploratory sample consists of 4,096 generic four-dimensional polytopes with facet counts $F = 5, \ldots, 12$ and 10,240 Lagrangian products $P \times Q$ with factor side counts $3 \le k \le m \le 6$. Each facet count has the same number of generic bodies, and each side-count pair has the same number of products. In coordinates (q_1, q_2, p_1, p_2), P lies in the q-plane and Q in the p-plane. A fixed facet count or pair of side counts defines a sample group.

The unit facet normals are independent and uniform on the appropriate sphere, and the support heights are independent and uniform in $[0.8, 1.2)$. Boundedness and the presence of every proposed facet are required. Thus these are specified distributions of bodies; the precise admission rules are in Appendix A.2. The largest numerical ratio in the exploratory sample is 0.863. Its retained values and current reevaluation are described in Appendix A.3.

The pooled rank correlation between R and the numerical systolic ratio is about -0.94. Rank correlation compares the orderings of the bodies by the two quantities; a value of -1 would mean exactly reversed orderings. The ordinary linear correlation is much weaker, about -0.21. Thus the principal observation concerns which bodies have larger systolic ratio, rather than how much the systolic ratio changes per unit of ridge area. The inverse association also occurs within every sampled facet-count or polygon-size group: the rank correlations range from about -0.84 to -1.00. Thus it is not solely a consequence of pooling these groups. Figures 8 and 9 show the separate populations.

Conditioning the original sample on either low R or high numerical systolic ratio weakens this association. If the lowest ten percent by R are retained within each group, the pooled correlation of within-group ranks is -0.222; retaining the highest ten percent by ratio gives -0.286, compared with -0.944 before restriction. Figure 7 shows the progressive attenuation. At the five-percent cutoff the high-ratio panel still has a negative association, so attenuation should not be confused with independence. The smallest panels contain only six generic or eleven product bodies per group.

Restricting the range of a variable can itself weaken a correlation. To assess this effect in a simple model, replace each group's measurements by pairs drawn from a bivariate Gaussian with the same full-sample rank correlation, then apply the same selections. At the five-percent cutoff, this reference gives mean correlation -0.695 for either selection, versus the observed -0.239 for high ratios and -0.113 for low R. The attenuation in the data is substantially stronger than in this reference. Other dependence structures can behave differently under selection, so

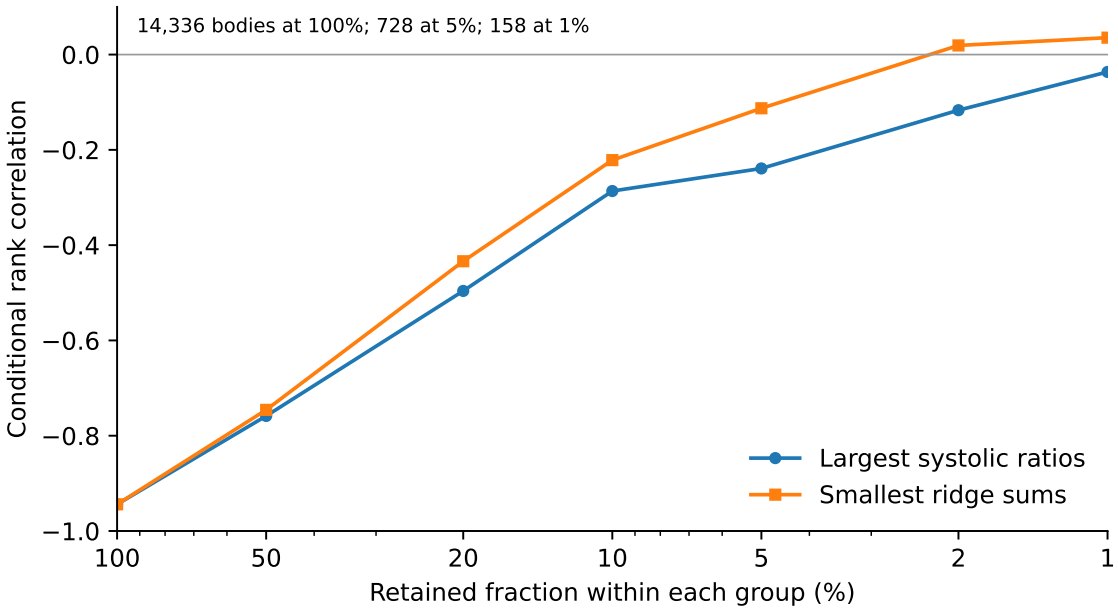


Figure 7: Restricting the original sample by either ridge sum or systolic ratio weakens their inverse association. Within each facet-count or factor-side-count group, retain the indicated fraction, rerank both quantities within that retained group, and compute the pooled Spearman correlation of the group-relative ranks. This is a descriptive conditional analysis of the original data. It does not evaluate a fresh selection rule or identify a geometric cause.

this comparison does not exclude range restriction as a contributor. The construction is in Appendix A.5.

Ridge measurements also contain predictive information beyond face counts and incidences. When predicting the numerical systolic ratio with gradient-boosted regression trees under a common training/test split, ridge-area features alone gave test $R^2 = 0.887$, whereas combinatorial counts alone gave 0.043. Here R^2 compares squared prediction error with prediction by the test-set mean. The feature lists, additional models and split definitions are in Appendix A.4.

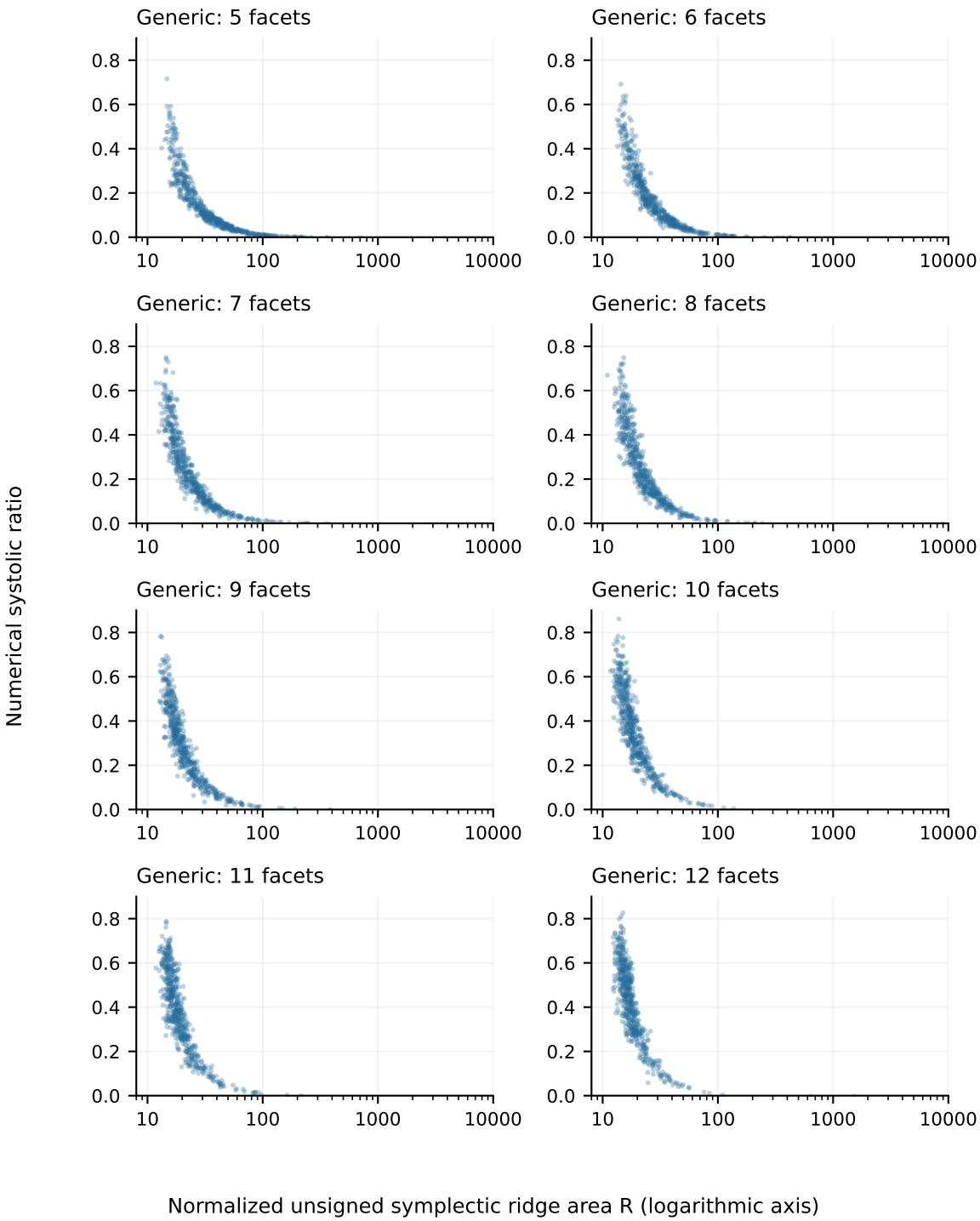


Figure 8: Ridge area and numerical systolic ratio for the 4,096 generic polytopes, separated by facet count. Each point is one body; the horizontal axis is logarithmic. All original numerical values are retained (Appendix A.3).

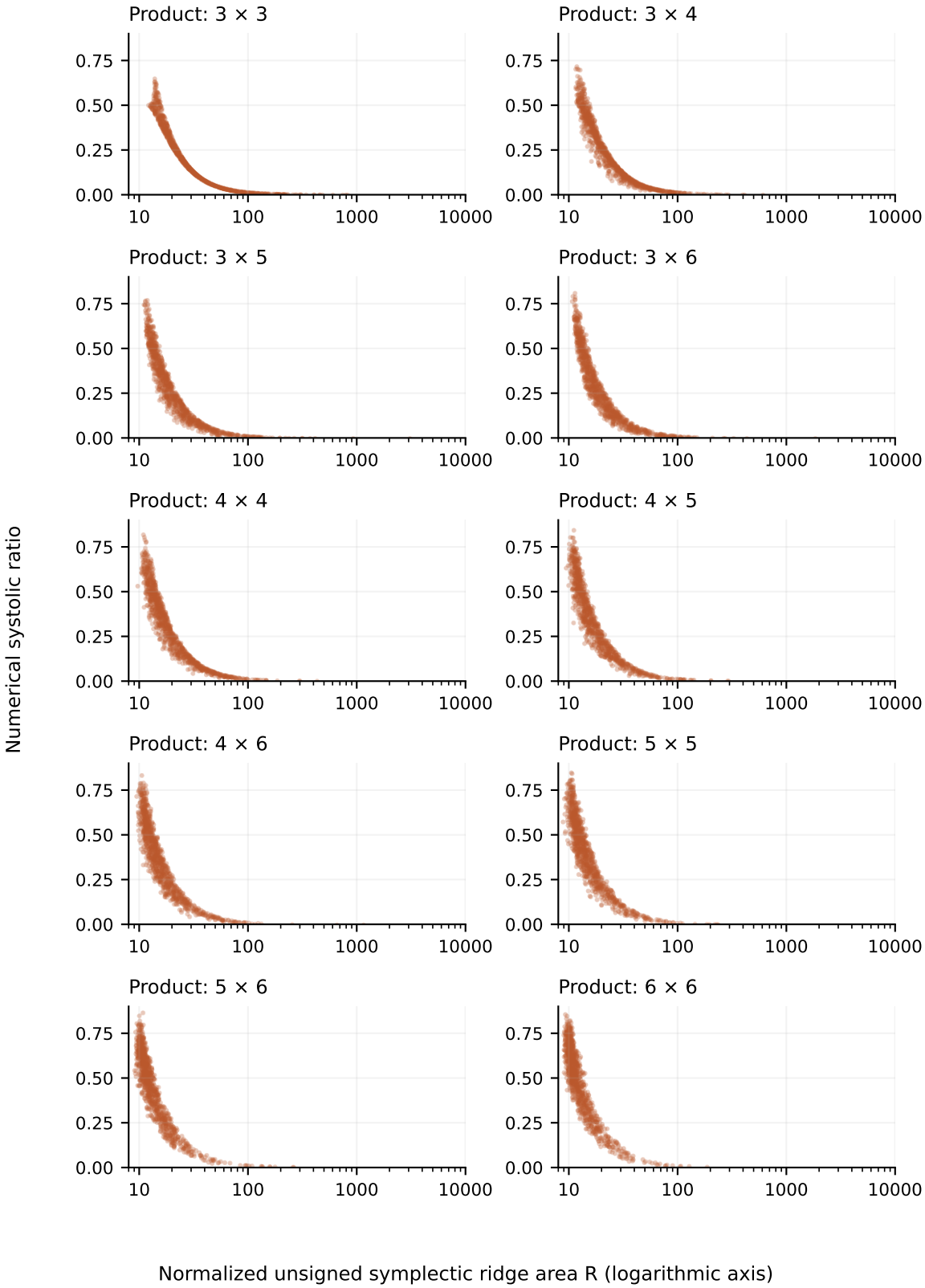


Figure 9: The same comparison for the 10,240 Lagrangian products, separated by factor side counts. Each group contains 1,024 bodies.

8.3 Coarse selection succeeds; stronger selection need not

Selecting products with small R raises the mean numerical systolic ratio, but further restriction within that selection need not raise it again. In a fresh pool of 100,000 Lagrangian products, selecting the ten lowest-R bodies per side-count group gave mean ratio 0.626, compared with 0.316 for the same number of comparison bodies. Selection used only geometry, before capacity was evaluated. Each factor-side-count group contributed equally. The comparison bodies came from the same pool, excluded the selected bodies, and were chosen without evaluating their capacities.

The distribution of area among the ridges supplies a second comparison. When the total symplectic ridge area is positive, define

$$C(K) = \frac{\max_F A_\omega(F)}{\sum_F A_\omega(F)}.$$

This is the share of the total carried by the largest ridge; like R, it is invariant under translation, common dilation and linear symplectic maps. Among products in the lowest one percent by R in a separate fresh sample, the half with lower C had a mean advantage of 0.034 over the other half, positive in eight of ten groups. Thus largest-ridge dominance distinguishes mean ratios within this selected pool, even when the total ridge area is small.

The strong ordering in the broad samples is not retained within the low-R panels themselves. Across 1,000 evaluated products in the lowest one percent by R, correlations within the ten side-count groups range from -0.315 to $+0.301$; pooling ranks computed within each group gives $+0.032$. The most extreme tenth has a larger mean ratio than the rest in only two groups. Small R distinguishes this population from the comparison bodies, but does not reliably order the ratios within it.

For generic ten-facet bodies, the related score R/N_2, where N_2 is the number of ridges, measures normalized area per ridge. Unlike products at fixed side counts, these bodies can have different ridge counts. Selecting the lowest one percent by R/N_2 from a fresh pool of 10,000 bodies gave mean ratio 0.626, against 0.339 for 100 other bodies chosen from the same pool without using their capacities. Within that selected panel, however, the ten most extreme scores had a mean ratio 0.032 below the remaining ninety. This is a failure to obtain further gain in that sample, not an established deterioration under every stronger filter. The comparison rules and conditional resampling summaries are retained in Appendix A.5.

The two geometric components of R/N_2 also behave differently under stronger selection. Let $E(F)$ be Euclidean ridge area and $\kappa(F) = A_\omega(F)/E(F)$, the alignment of its plane with the symplectic form relative to the Euclidean metric. Define the body-level quantities

$$\overline{E}(K) = \frac{\sum_F E(F)}{N_2\sqrt{\mathrm{vol}_4(K)}}, \qquad \overline{\kappa}_E(K) = \frac{\sum_F E(F)\kappa(F)}{\sum_F E(F)}.$$

Then $R(K)/N_2 = \overline{E}(K)\overline{\kappa}_E(K)$ exactly. Neither factor is invariant under all linear symplectic maps. In the generic selection both population means are smaller than in the comparison panel. Within the selected panel, however, the ten lowest scores have slightly larger $\overline{E}$ and smaller $\overline{\kappa}_E$ than the other ninety, without a larger mean ratio. Thus smaller alignment alone does not order these selected bodies by ratio. Appendix A.5 gives the population means; the identity above holds for each body and does not equate a mean product with the product of means.

8.4 Exact geometry: two families with opposite ordering

The broad inverse association has an exact counterpart in the rotated regular-pentagon products: their systolic ratio is proportional to R^{-2}. It is not a universal ordering. Equilateral triangle products have the opposite ordering, and the cube minimizes R among polygon products without

maximizing the systolic ratio. These exact statements constrain a proposed explanation of the statistical association; the counterexamples alone do not explain it.

For products the normalized ridge sum has an elementary edge formula. Let u_i and v_j be the counterclockwise boundary edge vectors of P and Q. A ridge lying entirely in a q- or p-plane has zero symplectic area. A mixed ridge is the rectangle formed from one edge in each factor, and its symplectic area is $|u_i \cdot v_j|$. Consequently

$$R(P \times Q) = \frac{\sum_{i,j} |u_i \cdot v_j|}{\sqrt{\operatorname{area}(P) \operatorname{area}(Q)}}.$$

For a fixed u_i, the sum over the edges of Q is the total variation of the projection of its boundary onto the direction of u_i. Convexity gives one increasing and one decreasing chain between the extreme projections, so

$$\sum_j |u_i \cdot v_j| = 2\|u_i\| \operatorname{width}_Q(u_i/\|u_i\|).$$

The normalized ridge sum therefore combines the edge lengths of one factor with the widths of the other in the edge directions.

For rotated regular pentagons, there is an exact inverse-square law. Let $K_\theta = P_5 \times \operatorname{Rot}(\theta)P_5$, where P_5 has circumradius one, and let $d(\theta) = \operatorname{dist}(\theta, (\pi/5)\mathbb{Z})$. Then

$$\operatorname{sys}(K_\theta) R(K_\theta)^2 = 16(3 + \sqrt{5}). \tag{35}$$

Thus distinct ridge sums within this family have strictly reversed systolic ratios. More precisely, R strictly decreases and sys strictly increases as d runs from 0 to $\pi/10$. Periodicity and reflection symmetry prevent strict monotonicity in the unrestricted angle θ.

For the proof, the area of each pentagon is $5\sin(\pi/5)\cos(\pi/5)$. Its edges have length $2\sin(\pi/5)$, so the mixed-edge formula gives

$$R(K_\theta) = 4\tan(\pi/5) \sum_{k=0}^{4} |\cos(\theta + 2\pi k/5)| = 16\sin(\pi/5)\cos d(\theta).$$

For the second equality, the absolute-cosine sum is even and has period $\pi/5$. On $[-\pi/10, \pi/10]$, the terms indexed by $0, 1, 4$ are nonnegative and those indexed by $2, 3$ are nonpositive. Pairing opposite indices reduces the sum to $[1 + 2\cos(2\pi/5) - 2\cos(4\pi/5)]\cos\theta = 4\cos(\pi/5)\cos\theta$. Combining this with the capacity profile in Theorem 9.1 proves (35).

Equilateral triangles give the opposite exact ordering. Let $T_\theta = P_3 \times \operatorname{Rot}(\theta)P_3$, where P_3 has circumradius one, and put $d_3 = \operatorname{dist}(\theta, (\pi/3)\mathbb{Z}) \in [0, \pi/6]$. Then

$$R(T_\theta) = 8\sqrt{3}\cos d_3, \qquad \operatorname{sys}(T_\theta) = \tfrac{1}{2}\sec^2(\pi/6 - d_3). \tag{36}$$

Both quantities strictly decrease with d_3, from $(8\sqrt{3}, 2/3)$ to $(12, 1/2)$. Thus the ridge sum and systolic ratio have the same strict ordering throughout this rotation family. Appendix A.1 proves the capacity formula by classifying the six-facet words in the quadratic program, and derives the ridge sum directly. Thus even the direction of a strict relation depends on the family. The pentagon law supplies exact positive evidence for an inverse relation on one family; the triangle law prevents extending that relation to all polygon products.

The minimum ridge sum for polygon products. Let P, Q be convex polygons with nonempty interior, set $D = P - P$, and write A_P, A_Q for their areas. Let $J_{\mathrm{ccw}}(x, y) = (-y, x)$, and normalize mixed area, for $t \geq 0$, by

$$\operatorname{area}(K + tL) = \operatorname{area}(K) + 2tV(K, L) + t^2 \operatorname{area}(L).$$

Then

$$R(P \times Q) = \frac{4V(D, J_{\mathrm{ccw}}Q)}{\sqrt{A_P A_Q}} \geq 8. \tag{37}$$

Equality holds precisely when P is centrally symmetric and $J_{\mathrm{ccw}}Q$ is positively homothetic to D, allowing translation.

Proof. Let h_D be the support function of D. The boundary-projection identity gives $\sum_i |u_i \cdot v_j| = 2h_D(v_j)$. The outward unit normal to the counterclockwise edge $J_{\mathrm{ccw}}v_j$ is $v_j/\|v_j\|$. The polygon mixed-area formula therefore gives

$$2V(D, J_{\mathrm{ccw}}Q) = \sum_j \|v_j\| h_D(v_j/\|v_j\|) = \sum_j h_D(v_j),$$

which proves the identity. Brunn–Minkowski and its equality case [Gar02, Sections 3 and 5] imply

$$V(D, J_{\mathrm{ccw}}Q)^2 \geq \mathrm{area}(D) A_Q, \qquad \mathrm{area}(D) \geq 4A_P.$$

The first inequality is an equality exactly when D and $J_{\mathrm{ccw}}Q$ are positively homothetic. The second is an equality exactly when P is centrally symmetric. Substitution proves the bound and equality statement. □

For two aligned unit squares, eight edge pairs have absolute scalar product one and the rest zero, so $R = 8$. Their product is the unit four-dimensional cube, whose capacity is one and whose systolic ratio is $1/2$.[2] HKO has the larger ridge score 8.944 and a ratio greater than one. This disproves a universal inverse ordering, while leaving the statistical association in the sampled distributions unexplained.

8.5 Covariance balance gives a second geometric signal

Ridge area is not the only summary associated with the ratio. In an exploratory analysis, more nearly equal symplectic covariance scales of the vertex set were associated with larger ratios. For vertices $v_1, \ldots, v_N$, define

$$\overline{v} = \frac{1}{N} \sum_i v_i, \qquad \Gamma = \frac{1}{N} \sum_i (v_i - \overline{v})(v_i - \overline{v})^T.$$

By Williamson normal form, a symplectic change of coordinates sends Γ to $\mathrm{diag}(\nu_1, \nu_2, \nu_1, \nu_2)$, with $0 < \nu_1 \leq \nu_2$. The ratio $\rho = \nu_2/\nu_1$ measures the imbalance between the two symplectic covariance scales.

Like R, this score is invariant under affine symplectic maps and common dilation. Centering removes translations. For symplectic S, $\Gamma(SK) = S\Gamma(K)S^T$, so with $J = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$,

$$J\Gamma(SK) = S^{-T}(J\Gamma(K))S^T.$$

The eigenvalues $\pm i\nu_1, \pm i\nu_2$ are unchanged. Dilation multiplies both ν_j by the same square factor. For a product with planar vertex covariances A, B, moreover, $\Gamma = \mathrm{diag}(A, B)$ and $(J\Gamma)^2 = \mathrm{diag}(-BA, -AB)$; thus ν_1^2, ν_2^2 are the eigenvalues of AB.

Selecting the lowest 0.5% by ρ within each side-count group gave larger mean ratios than comparison bodies in every group in each of two fresh pools of 50,000 products. Each group supplied 25 comparison bodies from the same pool, chosen without using their capacities. The

[2] For the centered unit cube the dual rows are $\pm 2e_i$. In the variational formula, closure forces equal weights x_j on the two q_j rows and equal weights y_j on the two p_j rows, with $\sum_{j=1}^{2}(x_j + y_j) = 1/2$. The order of the four rows in a coordinate pair gives a contribution at most $8x_jy_j$ to Q. Thus $Q \leq 8\sum_j x_jy_j \leq 1/2$. A cyclic word on one coordinate pair with its four weights equal to $1/4$ attains $Q = 1/2$, giving $c = 1/(2Q) = 1$.

twenty gains over comparison bodies ranged from 0.260 to 0.440, with mean 0.333. Against selection of the lowest 1% by R followed by its lower half by C, its mean advantage was only 0.014, and the differences changed sign across groups. Both geometric summaries therefore support selection in the tested populations; neither was uniformly better there.

Both selections also raised mean ratios under one change of sampling law. For tangential factors obtained by setting the support heights to one, the mean gains were 0.259 for covariance selection and 0.225 for ridge selection, averaged equally over the 4×6 and 6×6 groups. This is one conditioned source, requiring both original and changed polygons to be valid. It establishes a particular transfer, not independence from the sampling distribution. Details are in Appendix A.5.

8.6 Balancing the factors improves most tested products

Removing the planar vertex-covariance anisotropy of both factors raises the mean numerical ratio from 0.342 to 0.654 in a matched comparison of 320 products, 32 from each side-count group, with increases in 314 and decreases in six. At the same time, the pooled within-group rank correlation between R and the ratio weakens from -0.929 to -0.249. Thus affine shape variation of the factors contributes materially to the broad association in this sample.

The transformation has a precise geometric meaning. For a factor with vertex covariance A, set

$$L_A = (\det A)^{1/4} A^{-1/2}.$$

Then $\det L_A = 1$ and $L_A A L_A^T = \sqrt{\det A}\, I_2$. Applying L_A and L_B to the two factors preserves their areas and combinatorics and makes both vertex covariances scalar matrices. The product therefore has $\rho = 1$. These independent factor maps need not combine to a symplectic map, so capacity can change even though volume is preserved.

The mean ridge sum falls from 23.739 to 11.652, decreasing in 306 of the 320 products. The remaining relation depends on the side counts: the balanced 3×3 group has positive rank correlation, while the other nine groups retain negative correlations. This reversal has an exact geometric explanation. A triangle with isotropic vertex covariance is equilateral (Appendix A.1), so ideal balancing places triangle products in the family with the same-order relation (36). The transformation changes several geometric properties together. The paired changes support a contribution from affine factor shape to the bulk association, but do not identify covariance imbalance as its sole cause. Nor do its six decreases permit a universal improvement rule. The selection of original bodies without using their ratios, complete paired evaluations and numerical checks are in Appendix A.7.

8.7 Opposite changes under a support-height deformation

A second deformation gives predominantly opposite changes in ridge area and systolic ratio in a small paired sample. Keeping polygon normals fixed and setting both factors' support heights to one does so in fifteen of sixteen tested products, drawn equally from the 4×4 and 4×6 groups. In ten pairs R decreased and the ratio increased; in five the reverse happened. Both quantities increased in the remaining pair. Making the factors tangential therefore does not always raise the ratio.

Separate normalization of the factors preserves the ratio: for $a, b > 0$,

$$\operatorname{diag}(aI_2, bI_2) = \sqrt{ab}\, \operatorname{diag}(\sqrt{a/b}\, I_2, \sqrt{b/a}\, I_2)$$

is a dilation followed by a symplectic map. Thus replacing unequal support heights is the geometric change being tested.

The evidence is stronger for direction than for size of effect. The Spearman correlation between the two changes is -0.55 overall and only -0.095 in the 4×6 group. All tested

ratios are at most 0.742, so this is not a deformation test of the high-ratio tail. It is consistent with shared geometric variation producing the bulk association, but does not isolate a causal role for ridge area or explain the whole pattern. The paired constructions and checks are in Appendix A.6.

8.8 What could explain the bulk association?

One possible explanation is that much of the broad association comes from shape variations that lower the systolic ratio and raise ridge area together. Reversing such variations would then improve many bodies in the bulk of the sample, while offering less scope for improvement among bodies already selected for large ratios or small ridge area. Factor balancing supports this possibility: it substantially raises most ratios and weakens the inverse association. The support-height deformation supplies a second example of predominantly opposite changes. Neither experiment, however, measures distance to a local optimum or establishes that this mechanism explains the attenuation under selection.

A different explanation is an underlying relation involving a geometric quantity that R measures only imperfectly. Extreme R can then reflect both favorable geometry and deviations specific to the proxy. Here noise would mean differences between bodies not captured by the relation, not numerical error in a fixed-body evaluation. Selection for extreme R would then select partly for these deviations, weakening its association with the ratio. This account is compatible with the residual associations after balancing, but no such underlying quantity or law has been identified.

These explanations are compatible, and the results do not choose between them. The observed attenuation is stronger than in the fitted Gaussian reference, but other dependence structures can behave differently under selection. Neither convenient account need describe the full phenomenon. The opposite exact laws for pentagons and triangles, the global counterexample and the paired shape changes constrain any broader explanation.

8.9 Using evaluated ratios to guide search

Geometric selection and subsequent improvement are separate search tasks. The ridge descriptors identify promising initial populations, but they do not supply a general ascent direction. Direct evaluations can instead guide changes to the sampling distribution or local proposals, while endpoint probes test whether a search has missed further improvements.

The tested adaptive sampling procedure attained larger maxima and higher top-eight medians than independent sampling in three paired runs. For 5×5 products, the cross-entropy method [Boe+05] shifts a fitted distribution towards the quarter of the current population with largest ratios, then samples again. Under equal evaluation counts, the results were:

	Maximum		Top-eight median	
Seed	IID	Adaptive	IID	Adaptive
1	.760	.799	.729	.782
2	.811	.831	.754	.776
3	.760	.856	.722	.814

The top-eight statistic is the median of the eight highest ratios after removing repeated exact dual-vertex arrays. Both arms share their initial population. The comparison does not isolate the benefit of adaptation: a coordinate-dependent acceptance condition filters the two representations differently. Independent samples retain their original coordinates, while adaptive proposals use a normalized representation; the evaluator imposes coordinate-size bounds on both. Thus the table compares procedures that differ in both sampling and admission. The fitted coordinates, updates and admission rules are in Appendix A.10.

In the tested finite schedules, rotating a fixed starting product by $K \mapsto UK$, $U \in SO(4)$, attained a smaller maximum numerical ratio than fresh product sampling at the same evaluation budget (Appendix A.6).

8.10 Competing branches in local search

The branch model in Section 6.4 supplies a more local source of proposed improvements. Its lower-envelope structure explains why increasing one currently minimizing branch can expose another as the new minimum. The finite-gap model therefore retains nearby branches, and history-based variants also retain previously active ones. A numerically chosen branch window is not guaranteed to contain every relevant branch.

Retaining branches from earlier points gave the largest median best-so-far numerical ratios in the comparison. Seven fixed implementations shared the same 64 generic ten-facet starting bodies, a cap of 128 full evaluations and a nominal one-second accumulated evaluator-plus-optimizer budget:

Implementation	Median best ratio	Across-start 10–90% range
Four-anchor history	.985	.957–.999
History with transition prediction	.982	.956–.994
Finite-gap model	.977	.948–.995
Literal gradient	.881	.790–.924
Safeguarded gradient	.855	.780–.916
CMA-ES	.818	.718–.869
Coordinate poll	.601	.358–.787

The finite-gap method models branches near the current minimum. The two history methods also retain earlier branches; one predicts additional branches from transitions. CMA-ES [Han16] and coordinate polling supply population and direct-search comparisons. These are rankings of the fixed implementations under this allocation, not of optimally tuned algorithm families. The historical numerical objective uses floating-point geometry and volume, and not every original executable or proposal trace was recovered. Appendix A.8 gives the retained steps, provenance and budget interpretation. The ranges in the table show variation across starting bodies; all 448 best-so-far ratios are below one.

8.11 A failed probe does not establish a local maximum

The endpoint diagnostics show that a high terminal ratio need not mean local convergence. One optimizer endpoint decreased along every tested signed basis direction, yet a continuation increased its ratio by 0.00223 along oblique moves of total normalized path length 0.01. All three selected endpoints in that continuation accepted ten improving steps, while HKO accepted none. HKO's local status is established by its theorem; the finite test supplies a consistent control observation.

A regular-pentagon control on a known increasing rotation path illustrates the same problem independently. In a suite of basis, random and prescribed rotation probes, its only positive changes were the twelve rotation probes; the generic directions missed the available increase. The other controls, probe design and numerical limitations are in Appendix A.9.

The chapter therefore supports a strong coarse association, opposite exact laws on two rotation families, and improvements under fixed search budgets. It does not supply a universal capacity law or a general route to local optimality. None of the random samples or search comparisons reported here produced a numerical ratio above one. For products with a triangle or quadrilateral factor, the Viterbo inequality is already known [BMP26, Theorem 1.2 and Section 4]; such products can illuminate the pattern but cannot yield counterexamples while remaining in that class. The remaining product families and generic polytopes leave open

both questions: what geometric relation explains the signal, and what search directions remain effective near large ratios?

9 The rotated-pentagon profile

The Haim–Kislev–Ostrover counterexample is a Lagrangian product of two regular pentagons. We determine how its capacity changes when one pentagon rotates relative to the other. The volume stays constant, so this also determines which relative positions maximize the systolic ratio.

Let $P_5 \subset \mathbb{R}^2$ be a regular pentagon centred at the origin with circumradius one. Write $R(\theta)$ for counterclockwise rotation through angle θ, and put

$$K_\theta = P_5 \times_L R(\theta)P_5 \subset \mathbb{R}^2_q \oplus \mathbb{R}^2_p.$$

The Lagrangian product here is the Cartesian product in the two coordinate planes, with symplectic form $\omega_0((q,p),(q',p')) = q \cdot p' - p \cdot q'$. Simultaneous rotation of both planes is symplectic, so the choice of initial orientation for P_5 does not affect the capacity. We use

$$\mathrm{sys}(K) = \frac{c_{\mathrm{EHZ}}(K)^2}{2\,\mathrm{vol}_4(K)}.$$

The sampled ratios in Figure 10 rise from the aligned position to a maximum at $\theta = \pi/10$, then fall symmetrically. They suggested a profile proportional to $\sec^2\theta$ on $[0, \pi/10]$. We prove this formula by comparing the quadratic programs at different angles. Their feasible weights remain the same under rotation, and each fixed choice has a particularly simple angular dependence. The known capacity at the Haim–Kislev–Ostrover position will then bound every such choice throughout the interval.

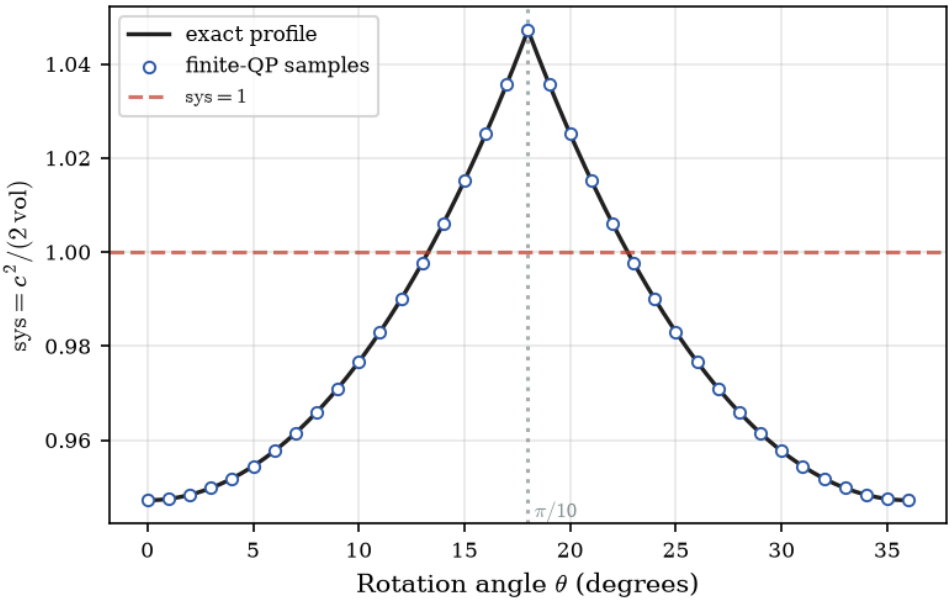


Figure 10: Systolic ratios for the rotating pentagon product. The blue markers are numerical evaluations of the finite quadratic programs; the black curve is the formula proved in this chapter. The horizontal line marks sys = 1.

To state the profile for all angles, define

$$d(\theta) = \min_{k\in\mathbb{Z}} \left|\theta - \frac{k\pi}{5}\right|.$$

This is the distance to the nearest multiple of $\pi/5$, and takes values in $[0, \pi/10]$.

Theorem 9.1 (Rotated-pentagon profile)**.** *For every real θ,*

$$c_{\mathrm{EHZ}}(K_\theta) = \bigl(1 + \cos(\pi/5)\bigr)^2 \sec d(\theta), \tag{38}$$

$$\mathrm{sys}(K_\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta). \tag{39}$$

The maximum systolic ratio is $(3 + \sqrt{5})/5 > 1$, attained exactly at $\theta = \pi/10 + k\pi/5$, $k \in \mathbb{Z}$.

9.1 Symmetries and the Haim–Kislev–Ostrover position

We first reduce the angles to $[-\pi/10, \pi/10]$ and identify the capacity at its endpoints. The pentagon's rotational symmetry gives $K_{\theta+2\pi/5} = K_\theta$. If S is reflection in a symmetry axis of P_5, the simultaneous reflection $(q, p) \mapsto (Sq, Sp)$ is symplectic and sends K_θ to $K_{-\theta}$. Thus capacity is even as a function of θ.

Exchanging the two factors gives a further symmetry. The symplectic map $(q, p) \mapsto (p, -q)$ sends K_θ to $R(\theta)P_5 \times_L (-P_5)$. Rotate both factors through $-\theta$ and use $-P_5 = R(\pi/5)P_5$ to obtain $K_{\pi/5-\theta}$. Together with evenness, this proves that capacity has period $\pi/5$. Every angle can therefore be reduced to $[-\pi/10, \pi/10]$ without changing capacity; evenness will give the final expression in terms of $d(\theta)$.

Haim–Kislev and Ostrover computed the capacity of the product with relative angle $-\pi/2$. Their convention for the symplectic form is $\sum_i dp_i \wedge dq_i = -\omega_0$. The map $(q, p) \mapsto (q, -p)$ changes this convention to ours and sends their body to $K_{\pi/2}$. Since $\pi/2 = \pi/10 + 2\pi/5$, their result [HO26, Proposition 1.4], together with the reflection symmetry, gives

$$c_{\mathrm{EHZ}}(K_{-\pi/10}) = c_{\mathrm{EHZ}}(K_{\pi/10}) = 2\cos(\pi/10)\bigl(1 + \cos(\pi/5)\bigr). \tag{40}$$

9.2 Feasible weights under rotation

The quadratic capacity formula lets us compare different angles using the same facet order and weights. We spell out its variables for this family. Set

$$h = \cos(\pi/5), \qquad n_i = \left(\cos\left(\frac{\pi}{2} + \frac{2\pi i}{5}\right), \sin\left(\frac{\pi}{2} + \frac{2\pi i}{5}\right)\right), \quad 0 \le i < 5.$$

Choose the orientation $P_5 = \{q : n_i \cdot q \le h,\ 0 \le i < 5\}$. The ten facets of K_θ then have normalized normals

$$a_i(\theta) = h^{-1}(n_i, 0), \qquad a_{i+5}(\theta) = h^{-1}(0, R(\theta)n_i), \qquad 0 \le i < 5,$$

so that their inequalities are $a_i(\theta) \cdot x \le 1$.

A facet word $\sigma = (\sigma_1, \dots, \sigma_m)$ is an ordered list of distinct indices from $\{0, \dots, 9\}$. Its feasible weights satisfy

$$\beta_k \ge 0, \qquad \sum_{k=1}^{m} \beta_k = 1, \qquad \sum_{k=1}^{m} \beta_k a_{\sigma_k}(\theta) = 0. \tag{41}$$

The last equation is closure. For these weights, define

$$Q_{\sigma,\beta}(\theta) = \sum_{1 \le j < k \le m} \beta_j \beta_k \omega_0\bigl(a_{\sigma_j}(\theta), a_{\sigma_k}(\theta)\bigr). \tag{42}$$

This uses the traversal order of the quadratic-program chapter. The Haim–Kislev formula [Hai19] states that

$$c_{\mathrm{EHZ}}(K_\theta) = \frac{1}{2Q_{\max}(\theta)}, \qquad Q_{\max}(\theta) = \max_{\sigma,\beta} Q_{\sigma,\beta}(\theta) > 0, \tag{43}$$

where the maximum is over all words and their feasible weights. There are finitely many words, and each feasible set is a closed subset of a simplex, so the maximum exists. Zero weights allow facets to be omitted.

For this product, closure separates into equations in the two planes:

$$\sum_{\sigma_k<5} \beta_k n_{\sigma_k} = 0, \qquad R(\theta) \sum_{\sigma_k \geq 5} \beta_k n_{\sigma_k - 5} = 0.$$

Removing the invertible rotation from the second equation shows that feasibility is independent of θ. Thus a word and weights chosen at one angle remain feasible at every other angle.

We first exhibit a choice that gives the proposed capacity as an upper bound. Take

$$\sigma = (3, 8, 1, 0, 5, 6), \qquad \beta = \frac{1}{4(1+h)}(2h, 2h, 1, 1, 1, 1).$$

These weights are positive and sum to one. The normal n_3 points opposite the bisector of n_0 and n_1, and hence

$$n_0 + n_1 = -2h n_3.$$

This identity gives closure in both factors. To evaluate Q, group the consecutive entries $(1, 0)$ and $(5, 6)$ of the word. Normals from the same factor have zero symplectic pairing, so bilinearity lets us replace each group by its weighted sum. Writing

$$u = \frac{n_3}{2(1+h)}, \qquad v = R(\theta)u,$$

the four weighted sums, in their order, are

$$(u, 0), \quad (0, v), \quad (-u, 0), \quad (0, -v).$$

The two same-factor pairings vanish. The other four are

$$\begin{aligned} \omega_0((u,0),(0,v)) &= u \cdot v, & \omega_0((u,0),(0,-v)) &= -u \cdot v, \\ \omega_0((0,v),(-u,0)) &= u \cdot v, & \omega_0((-u,0),(0,-v)) &= u \cdot v. \end{aligned}$$

Their sum is $2u \cdot v$. Therefore

$$Q_{\sigma,\beta}(\theta) = \frac{\cos\theta}{2(1+h)^2}. \tag{44}$$

For $|\theta| \leq \pi/10$ this is positive, and the capacity formula yields

$$c_{\mathrm{EHZ}}(K_\theta) \leq (1+h)^2 \sec\theta. \tag{45}$$

To prove equality, we must show that no other feasible choice gives a larger quadratic value.

9.3 Bounding all choices by their endpoint values

Fix any facet word σ and feasible weights β. Same-factor pairings in (42) vanish, and each mixed pairing is, up to sign, $h^{-2} n_i \cdot R(\theta) n_j$. There are consequently constants $A_{\sigma,\beta}$ and $B_{\sigma,\beta}$ such that

$$Q_{\sigma,\beta}(\theta) = A_{\sigma,\beta} \cos\theta + B_{\sigma,\beta} \sin\theta.$$

This angular dependence determines the value inside our interval from the two endpoint values. The sine addition formulas give

$$\begin{aligned} Q_{\sigma,\beta}(\theta) = {} & \frac{\sin(\pi/10 - \theta)}{\sin(\pi/5)} Q_{\sigma,\beta}(-\pi/10) \\ & + \frac{\sin(\pi/10 + \theta)}{\sin(\pi/5)} Q_{\sigma,\beta}(\pi/10). \end{aligned} \tag{46}$$

For $-\pi/10 \le \theta \le \pi/10$, both coefficients are nonnegative and their sum is $\cos\theta/\cos(\pi/10)$.

The fixed choice remains feasible at both endpoints, so its quadratic value there is at most the corresponding maximum. By (40) and (43),

$$Q_{\sigma,\beta}(\pm\pi/10) \le Q_{\max}(\pm\pi/10) = \frac{1}{4\cos(\pi/10)(1+h)}.$$

We may substitute these upper bounds into (46) because its coefficients are nonnegative. This gives

$$Q_{\sigma,\beta}(\theta) \le \frac{\cos\theta}{4\cos^2(\pi/10)(1+h)} = \frac{\cos\theta}{2(1+h)^2},$$

using $2\cos^2(\pi/10) = 1+h$. The bound holds for every feasible choice, and the choice in (44) attains it. We have proved

$$Q_{\max}(\theta) = \frac{\cos\theta}{2(1+h)^2}, \qquad c_{\mathrm{EHZ}}(K_\theta) = (1+h)^2\sec\theta, \qquad |\theta| \le \pi/10.$$

Evenness and $\pi/5$-periodicity give (38) for all real angles.

Finally, the five triangles joining the centre of P_5 to its edges give

$$\mathrm{area}(P_5) = \frac{5}{2}\sin(2\pi/5), \qquad \mathrm{vol}_4(K_\theta) = \left(\frac{5}{2}\sin(2\pi/5)\right)^2.$$

Substitution into the definition of the systolic ratio yields

$$\mathrm{sys}(K_\theta) = \frac{(1+\cos(\pi/5))^4}{2\left(\frac{5}{2}\sin(2\pi/5)\right)^2}\sec^2 d(\theta) = \frac{5+2\sqrt{5}}{10}\sec^2 d(\theta).$$

Since $\sec^2 t$ is strictly increasing on $[0,\pi/10]$, the maximum occurs precisely when $d(\theta) = \pi/10$. At those angles its value is $(3+\sqrt{5})/5$. This completes the proof of Theorem 9.1.

10 Independent linear deformations of the pentagon factors

The rotation formula leaves open what happens when the two pentagons are stretched or sheared independently. Such changes generally alter both capacity and volume. We show that they cannot increase the systolic ratio beyond the Haim–Kislev–Ostrover value. Translations do not affect either quantity, so linear maps suffice to describe the extension to affinely regular pentagons.

We use a regular pentagon with circumradius one and outward facet normals

$$n_i = (\cos(2\pi i/5), \sin(2\pi i/5)), \qquad P = \{q : n_i \cdot q \le h,\ 0 \le i < 5\}, \qquad h = \cos(\pi/5).$$

This normal convention differs by a common rotation from the preceding rotation proof; applying that rotation to both coordinate planes is symplectic. Its area is $S = \frac{5}{2}\sin(2\pi/5)$. Throughout this section the product is Lagrangian, with form $\omega_0((q,p),(q',p')) = q\cdot p' - p\cdot q'$.

Theorem 10.1. *Let $G, H \in GL(2,\mathbb{R})$, and set*

$$M = G^{-1}H^{-T}, \qquad m(M) = \max_{0\le i,j<5} |n_i^T M n_j|.$$

Then

$$c_{\mathrm{EHZ}}(GP \times HP) = \frac{(1+h)^2}{m(M)}.$$

Moreover,

$$\operatorname{sys}(GP \times HP) := \frac{c_{\mathrm{EHZ}}(GP \times HP)^2}{2\operatorname{area}(GP)\operatorname{area}(HP)} \le \frac{3+\sqrt{5}}{5}.$$

Write $D = \operatorname{conv}\{\pm n_i : 0 \le i < 5\}$ *and* $\alpha = \pi/10$. *Equality holds precisely when* $M = tR_\alpha U$, *where* $t > 0$ *and* U *is an orthogonal symmetry of the regular decagon* D.

The formula requires only twenty-five absolute pairings, even though the capacity formula initially ranges over all facet orders and feasible weights. The reason is a containment between convex sets of matrices. Unlike the endpoint comparison for rotations, this containment is preserved by independent linear changes in the two factors.

10.1 Reducing the possible facet orders

Write a polygon as $\{q : u_i \cdot q \le 1\}$. Its normalized closure polytope is

$$C = \{\gamma \ge 0 : \ \textstyle\sum_i \gamma_i = 1, \ \sum_i \gamma_i u_i = 0\}.$$

For the pentagon, $u_i = n_i/h$. The Haim–Kislev formula [Hai19, Theorem 1.1], with earlier word entries as the first argument of ω_0, reads

$$c_{\mathrm{EHZ}} = \frac{1}{2Q_{\max}}, \qquad Q = \sum_{j<k} \beta_j \beta_k \omega_0(a_{\sigma_j}, a_{\sigma_k}),$$

where the product covectors a_i are support-normalized, the weights have sum one, and their weighted covector sum vanishes.

For a product, same-factor pairings vanish. If the total weight in the first factor is t, the objective therefore has the form $t(1-t)B(\gamma,\eta)$, where B is bilinear in the normalized closure weights of the factors. At a positive global maximum, both factors have nonzero mass, so $0 < t < 1$ and $B(\gamma,\eta) > 0$. Extend its word to a full facet order by assigning zero weight to unused facets. With this order and η fixed, B is linear in γ and attains its maximum at a vertex of the first closure polytope. Replacing γ by that vertex cannot decrease the value; since we started at a global maximum, the value is unchanged. Apply the same argument to η with the new γ fixed. Thus both closure weights may be chosen at vertices.

A vertex has at most three positive entries. Otherwise the corresponding columns of the three equations $\sum \gamma_i = 1$ and $\sum \gamma_i u_i = 0$ have a nonzero dependence δ. Extend δ by zero outside the positive support. For sufficiently small $\varepsilon > 0$, both $\gamma + \varepsilon\delta$ and $\gamma - \varepsilon\delta$ remain nonnegative and satisfy the same equations, contradicting extremality. Finally, keeping the two closure vertices and the word fixed while varying t preserves feasibility. Since $B > 0$, its factor $t(1-t)$ is maximal at $t = 1/2$.

Thus some maximizer uses at most three facets of each factor. This is an existence statement; it does not assert that all maximizing words are sparse. Moving the first weighted covector cyclically to the end preserves Q, since the change is its pairing with minus twice the sum of all the other weighted covectors, and closure makes that pairing zero. We may consequently group adjacent entries from the same factor cyclically. There are equally many groups of each factor, and each uses at most three nonzero facets. Thus the number of alternating groups is two, four, or six. With two groups, each group contains every step of one factor, so both grouped sums are zero by factorwise closure and $Q = 0$.

Put $x_i = \gamma_i u_i$ and $y_j = \eta_j v_j$. These are factor steps with sum zero; their actual product weights carry an additional factor $1/2$. A four-group order $x, y, -x, -y$ has value $x \cdot y/2$: its unscaled pairing sum is $2x \cdot y$, and the two half-masses contribute a factor $1/4$. A six-group order $x_1, y_1, x_2, y_2, x_3, y_3$ has value

$$Q = \frac{1}{2}(x_1 \cdot y_1 - x_3 \cdot y_2).$$

To see this, let P_j be the sum of the q-steps preceding y_j. The following q-steps sum to $-P_j$ by closure. The mixed terms involving y_j therefore contribute $\frac{1}{4}(P_j \cdot y_j - y_j \cdot (-P_j)) = \frac{1}{2} P_j \cdot y_j$. Here the three prefixes are $x_1, x_1 + x_2 = -x_3, 0$, which gives the formula. Represent these expressions by matrices

$$\frac{1}{2} x y^T, \qquad \frac{1}{2}(x_1 y_1^T - x_3 y_2^T),$$

whose Frobenius pairing with the identity matrix is the objective. Under independent changes G, H, the support-normalized covectors become $G^{-T} u_i$ and $H^{-T} v_j$. Their closure weights are unchanged, and both matrix types transform by the same linear map $T \mapsto G^{-T} T H^{-1}$. It remains to show that every six-group matrix for regular pentagons lies in the convex hull of the four-group matrices.

10.2 Two identities cover all six-group competitors

Set

$$\lambda = \frac{\sqrt{5} - 1}{2} = \frac{1}{2h}, \qquad s = \frac{1}{1 + h}, \qquad \rho = \frac{s^2}{2}, \qquad \mathcal{D} = \operatorname{conv}\{\pm n_i n_j^T : 0 \le i, j < 5\}.$$

A positive closure triple places the origin in the interior of the triangle of its three normals. Indeed, its positive closure weights are barycentric coordinates of the origin, and three distinct points on the circle are noncollinear. If a cyclic angular gap were at least π, the three normals would lie in a closed semicircle and hence in a closed half-plane through the origin. Their triangle could not contain the origin in its interior. Thus every cyclic angular gap is less than π. Its three positive integer index gaps therefore sum to five and are at most two, so they are cyclic permutations of $(1, 2, 2)$. There are exactly five triples, all rotations of $\{n_0, n_2, n_3\}$, and no opposite pairs. The normalized steps for this representative triple are

$$se, sa, sb, \qquad e = n_0, \quad a = \lambda n_2, \quad b = \lambda n_3, \qquad e + a + b = 0.$$

Indeed the closure weights are $h/(1+h), 1/[2(1+h)], 1/[2(1+h)]$. Every proper grouped sum of a triple is one step or its negative. The convex hull of all such sums, over the five triples, is sD: all $\pm s n_i$ occur, and the remaining steps are their contractions by $\lambda < 1$. In particular, writing $x = s \sum_r a_r d_r$ and $y = s \sum_k b_k e_k$ with $d_r, e_k \in \{\pm n_i\}$ and probability weights a_r, b_k, bilinearity gives $\frac{1}{2} x y^T = \rho \sum_{r,k} a_r b_k d_r e_k^T \in \rho \mathcal{D}$. Conversely, each generator $\pm \rho n_i n_j^T$ is a four-group matrix. Thus the convex hull of the four-group matrices is exactly $\rho \mathcal{D}$. Every generator is feasible: choose a triple with the specified singled-out step, and group its other two steps together.

Lemma 10.2. *The two matrices $T_1 = e e^T - a a^T$ and $T_2 = e b^T - a e^T$ belong to $\mathcal{D}$.*

Proof. They have the identities

$$T_1 = \lambda^2 e e^T - \lambda^2 e n_3^T - \lambda^3 n_1 n_2^T,$$
$$T_2 = \lambda^2 n_1 n_3^T - \lambda^2 n_2 n_4^T.$$

To verify them, substitute $n_2 + n_3 = -\lambda^{-1} e$, $\lambda n_1 = e + n_2$, and $\lambda n_4 = e + n_3$. The first right-hand side becomes $(\lambda^2 + \lambda) e e^T - \lambda^2 n_2 n_2^T = T_1$; the second becomes $\lambda(e n_3^T - n_2 e^T) = T_2$. The unsigned coefficients in the first identity sum to $2\lambda^2 + \lambda^3 = 1$. Those in the second sum to $2\lambda^2 < 1$, with the remaining mass assigned to $0 \in \mathcal{D}$. The signs are part of the defining generators, so these are convex combinations, not signed combinations with negative weights. □

Lemma 10.3. *Every six-group matrix belongs to the four-group convex hull $\rho \mathcal{D}$.*

Proof. The set $\mathcal{D}$ is invariant under independent pentagon rotations, so we may use the representative triple above in both factors. A cyclic shift preserves the coefficient matrix as well as Q: moving a q-step x from first to last changes the unscaled mixed matrix by $-2x(\sum_j y_j)^T = 0$, and moving a p-step gives the analogous zero change by q-closure. Cut the word at the q-step se, meaning that this step becomes the first entry in the cyclic order. In the notation of the six-group formula, this means $x_1 = se$. Write $x_3 = sw$, $y_1 = sv$ and $y_2 = sz$; the remaining steps are fixed by closure. After division by ρ, the matrix is therefore $ev^T - wz^T$, where $w \in \{a, b\}$ and (v, z) is an ordered pair of distinct members of $\{e, a, b\}$. These are the two possible orders of the remaining q-steps and the six orders of the p-steps. Reflection $F = \operatorname{diag}(1, -1)$ on the left exchanges a, b and fixes e, so it suffices to take $w = a$. Its six possibilities are

(v, z)	(e, a)	(e, b)	(a, e)	(b, e)	(a, b)	(b, a)
$ev^T - az^T$	T_1	T_1F	T_2F	T_2	$-FT_1F$	$-FT_1$

The last two equalities also follow from $e + a + b = 0$. The set $\mathcal{D}$ is invariant under sign, independent left and right reflections, and independent pentagon rotations. Lemma 10.2 therefore covers all twelve cyclic orders for every pair of closure triples. □

10.3 Capacity and the sharp ratio

Proof of Theorem 10.1. The matrix containment survives the map $T \mapsto G^{-T}TH^{-1}$. Thus the maximum is attained among transformed four-group matrices, giving

$$Q_{\max} = \rho \max_{i,j} |n_i^T G^{-1} H^{-T} n_j|.$$

Taking $1/(2Q_{\max})$ proves the capacity formula. Six-group words may tie this value; the argument only excludes a strictly larger value.

Since areas multiply by absolute determinants, the systolic ratio becomes

$$\frac{(1+h)^4}{2S^2} \frac{|\det M|}{m(M)^2}.$$

Put $L = M/m(M)$. The inequalities defining $m(M)$ say exactly that $LD \subseteq D^\circ$: they test every vertex of LD against every vertex of D, and bilinearity extends them to both convex hulls. The vertices of D have angular spacing $\pi/5 = 2\alpha$. Two adjacent polar boundary lines $d \cdot x = 1$ therefore meet on their angular bisector at distance $\sec\alpha$ from the origin. It follows that

$$D^\circ = \sec\alpha \, R_\alpha D.$$

Comparison of areas now yields

$$|\det L| \le \sec^2\alpha.$$

Now

$$\frac{(1+h)^4}{2S^2} = \frac{5+2\sqrt{5}}{10}, \qquad \frac{5+2\sqrt{5}}{10}\sec^2(\pi/10) = \frac{3+\sqrt{5}}{5},$$

which proves the bound. Equality of areas under convex-body containment holds precisely when $LD = D^\circ$. Then $\cos\alpha R_{-\alpha} L$ is a linear automorphism of D. Such an automorphism permutes vertices and preserves or reverses cyclic adjacency, since it sends edges to edges. There is an orthogonal dihedral symmetry with the same images of any chosen adjacent vertex pair. Those two vertices are linearly independent, so the two linear maps agree everywhere. Consequently $L = \sec\alpha R_\alpha U$, and multiplying by $m(M) > 0$ gives the stated form $M = tR_\alpha U$. Conversely, $UD = D$ and the polar identity imply $m(tR_\alpha U) = t\cos\alpha$, so every listed matrix attains equality. □

Every equality body in this family is a positive dilation of a linear symplectic image of the HKO body. Indeed, when $M = tR_\alpha U$, the symplectic map $\mathrm{diag}(G^{-1}, G^T)$ sends the product to

$$P \times M^{-T}P = P \times t^{-1}R_\alpha UP.$$

The symplectic rescaling $(q,p) \mapsto (t^{-1/2}q, t^{1/2}p)$ then gives $t^{-1/2}(P \times R_\alpha UP)$. The five normals of P are every other vertex of D. A dihedral symmetry preserves or exchanges the two alternating vertex sets, so every such U sends P to P or $-P$. The remaining regular product therefore has relative angle α or $\alpha + \pi$. Modulo the pentagon period $2\pi/5$, both $\alpha + \pi$ and the HKO angle $-\pi/2$ equal $-\alpha$; simultaneous reflection in both planes exchanges α and $-\alpha$ symplectically. Conversely, dilation and linear symplectic maps preserve the systolic ratio, so every such HKO image in this family attains equality.

For $G = I$ and $H = R_\theta$, the formula reduces to the rotation profile: $m(R_\theta) = \cos d_5(\theta)$, where $d_5(\theta) = \min_{k\in\mathbb{Z}} |\theta - k\pi/5|$. The shorter endpoint proof remains useful because it establishes that profile without the matrix classification. The original exhaustive SageMath argument is retained as computational provenance in the repository's `topics/products/regular-products/` directory, under `pentagon-rotation-formula-proof/`; it is not an input to either analytic argument presented here.

The affine theorem concerns all independent linear images of regular pentagons, including arbitrarily large deformations. The separate local maximality theorem concerns nearby ten-facet polytopes and includes changes that destroy the product structure. The two results control different sets of perturbations. Neither is a bound for all ten-facet polytopes or for products of arbitrary pentagons.

11 A regular pentagon with a centrally symmetric partner

The preceding section allowed the two pentagons to deform independently. We now keep one regular pentagon fixed and allow its partner to be any centrally symmetric planar convex body, including a smooth body. Capacity becomes the largest scale of one prescribed regular decagon that fits inside the partner. The same decagon uniquely maximizes the systolic ratio in this class.[3]

Keep the preceding normal convention, and put

$$\theta = \frac{\pi}{5}, \qquad h = \cos\theta, \qquad n_i = (\cos(2\pi i/5), \sin(2\pi i/5)), \qquad Q = \{p : n_i \cdot p \le h,\ 0 \le i < 5\}.$$

Thus Q has circumradius one. Define

$$R = \frac{1}{2(1+h)}, \qquad B = \mathrm{conv}\{\pm Rn_i : 0 \le i < 5\}.$$

The regular decagon B has its orientation fixed relative to Q: its vertices lie on the pentagon's normal lines. For a centrally symmetric convex body K with centre z, write

$$\lambda_B(K) = \max\{r \ge 0 : z + rB \subseteq K\}.$$

This is also the largest scale of any translated copy of B contained in K. Indeed, averaging a fitting copy with its reflection through z centres the copy without changing its scale.

Theorem 11.1. *For every centrally symmetric planar convex body K with nonempty interior,*

$$c_{\mathrm{EHZ}}(K \times_L Q) = \lambda_B(K), \qquad \mathrm{sys}(K \times_L Q) \le \frac{10 + 6\sqrt{5}}{25} < 1.$$

Equality in the ratio holds exactly for translates and positive dilates of B, with its specified orientation.

[3]The reference-cover argument in this section was developed and independently checked by Codex agents. This records agent review, not completed personal verification or publication novelty.

The proof calibrates one reference body, B, and transfers the result to all symmetric partners by containment. Its geometric input is a translation-cover characterization of capacity. For an oriented polygonal curve with edge vectors v_j, its Q-length is $\sum_j s_Q(v_j)$, where $s_Q(v) = \max_{p\in Q} v \cdot p$. This length can depend on the direction of traversal. Balitskiy–Mitrofanov–Polyanskii show that $c(K \times_L Q) \geq 1$ exactly when every closed curve of unit Q-length fits into a translate of K [BMP26, Theorem 2.5]. For polygonal Q, it suffices to check the *normal triangles*: unit-length triangles whose oriented edges follow outward facet normals [BMP26, Definition 3.2 and Theorem 3.3]. Degenerate triangles are included when opposite normals occur.

Our pentagon has no opposite normals. Three normal directions can close with positive edge lengths only when their cyclic gaps are less than π. Those gaps must therefore be $2\theta, 4\theta, 4\theta$, so the triangle's angles are $\theta, \theta, 3\theta$. There are five such direction triples and two cyclic edge orders, giving ten translation classes. Each has its longest side parallel to a pentagon normal. These orientations are fixed: the triangles must fit by translation alone.

Proof. Write c for the Ekeland–Hofer–Zehnder capacity, and translate K so that $K = -K$. We first obtain an upper bound from five segments. Since $s_Q(n_i) = h$ and $s_Q(-n_i) = 1$, the segment joining $-Rn_i$ to Rn_i, traversed out and back, has Q-length $2R(h+1) = 1$. If $c(K \times_L Q) \geq r > 0$, apply the curve-cover characterization to K/r. Each segment $[-rRn_i, rRn_i]$ then translates into K. Symmetry and convexity put its centred copy in K: average a fitting translation with its negative. Their convex hull is rB, so

$$c(K \times_L Q) \leq \lambda_B(K).$$

To prove $c(B \times_L Q) \geq 1$, consider a normal triangle with longest side L. Its other sides both have length $L/(2h)$. Its directed edges follow outward normals, so their Q-lengths are h times their Euclidean lengths. Unit normalization gives

$$1 = h(L + L/h) = L(1 + h), \qquad L = 2R.$$

Translate this side's midpoint to zero. Its endpoints become opposite vertices of B, and its apex has distance $R\tan\theta$ from zero. This lies strictly inside the incircle of B, whose radius is $R\cos(\theta/2)$, because

$$\tan^2(\pi/5) = 5 - 2\sqrt{5} < \frac{5+\sqrt{5}}{8} = \cos^2(\pi/10).$$

Convexity puts the entire triangle in B (Figure 11). The placement works for all ten normal triangles, giving $c(B \times_L Q) \geq 1$. The upper bound already proved gives equality.

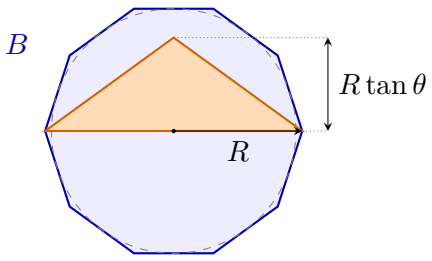


Figure 11: A normal triangle translated into the reference decagon B. Its longest side joins antipodal vertices, and its apex lies inside the dashed incircle, of radius $R\cos(\theta/2)$. The other templates admit the same placement in their prescribed orientations.

Now $\lambda_B(K)B \subseteq K$. Monotonicity and homogeneity in the first factor give

$$c(K \times_L Q) \geq \lambda_B(K)c(B \times_L Q) = \lambda_B(K),$$

proving the capacity formula. The same containment yields $\operatorname{area}(K) \geq \lambda_B(K)^2 \operatorname{area}(B)$. Since

$$\operatorname{area}(B) = 5R^2 \sin\theta, \qquad \operatorname{area}(Q) = \tfrac{5}{2}\sin(2\theta), \qquad \frac{1}{2\operatorname{area}(B)\operatorname{area}(Q)} = \frac{10+6\sqrt{5}}{25},$$

this proves the ratio bound. A proper inclusion between full-dimensional convex bodies strictly increases area, so equality forces $K = \lambda_B(K)B$ after translation. These partners also attain equality. □

The capacity formula can equivalently be expressed using the difference body. Since $B = \frac{1}{2}(Q-Q)^\circ$ and $K - K = 2(K - z)$ for a symmetric partner,

$$c(K \times_L Q) = W_Q(K), \qquad W_Q(K) := \max\{r \geq 0 : r(Q-Q)^\circ \subseteq K - K\}.$$

For an arbitrary partner K, a segment translates into K exactly when its displacement lies in $K - K$. The same curve-cover argument gives $c(K \times_L Q) \leq W_Q(K)$. Its central symmetrization $S = (K-K)/2$ satisfies $S - S = K - K$, so

$$W_Q(K) = c(S \times_L Q).$$

Thus the width ceiling is precisely the capacity of the symmetrized partner. The theorem shows that a partner violating the volume–capacity inequality with this fixed pentagon must be asymmetric; equality with the width ceiling for arbitrary asymmetric partners does not follow.

12 Rotating a product out of symplectic position

The capacity of a product of two planar convex bodies depends on how its factor planes sit in symplectic space. If the planes are symplectically orthogonal, the capacity is the smaller factor area. We show that this arrangement never maximizes capacity among rigid rotations of the product: a small rotation increases capacity while preserving volume.

Let $A, B \subset \mathbb{R}^2$ be convex bodies with nonempty interiors, with areas $a \leq b$. We keep the Euclidean body $A \times B$ fixed and vary the symplectic structure. This is equivalent to rotating the body while keeping the symplectic structure fixed. In coordinates adapted to the two factor planes, define the following family of matrices:

$$J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \mathcal{J}_\theta = \begin{pmatrix} \cos\theta\, J_2 & \sin\theta\, I_2 \\ -\sin\theta\, I_2 & -\cos\theta\, J_2 \end{pmatrix}, \qquad 0 \leq \theta \leq \frac{\pi}{2}.$$

These matrices represent symplectic forms by

$$\Omega_\theta(x,y) = x^{\mathsf{T}} \mathcal{J}_\theta y.$$

Thus $\mathcal{J}_{\pi/2} = -J_0$ and $\Omega_{\pi/2} = \omega_0$ in the earlier convention $\omega_0(x,y) = \langle J_0 x, y\rangle$. Write c_θ for the Ekeland–Hofer–Zehnder capacity of $A \times B$ with respect to Ω_θ. At $\theta = 0$ the factor planes are symplectic, with opposite orientations, and $c_0 = a$. At $\theta = \pi/2$ both factor planes are Lagrangian; write $\ell = c_{\pi/2}$. The relative coordinates of A and B are fixed throughout this path. The parameter θ describes the symplectic form: the corresponding rigid rotation has angle $\theta/2$, as verified below.

Theorem 12.1. *For these factors,*

$$c_\theta = \frac{a}{\cos\theta} \quad \textit{whenever} \quad 0 \leq \tan\theta \leq \frac{\ell}{2a}.$$

Consequently, every sufficiently small positive rotation along this path increases both capacity and systolic ratio.

Proof. First suppose that A and B are polygons containing the origin in their interiors. For each facet write $u_i = n_i/h_i$ for its support-normalized normal. Haim-Kislev's formula [Hai19, Theorem 1.1] expresses reciprocal capacity as the maximum ordered symplectic pairing sum, with factor 2, over weights satisfying

$$\lambda_i \geq 0, \qquad \sum_i \lambda_i = 1, \qquad \sum_i \lambda_i u_i = 0.$$

For a product, the last equation holds separately in each factor. Let t be the total weight on facets from A. Dividing those weights by t gives a feasible planar configuration, so their contribution to reciprocal capacity is at most t^2/a. The corresponding bound for B is $(1-t)^2/b$. More explicitly, if the B-facets occur in the order $j_1, \ldots, j_r$, their pairing sum for $-J_2$ equals the sum for J_2 in the reversed order $j_r, \ldots, j_1$, with each weight attached to its facet. Thus the opposite sign in the second diagonal block does not change the planar maximum $1/b$. At zero mass the corresponding contribution vanishes, so no division is needed.

For the mixed contribution, divide the A weights by $2t$ and the B weights by $2(1-t)$. The resulting weights have total mass one, satisfy closure, and are feasible in the same facet order for the Lagrangian endpoint. Each mixed product of weights has been divided by $(2t)(2(1-t)) = 4t(1-t)$. Since the endpoint has no same-factor terms, its entire pairing sum bounds this rescaled mixed contribution. Its reciprocal capacity is $1/\ell$, so the original mixed contribution is at most $4t(1-t)/\ell$. The cases $t = 0, 1$ have no mixed contribution. The first two bounds correspond to the diagonal blocks $\cos\theta J_2$ and $-\cos\theta J_2$, and the third to the two off-diagonal blocks $\pm\sin\theta I_2$. The coefficients $\cos\theta$ and $\sin\theta$ are nonnegative on the stated interval. Linearity in the symplectic form therefore gives

$$\frac{1}{c_\theta} \leq \max_{0\leq t\leq 1} \left[\cos\theta \left(\frac{t^2}{a} + \frac{(1-t)^2}{b} \right) + \frac{4\sin\theta}{\ell} t(1-t) \right]. \tag{47}$$

Putting all weight in an optimizing planar configuration of A also gives $1/c_\theta \geq \cos\theta/a$.

Set $U = \cos\theta/a$, $V = \cos\theta/b$ and $W = 4\sin\theta/\ell$. Then $V \leq U$, and the expression in brackets minus U is

$$(1-t)\big[V - U + t(W - U - V)\big].$$

The inner bracket is affine in t; its endpoint values are $V - U$ and $W - 2U$. Both are nonpositive when $\tan\theta \leq \ell/(2a)$. Since the bracketed expression equals U at $t = 1$, its maximum is then U. We obtain the explicit squeeze

$$\frac{\cos\theta}{a} \leq \frac{1}{c_\theta} \leq \frac{\cos\theta}{a},$$

and hence the stated capacity formula for polygons.

For general convex bodies, translate each factor so that the origin is in its interior; translations preserve all the capacities under consideration. Choose polygons $A_m \to A$ and $B_m \to B$ in the Hausdorff metric. For a sequence $\varepsilon_m \downarrow 0$, these approximations satisfy

$$(1-\varepsilon_m)A \subseteq A_m \subseteq (1+\varepsilon_m)A,$$
$$(1-\varepsilon_m)B \subseteq B_m \subseteq (1+\varepsilon_m)B$$

for all sufficiently large m. This follows from uniform convergence of support functions and the positive inradii of A and B. In particular, $a_m = \mathrm{area}(A_m) \to a$ and $b_m = \mathrm{area}(B_m) \to b$. The same inclusions hold for the products. Writing $c_{\theta,m}$ for the capacity of $A_m \times B_m$ with respect to Ω_θ, monotonicity and quadratic homogeneity give

$$(1-\varepsilon_m)^2 c_\theta \leq c_{\theta,m} \leq (1+\varepsilon_m)^2 c_\theta.$$

This proves the required capacity continuity directly, as in Proposition 2.4; at $\theta = \pi/2$ it also gives $\ell_m := c_{\pi/2,m} \to \ell > 0$.

Pass to the limit in the two reciprocal bounds before imposing the condition on θ. The polynomial in t on the right of (47), with a_m, b_m, ℓ_m in place of a, b, ℓ, converges uniformly on $[0,1]$, so its maximum converges as well. Thus both bounds remain valid for A, B. The same algebra then gives the squeeze whenever $\tan\theta \le \ell/(2a)$, including equality at this threshold. This passage does not require the approximating polygons to satisfy that threshold themselves.

Finally, $\ell > 0$, so the interval contains positive angles. For each such angle, $a/\cos\theta > a$. The underlying rigid rotation preserves the four-dimensional volume ab, hence also increases $c_\theta^2/(2ab)$. □

To see the rotation explicitly, put

$$K = \begin{pmatrix} 0 & J_2 \\ J_2 & 0 \end{pmatrix}, \qquad R_\theta = \cos(\theta/2) I_4 + \sin(\theta/2) K.$$

Since $K^T = -K$ and $K^2 = -I_4$, the matrix R_θ is orthogonal with determinant one. Direct multiplication gives $\mathcal{J}_\theta = R_\theta \mathcal{J}_0 R_\theta^T$. Thus c_θ is the capacity of $R_\theta^T(A \times B)$ with respect to Ω_0. This rotation is not symplectic with respect to Ω_0 for positive θ in the displayed interval; symplectic rotations would preserve capacity.

Example 12.2. *The Lagrangian arrangement can have smaller capacity than the symplectic arrangement. Take*

$$A = \operatorname{conv}\{(-1,-1),(2,-1),(-1,2)\}, \qquad B = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} A.$$

Both areas are $9/2$*. To show that the Lagrangian capacity is smaller, it suffices to exhibit one feasible configuration in Haim-Kislev's formula. The support-normalized normals are*

$$u_0 = (0,-1), \quad u_1 = (1,1), \quad u_2 = (-1,0), \qquad v_0 = (0,-2), \quad v_1 = (1/2,2), \quad v_2 = (-1/2,0).$$

Give all six normals weight $1/6$ *and order them as* $(u_0, v_0, u_1, v_1, u_2, v_2)$*. Both triples sum to zero, so the configuration satisfies closure. Its signed cross-pairing sum is* 5*, giving*

$$\frac{1}{\ell} \ge 2\left(\frac{1}{6}\right)^2 5 = \frac{5}{18}, \qquad \textit{and hence} \qquad \ell \le \frac{18}{5} < \frac{9}{2} = c_0.$$

Remark 12.3. *If* $\ell \le a$*, Theorem 12.1 gives capacities larger than both endpoint values at small positive angles. By continuity, the maximum along this path is attained in its interior. The triangle example satisfies this condition.*

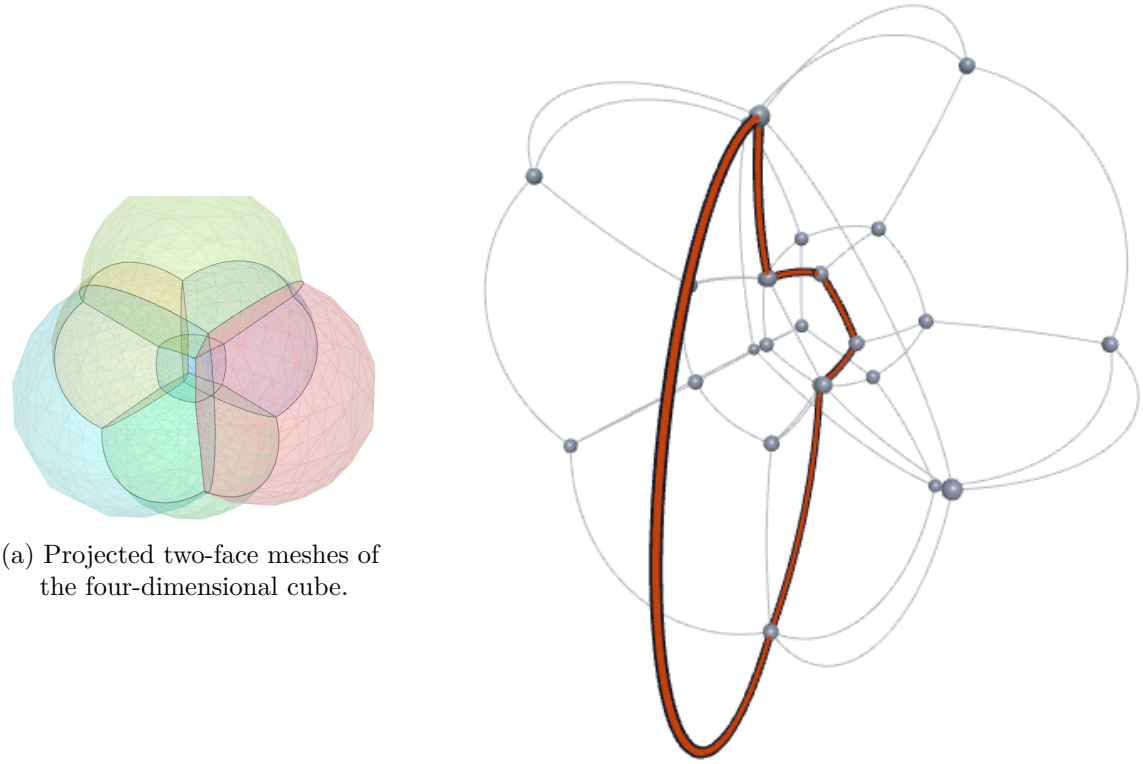
13 Viewing four-dimensional boundaries in three dimensions

A four-dimensional polytope cannot be drawn directly, but its boundary can be viewed in three dimensions after choosing a projection. For a polytope $K \subset \mathbb{R}^4$ with the origin in its interior, we first apply the radial map

$$\partial K \longrightarrow S^3, \qquad x \longmapsto \frac{x}{\|x\|},$$

and then stereographically project S^3 from the north pole $e_4 = (0,0,0,1)$ to $\mathbb{R}^3$. The viewer samples projected edges and Reeb segments along great-circle arcs. Its coloured two-face interiors are interpolated spherical meshes, included to make the boundary structure easier to see rather than to preserve metric geometry. The stereographic projection also distorts scale and clips arcs close to its pole.

Figure 12 shows projected two-faces of the four-dimensional cube and the projected one-skeleton of the HKO polytope K_{HKO} from Section 7 with one numerically recovered minimum-action generalized Reeb orbit. In panel (a), the coloured meshes help distinguish ridges and follow their common edges; their apparent areas cannot be used to compare the symplectic ridge areas studied in Section 8. In panel (b), the orange curve places a piecewise-linear orbit in relation to the polytope's boundary structure. Its six straight segments in four dimensions become curved arcs in the view. Their displayed lengths do not measure dwell times or action. The producer exports only recovered orbits passing numerical closure and containment checks within fixed tolerances. The orbit shown is therefore a computed example, not a proof of a capacity statement. The exporter, projection settings and figure-generation instructions are recorded in `experiments/visualization/README.md`.



(a) Projected two-face meshes of the four-dimensional cube.

(b) The HKO one-skeleton and a recovered minimum-action orbit.

Figure 12: Radial and stereographic views of four-dimensional polytopes. In (a), colour separates projected two-face meshes of the cube. In (b), the grey one-skeleton belongs to the Lagrangian product of two regular pentagons and the dark-outlined orange curve is a recovered six-segment closed Reeb orbit. Both panels use the pole e_4 and omit parts mapped beyond the viewer's clipping radius. They are qualitative explanatory views, not symplectic invariants or proof evidence.

The interactive views were also explored as a possible source of geometric patterns for our exploratory search. In practice, passing from four dimensions to a three-dimensional projection discarded too much visible Reeb-geometric information, and the impressions changed with the projection pole. We found no visual pattern that gave a reliable hypothesis or candidate-selection rule. For this exploration, the useful outcome is therefore limited to the explanatory role illustrated above.

14 Numerics

The computations in this thesis support two different kinds of claims. The HKO proof uses a separate SageMath packet over rational or ordered algebraic fields; the rotated-pentagon chapter uses an analytic proof. The Rust `capacity_4d` implementation computes the EHZ capacity of a four-dimensional polytope supplied in binary64 coordinates. It uses floating-point arithmetic for speed, but it does not accept an unverified floating-point sign or ordering decision: every such decision is enclosed by an outward error bound or resolved in exact rational arithmetic.

14.1 Input and output contracts

A finite binary64 number is a dyadic rational. The numerical capacity contract regards the supplied numbers as these exact rationals; it does not claim to recover unavailable source coordinates from which they may have been rounded. Before the QP search, exact geometry checks establish that

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x\rangle \le 1\}$$

is bounded, nonempty, full-dimensional, and described without redundant facets. The exact primal vertices and vertex–facet incidence are then constructed from the dyadic input. Floating-point filters may discard a subset only when a one-sided bound proves the exact conclusion; an indeterminate filter result is resolved exactly.

The production route accepts at most sixteen facets and requires every primal and dual vertex to have infinity norm in

$$[10^{-3}, 10^{3}].$$

These bounds are a deliberately narrow applicability policy, not a theorem about all polytopes. They exclude scales for which the current binary64 enclosures have not been justified and tested. The general candidate stream is additionally limited to 100 000 cyclic words. Violating any of these conditions returns an error rather than an uncertified capacity.

For a general polytope the result is an outward interval $[c_{\mathrm{L}}, c_{\mathrm{U}}]$ containing the exact capacity of the dyadic input. A scalar wrapper returns the midpoint only if this interval certifies the relative error requested by the caller. For a recognized Lagrangian product the specialized route returns the exact rational capacity of the dyadic input, together with outward binary64 endpoints for convenient downstream use.

14.2 Certified fixed-word decisions

For a word σ, write the equality-constrained stationary system in the symmetric form

$$\mathcal{K}\begin{pmatrix}\beta\\ \lambda\end{pmatrix} = \begin{pmatrix}H & C^{\mathsf{T}}\\ C & 0\end{pmatrix}\begin{pmatrix}\beta\\ \lambda\end{pmatrix} = \begin{pmatrix}0\\ d\end{pmatrix} =: b, \qquad d = (0,0,0,0,1)^{\mathsf{T}}.$$

Here $C\beta = d$ is closure plus normalization and H is twice the quadratic form in (19). A numerical factorization supplies an approximate solution $\widehat{x}$ and approximate inverse R. The following standard inverse-defect estimate is the central certificate.

Lemma 14.1 (Verified inverse defect)**.** *Suppose outward arithmetic establishes*

$$\overline{r} \ge \|\mathcal{K}\widehat{x} - b\|_\infty, \qquad \overline{e} \ge \|I - \mathcal{K}R\|_\infty, \qquad \overline{e} < 1.$$

Then $\mathcal{K}$ is invertible, and its exact solution x^* satisfies

$$\|\widehat{x} - x^*\|_\infty \le \rho := \frac{\|R\|_\infty\, \overline{r}}{1 - \overline{e}}.$$

Proof. The Banach lemma applied to $\mathcal{K}R = I - (I - \mathcal{K}R)$ gives $\|\mathcal{K}^{-1}\|_\infty \le \|R\|_\infty/(1-\overline{e})$. Apply this bound to the residual $\mathcal{K}\widehat{x} - b$. □

Thus $\widehat{\beta}_i - \rho > 0$ for every i certifies positive weights, while $\widehat{\beta}_i + \rho \le 0$ for some i certifies failure of strict positivity on this support. If ξ is the multiplier for normalization, stationarity gives

$$2Q(\beta^*) + \xi^* = 0.$$

Consequently the same radius yields

$$Q(\beta^*) \in \left[-\frac{\widehat{\xi} + \rho}{2}, -\frac{\widehat{\xi} - \rho}{2} \right].$$

For $0 < q_{\mathrm{L}} \le Q \le q_{\mathrm{U}}$, monotonicity then gives the action enclosure

$$\frac{1}{2q_{\mathrm{U}}} \le \mathcal{A} \le \frac{1}{2q_{\mathrm{L}}}.$$

The implementation computes the residual and inverse-defect bounds with explicit roundoff and subnormal-error terms. It first uses a cheap induced norm bound and retries an indeterminate system with a tighter entrywise bound. The proof assumes round-to-nearest binary64 arithmetic with gradual underflow and the inspected matrix-multiplication evaluation order. The runtime checks gradual underflow; if the contract is unavailable, or if either bound remains indeterminate, the decision is made by exact rational arithmetic. A compiler, dependency, target, or arithmetic-contract change therefore requires the floating-point certificate tests to be repeated.

14.3 Complete general and product routes

The general search constructs the exact transition graph, pruned by facet incidence and symplectic signs, then enumerates its simple cyclic words. Candidate correctness is not obtained merely by solving every displayed KKT system. At a fixed-word maximum with strictly positive weights, both signs of every sufficiently small tangent perturbation remain feasible. The quadratic form must therefore have nonpositive curvature in every tangent direction. Conversely, a certified vector $z \in \ker C_\tau$ with $z^{\mathsf{T}} H_\tau z > 0$ rules out τ as the positive support of a maximizer. If τ injects into a longer word σ while preserving cyclic order, zero-extending z preserves both its tangent constraints and positive curvature. Hence σ is ruled out as a positive support as well. Boundary maxima are represented on their smaller positive supports. Processing words by length allows these proved obstructions to skip many later factorizations.

For every word not excluded in this way, the route uses the verified fixed-word decision above, with exact fallback for every indeterminate case. The resulting objective intervals are aggregated outwards over the complete candidate family, and the reciprocal reverses the endpoints to enclose the capacity. Exact arithmetic is therefore a selective correctness fallback, not the default arithmetic for every word.

For a Lagrangian product, the inverse-defect route is correct but frequently indeterminate and unnecessarily expensive. The specialized implementation uses Algorithm 4.7. It encloses factor-closure weights and bilinear objectives in floating-point arithmetic, rejects a support or order only when the intervals prove it cannot win, and resolves ambiguous supports and every possible winner exactly. Because the final comparison is rational, this route returns the exact capacity and every tied sparse winner in the closure-vertex candidate family. It intentionally does not enumerate all minimizing geometric branches.

14.4 A certified calculation for the cube

Consider the exactly representable input

$$K = [-1, 1]^4, \qquad (a_0, \dots, a_7) = (e_1, e_2, -e_1, -e_2, e_3, e_4, -e_3, -e_4).$$

Each planar closure polytope has two vertices: the opposite pairs along the two coordinate axes, with normalized weights $(1/2, 1/2)$. Pairing the factor supports gives four choices, each with six cyclic orders. The product route therefore compares only 24 objective values.

For the word $(0, 4, 2, 6)$, the four QP weights are all $1/4$, so closure and normalization hold exactly. The nonzero ordered symplectic pairings are $1, -1, 1, 1$, giving

$$Q_{(0,4,2,6)} = \frac{1 - 1 + 1 + 1}{16} = \frac{1}{8}, \qquad \mathcal{A} = \frac{1}{2Q} = 4.$$

The complete closure-vertex comparison has two orders with $Q = 1/8$, two with $Q = -1/8$, and twenty with $Q = 0$. Thus the winning action is exactly four, with two tied sparse words. The retained numerical audit reports $Q_{\max} = 1/8$ and the exact rational capacity 4 for this input. The product API's outward conversion consequently returns

$$[c_{\mathrm{L}}, c_{\mathrm{U}}] = [4, 4], \qquad c_{\mathrm{EHZ}}([-1, 1]^4) = 4.$$

An action cutoff below four therefore excludes these winners, whereas a closed cutoff at four includes both. In this case input, optimizer weights and capacity are all exactly representable; the zero-width output does not describe the usual general-polytope interval.[4]

14.5 Empirical verification and limitations

The audits below concern separate development implementations of the general and product algorithms. Their reports and producers are under `topics/capacity-and-qp/development/tools/`. The two packets are `general_algorithm_ablation/` and `product_closure_route/`. A four-test comparison, described below, checks their outputs against the production implementation on fixed fixtures.

An audit of the general development implementation compared the floating-point predicates with exact rational arithmetic on dyadic inputs for 249 KKT systems: ordinary generated systems, scales from 10^{-2} through 10^{3}, and systems approaching singularity down to binary64 coincidence. It found no incorrect determinate decision and no violation of a β- or Q-radius. Fifteen cases were indeterminate and were resolved exactly. The largest observed ratios of actual error to certified radius were 0.00149 for β and 0.00081 for Q.

Complete comparisons for that implementation used 88 words from three generated general polytopes and 1271 words from three product controls. Exact aggregation found no missed capacity and no missed minimizer class; every exact capacity lay in the returned interval. On the general cases, the largest difference between the printed central value and exact capacity was $2.31 \cdot 10^{-14}$. These finite comparisons test the implementation and exercise fallback, but the soundness claim rests on the preceding error bounds and complete-candidate arguments. The 249-system and 88/1271-word counts are recorded in the general packet's `RESULTS.md`. That retained summary and its producer are available; the original per-system audit output is not included in the repository.

An audit of the product development implementation used 88 ordinary and adversarial products. Across 1835 closure weights and 173 496 objective intervals it found no interval violation,

[4] The input is `square_product_case` in `topics/capacity-and-qp/development/tools/product_closure_route/main.rs`; the tracked `sample5.jsonl` in that directory contains the row `adversarial:square_product_exact_zeros`. The outward rational conversion is implemented in `crates/symplectic/src/algorithms/capacity_4d/mod.rs`.

no incorrect determinate sign, and no disagreement between the hybrid and complete exact closure-vertex computations. Raw comparison with zero disagreed with the exact objective sign 23 559 times, mostly at exact zero or tiny values; the certified predicate returned indeterminate instead. These 88 per-body records are tracked in the product packet's `sample5.jsonl`. A separate retained population of 10 240 random products had a literal capacity winner of length six in every case. This supports the implementation and stress-tests Theorem 4.6; it is not its proof and does not exclude longer tied minimizers. The product packet's `RESULTS.md` records the join between source geometries and cached winning words. Its two source datasets belong to the registered external `polytope-datasets` artifact, rather than to the ordinary tracked files; Section 15 describes their availability.

The four-test correspondence check compares the development implementations with the production implementation on fixed fixtures: general capacity bounds, an exact action window, derivatives of the retained words, and product capacities with their sparse winning words. All four tests pass; this is finite regression evidence, not all-input equivalence.[5]

The guarantees above concern the polytope defined by the exact dyadic input rows. They do not propagate measurement error or prior rounding uncertainty in the coordinates, and they apply only within the stated facet, norm, and candidate-count limits. The HKO certificate instead represents its algebraic input in an exact ordered field. The analytic rotated-pentagon proof does not depend on these binary64 computations.

15 Availability of Code and Data

At the time of writing, the public development repository for this thesis is

`https://github.com/JoernStoehler/msc-math`.

The public repository was checked on 18 September 2026 at commit `1bec30637070`. The HKO certificate and original pentagon-rotation packet described below are present there. This manuscript and subsequent research additions are held in the working repository; they are not all included in that public commit. The repository URL follows ongoing development.

The HKO local-maximality proof and the original rotation computation have proof-facing packets in the public tree. The HKO packet, at `experiments/hko-local-maximum/theorem/` in that public commit and now maintained at `topics/variation-and-local-maxima/hko-local-maximum/theorem/`, contains the finite witness, the exact SageMath verifier, its recorded verification summary, and the Rust programs used to select the witness. Rust supplies finite choices; it is not trusted for theorem correctness. SageMath reconstructs the algebraic objects and checks the exact predicates stated in Section 7.4. Those predicates imply the theorem only together with the mathematical upper-function and slice argument given there. The annotated program printed in the appendix is maintained separately as `thesis/appendices/hko_core.py`. It reconstructs the same selected words, feasible sections, and exact derivative covectors; the appendix maps its shorter presentation to the public verifier `experiments/hko-local-maximum/theorem/verify.sage.py` (now at `topics/variation-and-local-maxima/hko-local-maximum/theorem/verify.sage.py`). The appendix file was not part of the public commit cited above.

The second ten-facet local maximum is maintained in `topics/variation-and-local-maxima/hko-local-maximum/non-hko/`. Its rational witness and Python/SymPy verifier check the complete product calculation, the 24 feasible upper sections, the rank and positivity predicates, and the exceptional quadratic direction. An independent

[5] The test source is `topics/capacity-and-qp/development/tests/selected_route_correspondence.rs`; its execution and fixture scope are recorded in `topics/capacity-and-qp/evidence/numerics-correspondence-check/README.md`.

checker in the same packet recomputes the geometry and derivatives by a different triangulation and direct section solve. These exact checks do not replace the analytic neighborhood argument or constitute a proof-assistant formalization.

The rotated-pentagon packet is stored under

`topics/products/regular-products/pentagon-rotation-formula-proof/`.

It contains the exact SageMath classifier and the recorded full-run transcript for all 3340 finite words left after the mathematical reduction. This is the original computational proof route; the analytic rotation and independent-linear-deformation proofs presented here do not use it. In the computational route, the enumeration theorem, the translation from checked signs to action comparisons, and the continuity argument remain mathematical inputs. The repository exposes the finite computation and its output for inspection, with SageMath's exact arithmetic and root isolation inside the software trust boundary. The maintained affine-pentagon argument and its source-convention check are in `topics/products/pentagon-affine-products/`.

The numerical audits in Section 14 have distinct evidence owners. The tracked report and producers for the 249 KKT systems and the complete 88- and 1271-word comparisons are in `topics/capacity-and-qp/development/tools/general_algorithm_ablation/`. Its `RESULTS.md` records the counts and outcomes; its `run.sh` produces new verification and numerical-audit transcripts. The original per-system output is not included as a tracked or registered artifact, so the retained report and a new execution are different evidence objects.

The 88-product audit is retained row by row in `topics/capacity-and-qp/development/tools/product_closure_route/sample5.jsonl`, with its interpretation in that directory's `RESULTS.md`. The external `polytope-datasets` artifact supplies `random-product.jsonl` and `shared-cache.jsonl` for the separate 10 240-product observation. These are registered files, materialized under `topics/empirical-studies/polytope-datasets/`; they are not part of a plain checkout. The product report identifies the historical geometry-to-cache join and its scope. These numerical packets concern finite tests, with the production-to-development comparison recorded separately in `topics/capacity-and-qp/evidence/numerics-correspondence-check/`.

Other non-proof evidence is tracked as ordinary repository files. The historical first-order experiment retains its 12-run finite-step panel. The bounded generated-candidate experiment retains 1675 rows selected before the systolic ratio was computed, the corresponding 1675 evaluated rows, its frozen configuration, and compact reports. These records support only the finite experiments described in the corresponding chapters. They neither prove a population statement nor supply evidence for local maximality.

The repository is not a self-contained copy of every dataset used during the project. Its artifact registry identifies external snapshots and their intended local paths; a plain checkout does not materialize those files. For example, the 14336-row random/product provenance table is tracked, while the source random/product datasets and the method-facing invariant table are not tracked at those paths. The original source geometries have been recovered locally from the registered snapshot and matched against all 14336 table rows. They can be materialized through the `polytope-datasets` artifact; this recovery does not supply missing historical execution receipts. Separate current computations restored the feature schema for all bodies and returned capacity intervals for 14,335 of the 14,336 original geometries; one body remains outside the input-size policy. These new intervals do not certify the historical numerical ratios, whose volumes are floating-point values. Recovery and reevaluation receipts are in `topics/empirical-studies/retrospective-revalidation/`. Consequently, cloning the repository alone does not provide every input needed to repeat all empirical analyses, regenerate every figure, or reproduce every reported run. Packet-local instructions describe available producers and checks, but compatible software, dependencies, compute resources, and untracked inputs may still be required; timing and other environment-dependent fields need not reproduce byte for byte.

No frozen release, public Zenodo record, or DOI for this thesis is claimed here. Any archival deposit created later would be a separate, immutable object whose own public record, inventory, and hashes would have to establish its contents. The public commit identified above fixes which material was accessible at the time of this check; later local additions require a separate handoff.

16 AI in the research process

As the agents became more capable, my work shifted from directing individual steps toward specifying what the finished work had to accomplish. They could develop a substantial mathematical argument from an incomplete suggestion, yet sometimes needed repeated intervention to pursue a much more ordinary objective. My main difficulty increasingly concerned what they treated as success. This section gives my interpretation of that experience; the separate disclosure records the extent and provenance of AI use in the thesis.

16.1 A mathematical objective with room for discovery

The local maximality proof illustrates how much could be delegated without specifying a procedure. I asked a GPT-5-series model to develop a local maximality argument for HKO in facet coordinates. I suggested looking for several upper bounds and somehow showing that together they control a neighborhood after quotienting out the symmetries. I had neither specified a finite certificate nor worked out how to construct the upper bounds.

The model developed essentially the entire argument from that starting point. It identified a sufficient configuration of twenty-six upper bounds in the twenty-five transverse dimensions. An initial derivative-based approach had provided only twenty-five nonsingular branches, too few for the certificate. The model then used the weights β in the quadratic objective to construct feasible sections: an invertible 5×5 closure minor lets five weights depend smoothly on the facet parameters while the remaining weights stay fixed. This also supplied upper bounds from singular optimizer sets that the initial approach could not treat. The model completed proof steps involving the volume and wrote the SageMath code for the exact computations.

16.2 A pursued criterion of success

Other tasks produced a more puzzling contrast. An agent could reason competently about a problem and still stop after a first step, answer a related question, or keep preparing work that it had been asked to finish. These failures sometimes looked oddly motivational: the next useful action seemed to be within the agent's capabilities, but its behavior was directed toward a different stopping point. An agent could correctly describe my goal and still pursue something else. I found it useful to distinguish understanding a goal from choosing the criterion of success that directs the work.

My hypothesis is that training with verifiable rewards encourages models to learn a concept of a *Grader*: a representation of what will judge the task completed successfully. Reinforcement learning with verifiable rewards uses outcomes that can be checked, such as a mathematical answer or a program's performance on test cases. For example, DeepSeek-R1-Zero was trained with rule-based accuracy and format rewards; its reasoning strategies were not supplied as complete human demonstrations [GYZ+25]. I suspect that such training can teach a model to adopt a criterion and organize its work around satisfying it. The training result motivates this interpretation, but does not establish that it explains the behavior of the agents I used.

This interpretation concerns more than reward hacking or literal obedience to an instruction. An agent need not exploit a loophole in a stated test, or misread a particular word, to pursue the wrong objective. A plan for improving a chapter can be a good answer to a conversational turn while leaving the requested chapter unwritten. Likewise, a useful preliminary calculation can become the apparent endpoint of a task that required a theorem. Each answer may be sensible on its own. The failure is that the criterion organizing the work covers only part of the intended result.

The practical requirement is sufficient overlap: satisfying the criterion an agent pursues must also produce a useful end result for me. The goals need not coincide. A clear objective makes completion evaluable: what result is needed, why it is needed, which constraints matter,

and what would leave the task unfinished. It does not require prescribing every intermediate action. This way of specifying the HKO task was effective because the mathematical objective constrained the result while allowing the agent to invent the method. I found it useful to think of prompts as both cueing learned behavior through their wording and supplying context for deliberate reasoning.

16.3 Keeping the whole task in view

Long conversations made the distinction between a useful answer and a completed task increasingly important. At checkpoints during such a task, I want to ask whether the current reasoning, evidence, and actions are advancing the intended goal as a whole within the available budget. This question should expose a change of destination: delivering only a first step, solving an easier related problem, or replacing completion by repeated planning. The same question belongs at the end. A successful evaluation of several components does not answer whether the requested result has been delivered.

The budget is part of that judgment. An investigation can produce useful information and still be a poor choice if it consumes the time needed to finish the work. When the task is to deliver a readable chapter, identifying another possible improvement has to be weighed against writing and revising the actual chapter. When the task is a mathematical search, the value of another computation depends on the question it can resolve and the resources left for following up. Progress has to be assessed against the intended outcome and its constraints, rather than the amount of activity generated.

This does not call for inspecting every step. An autonomous agent should choose and revise many of its own intermediate steps. The important human intervention is often to restore the purpose that those steps serve. More detailed instructions can help when a constraint was missing, but can also obscure the goal beneath local requirements. I increasingly needed to explain the result I wanted and the context that made it useful, then judge whether the work still addressed that result.

16.4 Why exposition remained difficult

For the mathematics required by this MSc project, I often found the agents' capabilities sufficient to make substantial progress with little detailed direction. Producing passable mathematical exposition was much harder to delegate. This was surprising to me given the amount of written material used in language-model training. The later work on the thesis became a prolonged struggle with writing and task-management quirks, even when the models could partly recognize those quirks in discussion.

One recurring problem was prose that presented itself as polished before it communicated its content clearly. Inflated claims of significance, colorful phrasing, and abstract summaries could make a passage sound complete while leaving its mathematical point difficult to recover. This is what I mean by *AI slop* in the present context. An agent's ability to decipher a dense paragraph did not show that a reader could comfortably follow it. The same problem arose with forward and backward references across several pages. An agent with those pages in context could follow the references, while a human reader had to remember deferred points or interrupt the argument to recover an earlier definition.

I suspect that preference training contributes to this tendency. In reinforcement learning from human feedback, rankings of model outputs can be used to train the model toward preferred responses, as in InstructGPT [OWJ+22]. If an impressive presentation is easier to recognize in a brief comparison than a clear explanation is to assess through sustained reading, that preference signal may favor the appearance of quality. This is my hypothesis about a possible source of the prose I encountered, not a demonstrated causal account of those models' training.

The practical difficulty persisted even after it had been named. A model could acknowledge that a passage needed a concrete argument and then return another description of the argument's intended role. It could recognize that the goal was a readable chapter while spending its effort on further diagnosis. I had to devise corrections tailored to recurring behaviors; there was no single instruction whose recognition reliably produced the desired writing. Understanding a criticism in one exchange and carrying it through a sustained piece of work proved to be different capabilities.

This experience also limits the lesson I draw from the mathematical success. The proof work in this project was tractable enough for the available models that their quirks often mattered less there. The difficulty of exposition was not captured by the formal sophistication of its subject. Choosing the right explanation, maintaining its purpose across a chapter, and removing language that obscured the mathematics remained demanding tasks.

17 Conclusion

The HKO body is a local maximum of the systolic ratio among ten-facet polytopes. Every sufficiently nearby body in this space has ratio at most $(3+\sqrt{5})/5$, and equality occurs precisely under translations, positive dilations and linear symplectic maps. This answers the local-improvement question within the fixed-facet space. Deformations that introduce new facets require a different local model.

The fixed-facet landscape is not unique. A rational hexagon–quadrilateral product with ten genuine facets has ratio one and is also a non-strict local maximum, with a two-parameter equality family modulo the usual symmetries. Its proof controls nonproduct directions in the full ten-facet chart. This example is below HKO's ratio and is therefore not a new Viterbo counterexample; it shows instead that the ten-facet local-maximality phenomenon is not confined to the HKO orbit.

A global result gives the same sharp bound for products of independently linearly deformed regular pentagons. Its equality cases are positive dilates of linear symplectic images of HKO. Together, the HKO local theorem and this affine-pentagon theorem rule out improvement by sufficiently small ten-facet perturbations and by arbitrary deformations within this family of affinely regular products. The latter can be large, while the former can destroy the product structure.

The symmetric-partner theorem removes a different restriction: with one regular pentagon fixed, its partner may be any centrally symmetric planar convex body. Capacity is the largest scale of a prescribed decagon contained in the partner, and the decagon, with its specified orientation, uniquely maximizes the ratio up to translation and positive dilation. The sharp value $(10+6\sqrt{5})/25$ is below one. The proof calibrates one reference body and then uses containment and area to control the whole class. Symmetry of the partner therefore rules out an HKO-sized ratio here; the theorem does not bound arbitrary asymmetric partners.

The placement of the factor planes matters even when the Euclidean product shape is fixed. Every symplectic product of full-dimensional planar convex bodies admits a capacity-increasing rigid rotation. The Lagrangian endpoint can nevertheless have smaller capacity; in the triangle example the maximum along the path lies strictly between the endpoints. Thus the familiar symplectic and Lagrangian arrangements need not contain the best position of a given product.

At HKO, the local proof continues feasible choices through a singular optimization problem and uses their values as upper bounds, without continuing stationary optimizers. Finitely many such bounds suffice: their derivatives force a strict decrease in every nonzero direction transverse to the symmetries. The algebraic calculation establishes the finite conditions, while the geometric proof identifies the upper bounds and relates their coordinates to nearby bodies.

The finite methods also support computation beyond these proofs. The Haim–Kislev formula gives a finite capacity problem for convex polytopes, and the product reduction shows that some maximizing word uses at most three facet directions from each planar factor. Its implementation supplies the capacity evaluations used to investigate broader populations. Our flow-graph construction uses the local passage geometry of Chaidez and Hutchings and provides a second capacity algorithm under explicit regularity assumptions.

In those populations, symplectic ridge area gives a strong coarse predictive signal for computed systolic ratios. Geometric selection raised mean computed ratios in fresh samples. The pooled inverse rank association weakens substantially when the original data are restricted within each sampling group to low ridge scores or high computed ratios, with ranks recomputed in the retained groups. Thus a strong ordering across the broad population becomes much weaker among selected candidates. Restricting the range can itself attenuate correlation, so this observation alone does not identify a geometric cause.

The balancing experiment goes further by changing the shapes. Applying determinant-one maps that make each factor's vertex covariance isotropic raises the mean numerical ratio from

0.342 to 0.654 in 320 paired products, with 314 increases and six decreases. The pooled within-group rank correlation weakens from -0.929 to -0.249. Affine changes in factor shape therefore contribute materially to the broad association in this sample. The intervention changes several geometric features together, so it does not identify covariance imbalance as the sole cause or give a universal improvement rule.

The residual relationship depends on the family. Relative rotations of regular pentagon factors satisfy the inverse-square law (35), whereas relative rotations of equilateral-triangle factors give the same strict ordering by ridge sum and systolic ratio, as proved in (36). Ideal covariance balancing makes triangles equilateral, which explains the opposite ordering in that balanced family. These exact laws show that even the direction of the relationship depends on geometric restrictions.

The local-search comparison supplies another practical conclusion. On the same 64 starts under common evaluation limits, the four-anchor history implementation attained median best ratio 0.985, compared with 0.881 for literal gradient ascent. Retaining previously active branches was useful in this finite comparison. Endpoint tests nevertheless found ascent directions missed by simpler probes, so high terminal values did not certify local maxima. None of the exploratory searches produced a new numerical ratio above one.

Two open questions follow directly. Does HKO remain locally maximal when the facet count changes? Which geometric features explain the residual ridge association after factor balancing, and how does that relation depend on the sampling distribution? The first asks how far the proved local extremality extends. The second asks which capacity laws extend beyond the solved rotation families.

A Data-science methods and additional results

This appendix records the population, prediction, selection and search methods supporting Section 8, together with the additional numerical diagnostics and their interpretation limits.

A.1 Exact profile of equilateral-triangle products

For the family in (36), the capacity is

$$c_{\mathrm{EHZ}}(T_\theta) = \frac{3\sqrt{3}}{4\cos(\pi/6 - d_3)}.$$

Here is a direct proof in the manuscript's quadratic-program convention. Choose the outward normals $n_i = (\cos(2\pi i/3), \sin(2\pi i/3))$, $i = 0, 1, 2$, so $P_3 = \{q : n_i \cdot q \le 1/2\}$, and write $m_j = \mathrm{Rot}(\theta) n_j$. The six dual rows are $a_{q_i} = (2n_i, 0)$ and $a_{p_j} = (0, 2m_j)$.

Closure forces a common weight x on the three q rows and a common weight y on the three p rows, with $x + y = 1/3$. Indeed, the kernel of the three-normal matrix is spanned by $(1, 1, 1)$. A positive quadratic objective requires $x, y > 0$, so all six rows occur. For their order,

$$Q = 4xyS, \qquad S = \sum_{i,j} \epsilon_{ij} n_i \cdot m_j,$$

where $\epsilon_{ij} = 1$ if q_i precedes p_j and -1 otherwise. Cyclically moving the first row to the end preserves S, since both normal sums vanish. Classify the cyclic words by their number of q runs, equal to their number of p runs.

With one run, $S = 0$. With two runs, the two q block sums are $u, -u$ with $u = \pm n_i$, and the two p sums are $v, -v$ with $v = \pm m_j$. The reduced word $u, v, -u, -v$ gives $S = 2u \cdot v \le 2$. With three runs the word alternates, say $q_a, p_b, q_c, p_d, q_e, p_f$, and

$$S = 2\big(n_a \cdot m_b + (n_a + n_c) \cdot m_d\big) = 2(n_a \cdot m_b - n_e \cdot m_d).$$

Set $c_k = \cos(\theta + 2\pi k/3)$. On $0 \le \theta \le \pi/6$ we have $c_0 \ge c_2 \ge c_1$, so all words satisfy

$$S \le 2(c_0 - c_1) = 3\cos\theta + \sqrt{3}\sin\theta.$$

The bound is at least three, so it covers the two-run case too. Equality is attained by $q_0, p_0, q_2, p_2, q_1, p_1$. Since $xy \le 1/36$, with equality at $x = y = 1/6$, we obtain

$$Q_{\max} = \frac{3\cos\theta + \sqrt{3}\sin\theta}{9} = \frac{2\sqrt{3}}{9}\cos(\pi/6 - \theta).$$

The capacity formula follows from $c = 1/(2Q_{\max})$. As $\mathrm{area}(P_3) = 3\sqrt{3}/4$ and $\mathrm{vol}_4(T_\theta) = 27/16$, the ratio is $\frac{1}{2}\sec^2(\pi/6 - \theta)$.

All edges have length $\sqrt{3}$, and their directions differ from the corresponding normals by a common quarter-turn. The mixed-edge sum is therefore $9\sum_{k=0}^{2}|c_k|$. On this sector $c_0 \ge 0$, $c_1, c_2 \le 0$ and $\sum_k c_k = 0$, giving $18\cos\theta$. Dividing by $3\sqrt{3}/4$ yields $R = 8\sqrt{3}\cos\theta$.

To reduce other angles, both quantities are even under simultaneous reflection in the factor planes and have period $2\pi/3$ by triangle symmetry. The symplectic factor exchange $(q, p) \mapsto (p, -q)$, followed by rotation through $-\theta$ in both planes, sends T_θ to $T_{\pi-\theta} = T_{\pi/3-\theta}$. Together with evenness this gives period $\pi/3$, proving the formulas with d_3. Both expressions strictly decrease as d_3 runs from zero to $\pi/6$.

The connection to covariance balancing is also exact. For a centered triangle with vertex matrix $X \in \mathbb{R}^{2\times 3}$, isotropic vertex covariance means $XX^T = 3\alpha I_2$, and $X\mathbf{1} = 0$. Hence

$$X^T X = 3\alpha(I_3 - \mathbf{1}\mathbf{1}^T/3).$$

All squared distances between different vertices equal 6α, so the triangle is equilateral. Independent factor dilations preserve both R and sys; thus ideal balancing of two triangles places their product in the family above. The rounded numerical outputs are close to this family, not asserted to have exactly equilateral binary64 coordinates.

A.2 Sampling and retained data

A polytope was specified by inequalities

$$K = \{x : \langle n_i, x\rangle \le h_i, \quad i = 1, \dots, F\}.$$

For the generic family, each n_i was independent and uniform on the unit sphere in $\mathbb{R}^4$: we normalized a vector of four independent standard Gaussian coordinates. The heights were independent and uniform on $[0.8, 1.2)$. A proposed body was retained only if it was bounded and every proposed inequality defined a facet. The construction also rejected zero or near-duplicate normals. These conditions are part of the sampling law; "random polytope" here does not mean a uniform choice from all convex bodies.

The original population contains 512 bodies for each generic facet count and 1,024 for each product side-count pair. The retained generator records identify seed 42; product seed/attempt metadata cannot be independently reconstructed from the recovered cache. The source geometries, row identifiers and stored numerical values are nevertheless matched for every body. The reproduction records are listed in Table 3.

A.3 Numerical evaluation and retrospective checks

The complete historical source geometry was recovered and matched to all 14,336 table rows. A geometry-only rebuild restored six later descriptors; all 645,120 overlapping table cells agreed. This retained the historical volume normalization and did not evaluate capacity.

A separate current-evaluator run attempted every body. It returned 14,245 successful evaluations, 54 input-policy failures and 37 scalar-tolerance failures. Among successful rows, the largest relative differences were $1.06 \cdot 10^{-14}$ in capacity, $3.14 \cdot 10^{-14}$ in volume and $3.23 \cdot 10^{-14}$ in the ratio. No row is dropped from the original statistical analysis. The original failures are retained alongside the separate follow-up below.

A subsequent interval-preserving evaluation recovered results for 90 of those refusals. The 37 scalar-tolerance refusals yielded capacity intervals without changing their input bodies. Exact power-of-two rescaling brought 53 of the 54 norm-policy failures into the admitted range; fresh geometry checks preceded evaluation, and homogeneity transferred their interval endpoints back to the original bodies. Across both runs, 14,289 bodies have accepted capacity scalars at tolerance 10^{-10} and another 46 have wider capacity intervals. One body, `random_F8_s3_45`, remains unresolved. No policy or tolerance was relaxed. The complete sample, including this body, remains in the historical statistical analysis.

The largest historical numerical ratio belongs to the 5×6 product `random_5x6_s3_168`. Its stored ratio is 0.86258589584944; the current evaluation gives 0.8625858958494405. Both capacity scalars round to 4.821447361515128, while the current capacity interval has binary64 endpoints printed as 4.821447361515127 and 4.821447361515128. The matched row and its source identifier are retained in

> `topics/empirical-studies/retrospective-revalidation/target-full/comparison.jsonl.gz` and `topics/empirical-studies/polytope-invariant-table/polytope-provenance-table.jsonl`.

The current evaluator reconstructs the geometry of the stored binary64 input and returns rational product capacities or outward capacity bounds. Its volume calculation remains floating-point. Agreement of central values does not mean that a narrow new interval contains an independently rounded old value: 5,583 old capacity scalars lie outside the new intervals despite the small differences above. The initial reevaluation corroborates the numerical values on its successful bodies; it does not certify the original execution or the complete ratio. Prospective selection experiments and optimizer traces have their own evidence records and are not validated by this baseline comparison.

A.4 Prediction models and feature families

Prediction experiments gave another view of the signal. A random forest [Bre01], which averages predictions from many decision trees, used 39 descriptors: 27 describing face counts and incidences, and 12 summarizing ridge areas. The test split held out groups defined by capacity-source label and facet count, leaving 8,192 training rows and 6,144 test rows. Its test coefficient of determination was $R^2 = 0.885$. Here

$$R^2 = 1 - \frac{\sum_{i \in \text{test}} (s_i - \widehat{s}_i)^2}{\sum_{i \in \text{test}} (s_i - \overline{s}_{\text{test}})^2}$$

compares squared prediction error with prediction by the test-set mean. Here s_i is the observed numerical systolic ratio and $\widehat{s}_i$ is the model prediction. The mean absolute error was 0.0495. This was one group split, not repeated cross-validation across independent datasets.

A separate feature-family comparison used histogram-based gradient-boosted regression [Sci], which successively fits trees to improve the prediction. Under its common training/test split, ridge-area features alone gave $R^2 = 0.887$, whereas combinatorial counts alone gave 0.043. The comparison points to information in the ridge areas beyond those counts. Its split and feature list differ from the random-forest experiment, so the two scores do not rank the learning algorithms against each other.

The 27 combinatorial descriptors are the four face counts; the simplicity indicator, simple-vertex fraction and edge density; and the mean, standard deviation, minimum and maximum

of five lists: incident facets per vertex, vertex degrees, ridge sizes, vertices per facet and facet-neighbour counts. The random forest's twelve ridge descriptors are the mean, standard deviation, minimum, maximum, lower and upper quartiles, median, 90th and 95th percentiles and sum of the ridge areas divided by $\sqrt{\mathrm{vol}_4(K)}$, together with the largest-ridge and largest-three-ridge shares. The boosted model adds the fractions at or below normalized-area thresholds $10^{-3}, 10^{-2}, 10^{-1}$, entropy, effective face count and normalized entropy, giving 18 ridge descriptors. The full field names are recorded, respectively, in

`topics/empirical-studies/sys-datascience/methods/prediction-ranking/artifacts/summary.json` and
`topics/empirical-studies/sys-datascience/methods/_shared/random_only.py`.

The boosted comparison used 8,704 training rows and 5,632 test rows. The held-out groups were products with 8, 10 or 12 facets and generic polytopes with 12 facets; all feature-family comparisons used this same split. Source labels determined groups but were not model covariates. The precise group counts, feature-family scores and current-schema input reconstruction are in `topics/empirical-studies/sys-datascience/methods/standard-baseline-p2/`.

A.5 Selection and comparison rules

The 100,000-product concentration experiment specified a positive aggregate mean difference and positive differences in at least seven of ten groups as its continuation criterion. It found eight positive groups and a mean difference of about 0.034. This is the descriptive design criterion; no significance claim is attached to passing it.

For the tangential-generator comparison, controls were selected by a fixed hash order, before capacity evaluation, from the eligible pool after removing the union of the ridge and covariance selections. They are unselected same-generator comparisons, not an asserted random sample of the full pool.

The opposing 3×6 and 4×4 rotation paths contain four recorded positions each. Their simultaneous decrease in ridge score and numerical ratio is a finite observation; monotonicity at every intermediate angle was not established.

Initial ridge selection. The preliminary pool contained 100,000 products. Thirty scalar selection sets and their baselines required 1,675 distinct evaluations; their union had maximum recorded ratio 0.868. The low-R rule selecting ten bodies per side-count group gave mean 0.6261166821, against 0.3155039434 for ten comparison bodies from each group. Comparison bodies were selected by a rule-specific fixed hash order excluding the selected set. Overlapping rules are not independent repetitions.

Restriction to the low-ridge tail. The conditional comparison uses 1,000 evaluated products from the lowest one percent by R, with 100 in each side-count group. Spearman correlations within groups range from -0.315 to $+0.301$; pooling ranks computed separately within each group gives $+0.03197$. Selecting the most extreme tenth increases the mean in only two of ten groups. The broad and conditional panels are different generated samples, so their comparison concerns a change in population and score range. The retained analysis is in `tail-dependence-feasibility/artifacts/current/`, under the methods directory in Table 3.

Conditioning the original dataset. A separate analysis retains the lowest-R or highest-ratio fraction within each of the eighteen original sample groups. At retained fractions $100\%, 20\%, 10\%, 5\%, 1\%$, both arms contain respectively 14,336, 2,874, 1,446, 728 and 158 bodies. There are no ties across these cutoffs. Each retained group is reranked independently;

its two rank columns are divided by the group's retained size, then pooled for a Spearman correlation. Thus the summary weights bodies rather than giving each group equal weight. The low-R values are $-0.944, -0.434, -0.222, -0.113, +0.035$, and the high-ratio values are $-0.944, -0.496, -0.286, -0.239, -0.036$. The smallest panels contain only six generic or eleven product bodies per group. These are descriptive statistics under conditioning, not evidence of independence, systematic reversal, or the effect of a shape intervention. Row identities, cutoffs and a second rank calculation were independently checked; the reproducible packet is `topics/empirical-studies/sys-datascience/methods/selection-mechanism/`.

Gaussian rank reference. Each original sample group is assigned a bivariate Gaussian reference with Pearson correlation $2\sin(\pi r_s/6)$, where r_s is its observed full-sample Spearman correlation. One thousand simulations preserve all eighteen group sizes and apply the same conditional ranking statistic. At the five-percent cutoff, the reference means are -0.695 for both selection directions. The central simulation intervals are $[-0.738, -0.649]$ for high-ratio selection and $[-0.736, -0.647]$ for low-R selection, against observed values -0.239 and -0.113. These are pointwise simulation references with fitted parameters held fixed. The model was considered after seeing the attenuation; this is not a prespecified model test or a confidence interval for the observed correlation. The reference, reproducible simulation and input hashes are in the same packet's `gaussian-result.md` and `gaussian-artifacts/`.

Covariance selection. Two fresh pools of 50,000 candidates compared the lowest 0.5% by ρ, the lowest 1% by R followed by its lower half by C, and 25 comparison bodies per side-count group. An ordinary covariance conditioning cutoff excluded three candidates across the two pools. Controls were selected by fixed hash order from the eligible bodies after removing both geometric selections. There were 500 memberships in each selection and 1,436 distinct evaluations. All twenty covariance-minus- control group effects were positive, ranging from 0.259639 to 0.440029. The mean effect was 0.333123, while the mean covariance-minus-ridge effect was 0.014488; thirteen of the latter twenty effects were positive.

The retained analysis also gives nominal Student-t intervals. They are $[0.310, 0.356]$ for covariance versus controls and $[-0.0149, 0.0439]$ for covariance versus ridge selection, using the twenty group effects and nineteen degrees of freedom. The groups have fixed, different side counts, and no common-mean sampling model or calibrated coverage for repeating the complete selection procedure was established. The main text therefore reports the observed consistency across groups.

Generic and tangential transfer. The generic ten-facet experiment selected 100 of 10,000 bodies by lowest mean normalized ridge area, with 100 disjoint, deterministically selected comparison bodies. The mean contrast was 0.287, with a percentile interval $[0.250, 0.324]$ from 20,000 resamples. The ten most extreme scores minus the other ninety had mean contrast -0.0318 and interval $[-0.0946, 0.0331]$. These resamples hold the selected panels fixed; they do not regenerate the pool, rerank its bodies or reselect its tail. The selected-panel rank correlation between mean ridge area and ratio was $+0.149$.

The tangential-source comparison required both original and changed factors to be admissible and normalized the changed factors separately to area one. One pool supplied 3,200 eligible bodies in each of the 4×6 and 6×6 groups. Each selection had sixteen memberships per group. Five overlapping selections gave 91 distinct evaluations. The equally weighted group effects were 0.259 for covariance versus controls and 0.225 for ridge versus controls; their within-panel resampling intervals were $[0.192, 0.329]$ and $[0.160, 0.296]$. These condition on the observed panels. Shared controls make the two comparisons dependent.

Euclidean area and alignment. For the generic selected and comparison panels, the respective means of $\overline{E}$ are 0.9167 and 1.3653, and the means of $\overline{\kappa}_E$ are 0.3769 and 0.5029. Within the selected panel, the ten lowest R/N_2 values have mean $\overline{E} = 0.9419$ and mean $\overline{\kappa}_E = 0.3400$, compared with 0.9139 and 0.3810 for the other ninety bodies. These are means of body-level quantities; they do not assert a factorization of the population mean of R/N_2. Definitions, per-ridge values and the decomposition check are retained in `topics/empirical-studies/sys-datascience/methods/ridge-tail-anatomy/`.

A.6 Paired height changes and ambient rotations

The paired support-height experiment used sixteen independently generated pairs of polygons, eight 4×4 and eight 4×6. For each, the original product was compared with equalizing heights in the first factor, the second, or both; all four constructions had to be valid before any ratio was evaluated. Each factor was scaled to area one. All 64 evaluations succeeded. Changing both factors improved eleven pairs and worsened five, with mean numerical change 0.0153. The group means were 0.0040 and 0.0266 respectively. This is the jointly admissible source, not an unconditioned statement about all polygon pairs.

Recomputing R from the retained paired geometries gives opposite signs of ΔR and $\Delta\widehat{\mathrm{sys}}$ in fifteen of sixteen both-factor changes. The one exception has R increasing from 11.074 to 11.218 while the numerical ratio increases from 0.419 to 0.436. The one-factor comparisons have opposite signs in thirteen and fourteen of sixteen pairs. All 64 bodies have $R \geq 11.065$ and numerical ratio at most 0.742. These results reuse the retained capacity evaluations; no new capacity calls enter the comparison. Reconstruction, row identity and hash checks are recorded in `topics/empirical-studies/sys-datascience/methods/selection-mechanism/controlled-result.md`.

For ambient rotations, the action is

$$P \times Q \longmapsto U(P \times Q), \qquad U \in SO(4).$$

It preserves Euclidean shape and volume. A nonsymplectic U can change capacity, and the image need no longer be a Lagrangian product for the fixed symplectic form. One allocation evaluated a source product, a symplectic rotation control and fourteen prescribed orthogonal rotations; the other evaluated sixteen fresh products, sharing the source body. There were four comparisons, two each for 3×3 and 4×4. The schedule used fourteen fixed matrices, not uniform sampling over $SO(4)$. The shared starts gave 124 distinct requests and 128 memberships; symplectic controls preserved numerical ratios within $4 \cdot 10^{-16}$.

Rotation improved three starts, but fresh sampling attained a larger maximum in all four comparisons. In one quadrilateral example, rotation raised 0.328 to 0.529, while fresh sampling reached 0.774. Rotation evaluations also took longer. These are observations about the prescribed schedules and starts, not a comparison of all ambient rotations with all products from the source law.

A.7 Factor covariance balancing

The paired comparison uses 32 original products from each of the ten side-count groups. The selection is fixed by hashes of row identifiers, without using the target ratios. For planar vertex covariances A, B, the maps $(\det A)^{1/4} A^{-1/2}$ and $(\det B)^{1/4} B^{-1/2}$ have determinant one and make each covariance scalar. There is one original and one transformed body for every selected row, evaluated through the same current product-capacity route. All 640 evaluations succeeded.

The mean numerical ratio changes from 0.341888 to 0.653683; 314 paired changes are positive and six are negative. The mean R changes from 23.738833 to 11.652347, with 306 decreases and fourteen increases. The pooled within-group rank correlation changes from -0.928922 to

-0.248680. The balanced triangle-product group has correlation $+1$; the other nine groups range from -0.571 to -0.098. Thus the pooled weakening does not mean that every transformed group has negligible or negative dependence.

The transformed covariance ratios differ from one by at most $5.1 \cdot 10^{-13}$, and relative factor-area changes are below $3.2 \cdot 10^{-13}$. The original current evaluations agree with their retained historical numerical ratios within $1.3 \cdot 10^{-15}$. These are paired transformations of retained source bodies, not a new independent population or a test isolating the scalar ρ from other changed geometric properties. The producer, frozen inputs, evaluator identity and complete paired results are retained in `topics/empirical-studies/sys-datascience/methods/selection-mechanism/balance-artifacts/`.

A.8 Finite-budget local optimizer comparison

The run configuration labels the 64 starting bodies as held out. The saved hashes identify the seven configurations used, but do not establish that these bodies were absent from earlier development or tuning. A later development configuration reuses 16 starts. Their earlier selection history is therefore unresolved; this does not show that the reported outcomes were used to select the configurations.

Every run used the same allocation rule: a nominal 1000 ms start-new-work cutoff for charged full-evaluator plus optimizer time and a cap of 128 full historical-evaluator calls. In-progress work could complete atomically past the cutoff, and methods could terminate early, so realized compute differs across runs. The 448 traces contain 28824 full calls. Using the historical numerical objective described in Section 8.10, four-anchor branch history has the largest comparison-block median best value, 0.9848, followed by transition prediction at 0.9816 and the finite-gap affine model at 0.9772. No run reaches a recorded numerical ratio of one.

The implemented optimizer steps. Except for the literal gradient, the methods remove the fifteen infinitesimal symmetry directions from the forty dual-normal coordinates. The local models and coordinate poll recompute this transverse space at their anchors; CMA-ES keeps its initial space. The literal gradient uses the ambient coordinates. Every outer proposal is checked by a fresh geometry reconstruction and the configured full orbit-word search. The implementations below are frozen comparison points; their parameters are not claimed to optimize the corresponding algorithm families.

Table 2: One outer iteration of each optimizer in the recorded comparison.

method	implemented step
literal and safeguarded branch gradients	Differentiate one currently minimizing branch. The literal method takes $a^+ = a + 0.01 D_a s_\sigma(a)$ even after a decrease. The safeguarded method projects and normalizes this gradient, accepts only a fully evaluated improvement, and expands or halves its distance after acceptance or rejection.
finite-gap affine model	Keep the current branch gaps in a 10% action window, linearize their values, and maximize the minimum affine value on a local ball. Fully evaluate the proposal and use the same expand-or-halve distance rule.
four-anchor branch history	Union candidate words from the four most recent full searches, using a 30% action window and normalized beta allowance 0.3. At a nearby point, re-solve the named KKT branches, discard transition-infeasible and beta-inadmissible solutions, linearize the remaining values, and maximize their affine minimum. Take at most two internal moves before the next full evaluation.

method	implemented step
history with transition prediction	Use the preceding method, but for a normalized internal move of length at least 0.08, also predict newly feasible transitions in the move direction and evaluate the words thereby generated.
CMA-ES	Fully evaluate a population of eight Gaussian perturbations, then update the mean, covariance, and normalized scale from their ranking.
coordinate direct search	Poll both signs of the 25 transverse basis axes. Move to the best improvement and double the radius, or halve the radius if no poll point improves.

The beta allowance applies when retaining candidate words at a discovery anchor: a raw KKT solution is kept only if

$$\frac{\min_i \beta_i}{\max_i |\beta_i|} \geq -0.3$$

as well as satisfying the action-window cutoff. This retains some words whose weights are negative at that anchor, in case they become feasible nearby. It does not relax orbit feasibility: each local model and target evaluation again checks the word's transitions and the usual nonnegative weight condition at the point being evaluated.

The two history methods accept any usable outer evaluation, including a decrease; their reported best-so-far value therefore differs from their current iterate. The safeguarded gradient and finite-gap method accept only improvements, multiplying the relative step length by 1.25 on success and by 0.5 on failure. The history methods use up to two internal model steps between full evaluations.

The companion endpoint packet does not test local maximality. Signed polls along the transverse basis find an improving direction at seven of 16 one-second branch-history endpoints and two of the corresponding five-second endpoints. Separate validated continuation paths show that even an endpoint with no improving basis direction can retain appreciable ascent between the tested axes. These observations diagnose insufficient finite budgets; they do not estimate the distance to a local maximum.

Detailed HKO calibration. Distances here use the Euclidean norm of the forty ordered dual-row coordinates a, with no further minimization over symmetries. Thus distance to HKO is $\|a - a_{\mathrm{HKO}}\|_2$, and the fraction removed is $1 - \|a_{\mathrm{after}} - a_{\mathrm{HKO}}\|_2 / \|a_{\mathrm{before}} - a_{\mathrm{HKO}}\|_2$. The four unit directions lie in the transverse space at HKO: two random directions, its first chosen basis vector (the slice-basis sentinel), and the projected, normalized tangent from rotating the pentagon's second factor (the pentagon-tangent sentinel). The source distances in the figure are absolute Euclidean distances along these rays.

On 16 known perturbations along four development directions from HKO, the multi-branch continuation accepted no move at HKO and improved every perturbation. One step recovered 93.4%–100% of the known evaluator gap and removed 77.9%–100% of the distance to HKO. In contrast, the gradient of one current minimizing branch recovered −22.1%–41.4% of the gap.

Selected finite-step failure audit. For the following probes and audits, a relative or normalized radius r instead scales a unit direction u by the current state norm: $\delta a = r\|a\|_2 u$. The normalized length of a continuation path is $\sum_k \|a_{k+1} - a_k\|_2 / \|a_k\|_2$.

At the eight highest four-anchor tuning endpoints, five admitted one validated improvement, by $1.57 \cdot 10^{-6}$ to $4.61 \cdot 10^{-5}$. Across the 80 multi-branch proposals, 52 decreased the recomputed evaluator value despite a represented target winner and positive affine prediction; 40 of these losses retained determinate, unchanged geometry. All 40 current-branch gradient proposals decreased.

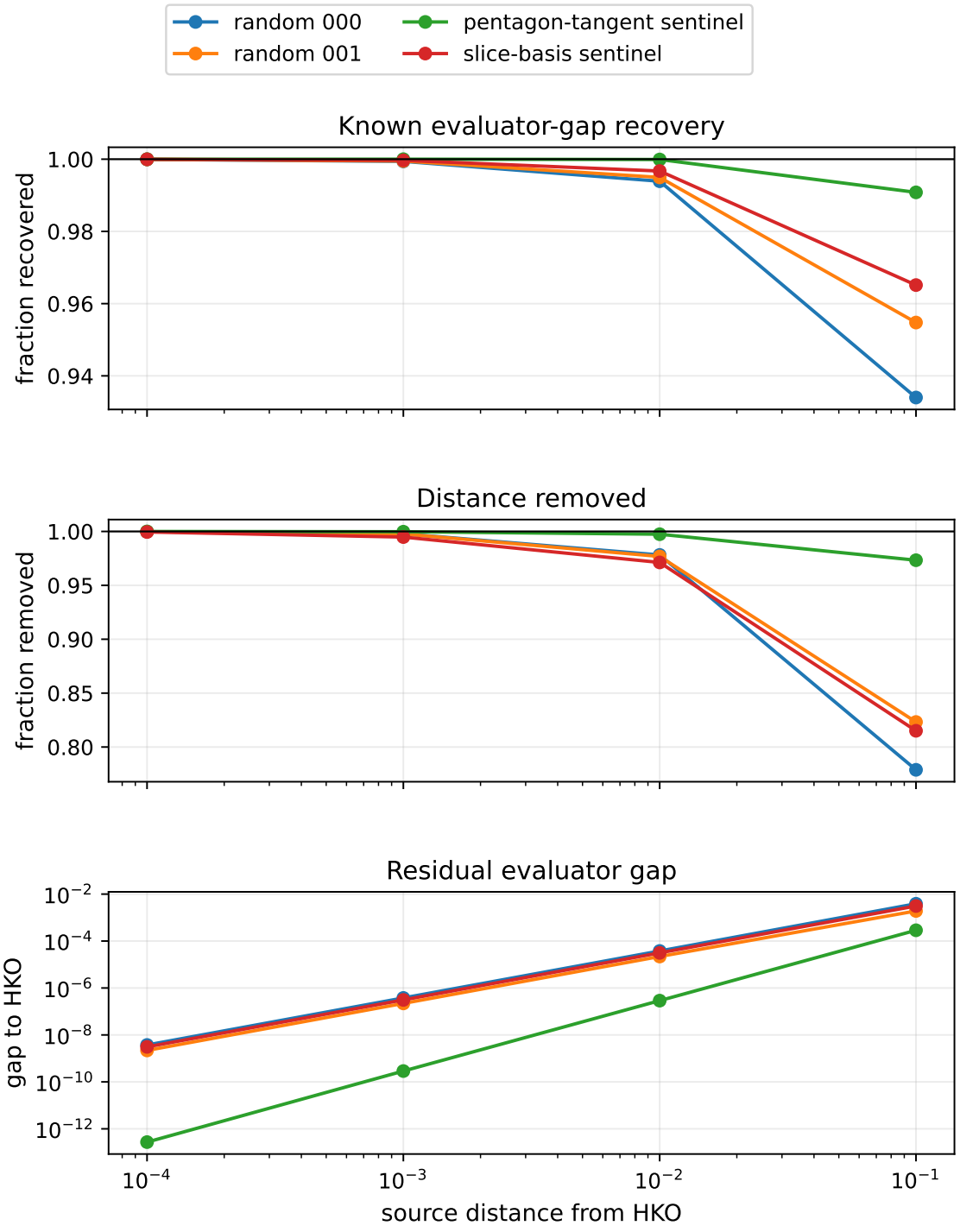


Figure 13: One-step calibration on known perturbations of HKO. From top to bottom, the panels show recovered evaluator gap, removed distance, and residual evaluator gap.

The three-proposal audit compared two such failures with one positive control. At normalized radius 10^{-5}, the failures perturbed their named KKT matrices by 218 and 67 times the smallest base eigenvalue magnitude and crossed zero; the control had ratio 1.53, did not cross zero, and kept the correct sign. At the smallest tested radius, 10^{-8}, central differences approximated the implemented analytic derivatives with relative errors below 1% for the named action and 1.9% for the named branch ratio. Binary64 and exact-arithmetic geometry reconstruction agreed at all 39 audited points, and relative volume discrepancies were below 10^{-15}. For these selected cases the observed failure is finite-distance nonlinearity of an ill-conditioned KKT branch.

These diagnostics use selected controls rather than population samples. They do not provide a convergence theorem, a population error rate, or a universal conditioning cutoff.

A.9 Endpoint probes and continuation

Sixteen history-method endpoints were chosen across the observed terminal performance range. Each received both signs of 25 transverse basis directions at relative radii 10^{-3}, 10^{-4} and 10^{-5}, giving 150 probes per endpoint. Seven endpoints had a raw positive change and nine had none. Rank-based endpoint selection prevents interpreting these fractions as a population convergence probability.

After a larger budget on those starts, continuation examined three selected endpoints and HKO. It tested branch-model and selected-gradient proposals at five decreasing step lengths, with fallback signed basis polling, and allowed ten accepted improvements. All three endpoints took ten improving steps; HKO took none. One endpoint with no improving signed basis direction gained 0.00223 along oblique moves of total normalized path length 0.01.

The separate triangle–hexagon, square–square, CH2021 and pentagon-control suite used basis, projected random and prescribed rotation directions at three radii. All 1,482 probes were valid. The three targets had no nominal gain above 10^{-12}; the pentagon control had twelve, all from rotations. Incomplete historical source provenance and broad capacity-derived bounds prevent interpreting the nominal misses as certified local maximality.

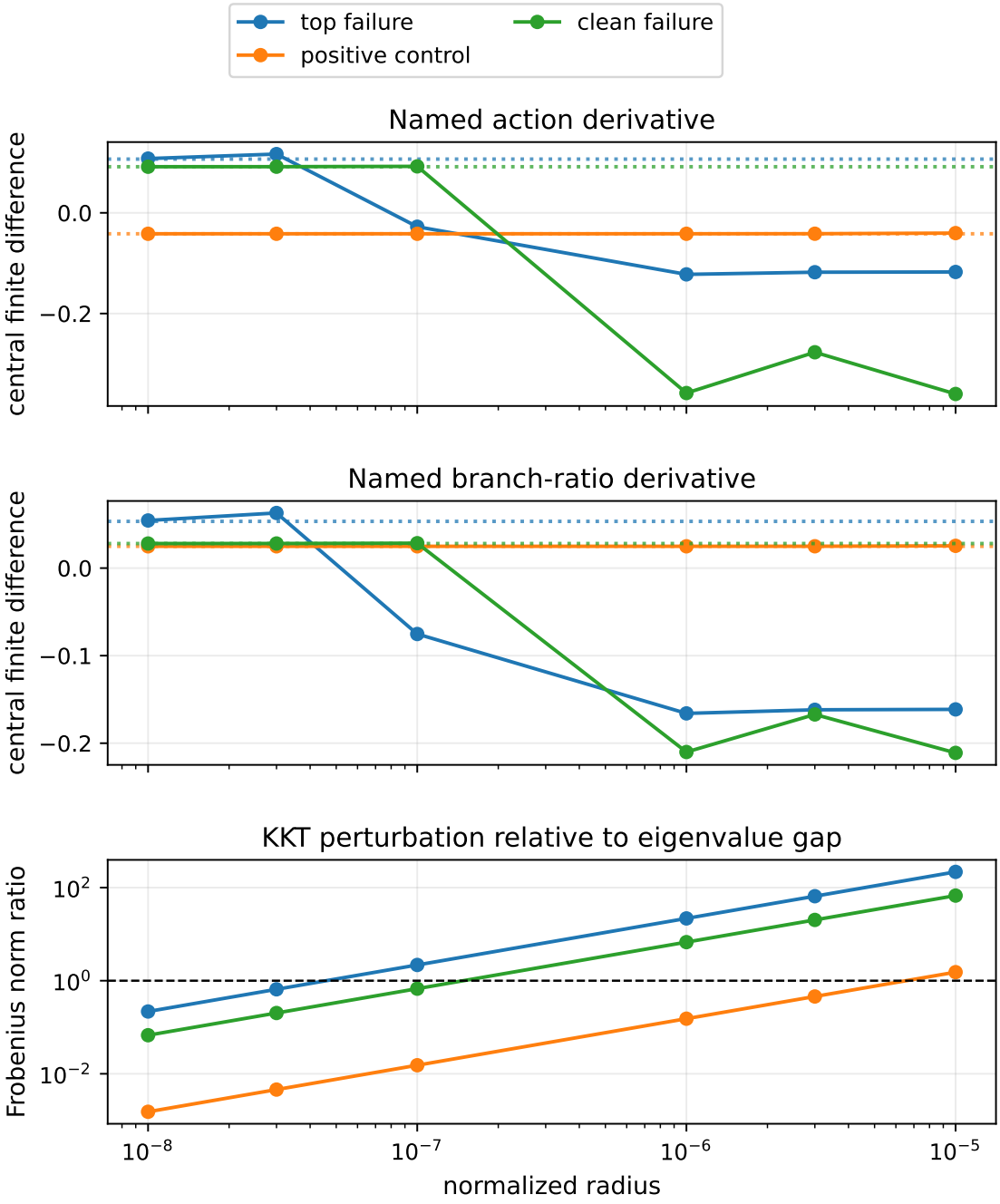


Figure 14: Finite differences for two failed affine proposals and one positive control. Prediction changes regime when the KKT perturbation becomes large relative to the smallest source eigenvalue magnitude (bottom). Dotted lines in the top two panels are the analytic derivatives.

A.10 Adaptive product sampling

Both arms began with the same 64 independently generated products. Four populations including initialization gave 256 charged evaluations per arm. The adaptive update used the sixteen bodies with largest ratios. Across three runs, shared initial populations gave 1,344 actual evaluations and 1,536 arm memberships; all actual evaluations succeeded and all recorded ratios were below one. The top-eight statistic removes repeated exact dual-vertex arrays, retaining different arrays with equal scores.

For reproducibility, let $g_1, \dots, g_5 > 0$ be the cyclic normal-angle gaps of a factor and let r_i be its dual-normal lengths. Set

$$\ell_i = \log(g_i/g_5) \quad (i = 1, \dots, 4), \qquad z_i = \log r_i - \frac{1}{5}\sum_{j=1}^{5} \log r_j.$$

We store the first four z_i and recover $z_5 = -\sum_{i=1}^{4} z_i$. Reconstruction uses

$$g_i = \frac{2\pi e^{\ell_i}}{1 + \sum_{j=1}^{4} e^{\ell_j}}, \qquad g_5 = \frac{2\pi}{1 + \sum_{j=1}^{4} e^{\ell_j}}, \qquad h_i = e^{-z_i}.$$

The first normal angle is zero in the first factor and ϕ in the second. The initial IID encoding chooses a cyclic origin by lexicographically minimizing the sequence of gap/length pairs. Sampled adaptive coordinates are retained for the updates without another cyclic relabelling.

For an ordinary coordinate j, write $\widehat{\mu}_{E,j}$ and $\widehat{v}_{E,j}$ for the mean and population variance of the sixteen elites. The update is

$$\mu_j^{+} = \tfrac{1}{2}\mu_j + \tfrac{1}{2}\widehat{\mu}_{E,j}, \qquad v_j^{+} = \max\{\tfrac{1}{2}v_j + \tfrac{1}{2}\widehat{v}_{E,j},\ 0.05 v_{0,j}\}.$$

Initialization uses the population moments of the shared 64 bodies. Coordinates are drawn independently with these Gaussian moments. The phase uses circular means, a unit-vector blend of old and elite means, and variance of angular deviations wrapped into $[-\pi, \pi)$. Samples are wrapped modulo 2π. A vanishing resultant retains the previous mean; initial fallback uses the smallest-ID candidate's phase. Ties in elite selection are broken by candidate ID.

Each population permits at most 640 construction attempts for 64 accepted bodies. Failed geometry and unsupported evaluator inputs are rejected before target evaluation; an incomplete population does not trigger an update. The evaluator admission includes finite, nonredundant bounded geometry and primal and dual infinity-norm bounds between 10^{-3} and 10^{3}. The three seeds are 202609140301, 202609140302 and 202609140303. The retained report records the stopped first version and the subsequent supported version separately.

The source restriction needs particular care. An initial version stopped when one body exceeded the evaluator's admitted primal-vertex infinity-norm range. The completed version applied the evaluator's existing admission conditions to both proposal arms before accepting a body for evaluation. This was a procedural revision after seeing the failure. Both arms reject unsupported inputs, but the admission test acts on the actual coordinate representation: IID bodies retain their original representation and adaptive bodies use the normalized reconstruction. It is not a symmetry-invariant condition on all pentagon products. The result is therefore for this implemented, admission-conditioned comparison.

The complete run took 215.8 seconds: 182.2 seconds for construction and admission checks, 32.9 seconds in evaluator processes, and 0.053 seconds for distribution updates. Including construction and charging initialization to each arm, the adaptive runs used about 35–36 seconds per seed and IID about 46–51 seconds. Equal evaluation counts did not imply equal cost. These measured costs favor the adaptive arm in this run; they are not the outcome of a separate equal-wall-time experiment.

A.11 Evidence and reproduction index

Paths in Table 3 are relative to the accompanying repository. Each packet identifies its inputs and analysis; the original historical optimizer executable and full proposal traces were not recovered.

Table 3: Source packets for the chapter.

Result	Repository source
Historical population and replay	`topics/empirical-studies/evidence-closure/README.md`; `topics/empirical-studies/polytope-datasets/retained-lineage.md`
Restored features and reevaluation	`topics/empirical-studies/retrospective-revalidation/geometry-full/README.md`; `topics/empirical-studies/retrospective-revalidation/target-full/README.md`; `topics/empirical-studies/retrospective-revalidation/repair/README.md`
Feature models	`topics/empirical-studies/sys-datascience/methods/prediction-ranking/`; `topics/empirical-studies/sys-datascience/methods/standard-baseline-p2/`
Fresh selection	`topics/empirical-studies/sys-datascience/methods/extreme-scalar-rejection-proposer/`; `topics/empirical-studies/sys-datascience/methods/extreme-scalar-rejection-proposer/artifacts/covariance-rho-frozen-validation/`
Transfer	`topics/empirical-studies/sys-datascience/methods/alternative-source-transfer/`; `topics/empirical-studies/sys-datascience/methods/generic-ridge-tail-stage1-target/`
Paired support heights	`topics/empirical-studies/sys-datascience/methods/paired-tangentialization/`
Orientation allocation	`topics/empirical-studies/sys-datascience/methods/orientation-allocation/`
Adaptive product sampling	`topics/empirical-studies/sys-datascience/methods/diagonal-cem-pilot/`
Local optimization and diagnostics	`experiments/dev-gradient-ascent/optimizer-comparison/`; `topics/empirical-studies/evidence-closure/search-account.md`
Conditional and controlled ridge analyses	`topics/empirical-studies/sys-datascience/methods/selection-mechanism/`
Conditional ridge correlations	`topics/empirical-studies/sys-datascience/methods/tail-dependence-feasibility/`
Product ridge geometry	`topics/empirical-studies/evidence-closure/ridge-mathematics.md`; `topics/empirical-studies/evidence-closure/triangle-profile.md`; `topics/empirical-studies/sys-datascience/methods/ridge-endpoint-path/`

B HKO computation: an annotated exact program

B.1 Executable exact verification of the HKO bounds

This appendix isolates the finite certificate used in Lemma 7.4. The listings below form a complete SageMath program. Facet labels and positions in each word are displayed one-based; the executable converts them to zero-based indices at the section boundary.

The verifier from the public packet identified in Section 15 is now maintained at `topics/variation-and-local-maxima/hko-local-maximum/theorem/verify.sage.py` and its `witness.json`. The printed program, `thesis/appendices/hko_core.py`, is a compact exposition of that calculation in the selected manuscript. It writes the exact fixed values directly in `rows`, whereas the packet verifier reconstructs them from its finite witness. The twenty-six words, solved and fixed positions, resulting exact weights, and derivative rows agree. The printed program is part of this manuscript's working tree; its inclusion in the historical public commit is not claimed.

The geometric lemmas in the main text establish that these ten labelled dual vertices form the relevant chart, that volume is smooth there, and that each positive feasible section supplies a touching upper bound. The finite CAS task is to instantiate 26 such sections, differentiate their upper bounds, and check the quotient certificate.

Exact field, geometry, and quadratic form. The first source block defines the ordered number field, the ten dual vertices, the symplectic form, the closure matrix, and the quadratic form.

```
from sage.all import *
R = PolynomialRing(QQ, "x")
x = R.gen()
f=x**4-10*x**2+5
F=NumberField(f,'t',embedding=AA.polynomial_root(f,RIF(0,1)))
t=F.gen(); s=(5-t**2)/2; al=(3-s)/2; be=t*(1+s)/2; d=s-1
a=[vector(F,v) for v in [(1,t,0,0),(-al,be,0,0),(-d,0,0,0),(-al,-be,0,0),(1,-t,0,0),(0,0,t,-1)
    ,(0,0,be,al),(0,0,0,d),(0,0,-be,al),(0,0,-t,-1)]]
def om(u,v): return u[0]*v[2]+u[1]*v[3]-u[2]*v[0]-u[3]*v[1]
def C(w): return matrix(F,[[a[i][j] for i in w] for j in range(4)]+[[F(1)]*len(w)])
def q(w,b): return sum(b[i]*b[j]*om(a[w[j]],a[w[i]]) for i in range(1,len(w)) for j in range(i)
    )
# (one-based word, solved positions I, fixed positions J, exact fixed values)
```

All 26 exact assignments. The following is the complete `rows` object, including every word, solved coordinate set, fixed coordinate set, and exact fixed value. Values written with the caret character are strings parsed by Sage's exact evaluator; they are data, not Python exponentiation.

```
rows=[
([1,2,8,4,10,6],[1,2,3,4,5],[6],['t^2/40+1/8']),
([1,2,8,7,4,10],[1,2,3,4,5],[6],['-t^2/20+1/4']),
([2,3,8,9,5,4,6],[1,2,3,4,5],[6,7],['6909830056250513/10^17','-t^2/20+1/4']),
([1,8,7,3,4,10],[1,2,3,4,5],[6],['-t^2/20+1/4']),
([2,9,8,5,4,6],[1,2,3,4,5],[6],['-t^2/20+1/4']),
([2,8,4,5,6,10],[1,2,3,4,5],[6],['t^2/40+1/8']),
([1,6,2,8,5,4,10],[1,2,3,4,5],[6,7],['36180339887499/(2*10^14)','t^2/40+1/8']),
([1,7,8,4,3,9,10],[1,2,3,4,5],[6,7],['(8090169943749473*t^2+23368810393753689)/(4*10^17)','
    18090169943749473/10^17']),
([1,5,7,6,3,10,9],[1,2,3,4,5],[6,7],['(4045084971874731*t^2+11684405196876883)/(2*10^17)','
    9045084971874731/(5*10^16)']),
([2,8,4,3,9,5,6],[1,2,3,4,5],[6,7],['1809016994374947/10^16','-t^2/20+1/4']),
([2,3,9,8,4,5,6],[1,2,3,4,5],[6,7],['2261271242968689/(125*10^14)','-t^2/20+1/4']),
([2,8,9,5,7,6,3],[1,2,3,4,5],[6,7],['18090169943749473/10^17','t^2/40+1/8']),
```

```
([1,7,8,4,10,2],[1,2,3,4,5],[6],['t^2/40+1/8']),
([1,7,4,3,9,10],[1,2,3,4,5],[6],['t^2/40+1/8']),
([2,9,5,6,7,3],[1,2,3,4,5],[6],['t^2/40+1/8']),
([2,8,3,9,5,6],[1,2,3,4,5],[6],['-t^2/20+1/4']),
([2,8,4,5,10,6],[1,2,3,4,5],[6],['t^2/40+1/8']),
([1,7,3,2,8,4,10],[1,2,3,4,5],[6,7],['1809016994374950 7/10^17','-t^2/20+1/4']),
([1,5,6,7,3,2,9],[1,2,3,4,5],[6,7],['6909830056250547/10^17','-t^2/20+1/4']),
([1,7,3,9,5,6,2],[1,2,3,4,5],[6,7],['t^2/40+1/8','863728757031319/(125*10^14)']),
([1,8,5,4,6,10,2],[1,2,3,4,5],[6,7],['t^2/40+1/8','7463176480767021/(5*10^16)']),
([1,6,2,8,7,4,10],[1,2,3,4,5],[6,7],['-t^2/20+1/4','1809016994374947/10^16']),
([1,5,7,4,3,9,10],[1,2,3,4,6],[5,7],['9045084971874741/(5*10^16)','t^2/40+1/8']),
([1,7,3,4,10,9,5],[1,2,3,4,5],[6,7],['t^2/40+1/8','3454915028125259/(5*10^16)']),
([1,7,6,3,10,9,5],[1,2,3,4,5],[6,7],['452254248593737/(25*10^14)','t^2/40+1/8']),
([2,8,9,5,4,6,3],[1,2,3,4,5],[6,7],['-t^2/20+1/4','12028993467659169/10^17'])]
def val(z): return F(sage_eval(z, locals={'t': t}))
```

Feasible sections and derivative rows. The next block is the actual section solver. It explicitly converts the one-based assignment data, solves $C_I\beta_I = e - C_J u$, restores $\beta_J = u$, and checks the minor's invertibility, closure, normalization, and positivity. The constants `V`, `A`, and `vr` are respectively the base volume, the known HKO capacity, and the volume derivative from Lemma 7.3. The derivative block computes $D\beta$, then differentiates Q and forms $r_j = DU_j(a_0)$ by the product rule for $U_j = A_j^2/(2V)$.

```
def section(W):
 w,I,J,u=W; w=[i-1 for i in w]; I=[i-1 for i in I]; J=[i-1 for i in J]
 assert len(set(w))==len(w) and all(0<=i<10 for i in w); assert sorted(I+J)==list(range(len(w))
    ); cm=C(w); assert cm[:,I].det()!=0; U=vector(F,[val(z) for z in u]); b=[None]*len(w)
 x=cm[:,I].solve_right(vector(F,[0,0,0,0,1])-cm[:,J]*U)
 for k,i in enumerate(I): b[i]=x[k]
 for k,j in enumerate(J): b[j]=U[k]
 b=vector(F,b); assert cm*b==vector(F,[0,0,0,0,1]) and all(z>0 for z in b)
 return w,I,J,b
V=QQ(25)*(5+s)/QQ(32); A=5*t-t**3/QQ(2); qm=1/(2*A)
vr=vector(F,sum((list(-(QQ(25)/QQ(32)+QQ(5)*s/QQ(16))*v) for v in a),[]))
def drow(w,I,J,b):
 cm=C(w); db=[[F(0)]*40 for _ in w]
 for z in range(40):
  facet,coord=divmod(z,4); rhs=vector(F,[0]*5)
  if facet in w: rhs[coord]=-b[w.index(facet)]
  xi=cm[:,I].solve_right(rhs); assert cm[:,I]*xi==rhs
  for k,i in enumerate(I): db[i][z]=xi[k]
 for j in J: assert all(x==0 for x in db[j])
 dq=[]
 for z in range(40):
  facet,coord=divmod(z,4); u=vector(F,[1 if k==coord else 0 for k in range(4)]); x=F(0)
  for i in range(1,len(w)):
   for j in range(i):
    x+=(db[i][z]*b[j]+b[i]*db[j][z])*om(a[w[j]],a[w[i]])
    if w[j]==facet: x+=b[i]*b[j]*om(u,a[w[i]])
    if w[i]==facet: x+=b[i]*b[j]*om(a[w[j]],u)
  dq.append(x)
 return vector(F,[A/V*(-x/(2*qm**2))-A**2/(2*V**2)*vr[i] for i,x in enumerate(dq)])
# Build the fifteen symmetry columns, then verify all rows, rank, and kernel.
```

Symmetry and finite conclusion. The final source block constructs the four translations, dilation, and ten $\mathfrak{sp}(4)$ columns, then checks symmetry annihilation, row rank 25, and a strictly positive normalized left-kernel relation. The check `q(w,b)==qm` establishes that each section gives the known HKO action at the base point, so its upper bound touches the systolic ratio there. Thus the explicitly assigned `rows`, together with the computed `rr` and `lam`, witness the

existence assertions of Lemma 7.4. The rank and positive relation supply the hypotheses of Lemma 7.5.

```
Jm=matrix(F,[[0,0,-1,0],[0,0,0,-1],[1,0,0,0],[0,1,0,0]])
T=[vector(F,sum((list(-v[k]*v) for v in a),[])) for k in range(4)]
T += [vector(F,sum((list(-v) for v in a),[]))]
G=[matrix(F,g) for g in [
[[1,0,0,0],[0,0,0,0],[0,0,-1,0],[0,0,0,0]],[[0,1,0,0],[0,0,0,0],[0,0,0,0],[0,0,-1,0]],
[[0,0,0,0],[1,0,0,0],[0,0,0,-1],[0,0,0,0]],[[0,0,0,0],[0,1,0,0],[0,0,0,0],[0,0,0,-1]],
[[0,0,1,0],[0,0,0,0],[0,0,0,0],[0,0,0,0]],[[0,0,0,1],[0,0,1,0],[0,0,0,0],[0,0,0,0]],
[[0,0,0,0],[0,0,0,1],[0,0,0,0],[0,0,0,0]],[[0,0,0,0],[0,0,0,0],[1,0,0,0],[0,0,0,0]],
[[0,0,0,0],[0,0,0,0],[0,1,0,0],[1,0,0,0]],[[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,1,0,0]]]
for X in G:
 assert X.transpose()*Jm+Jm*X==0
 T.append(vector(F,sum((list(-X.transpose()*v) for v in a),[])))
rr=[]
assert len(rows)==26
for W in rows:
 w,I,J,b=section(W); assert q(w,b)==qm; rr.append(drow(w,I,J,b))
M=matrix(F,[list(r) for r in rr]); assert M.rank()==25 and matrix(F,[list(c) for c in T]).rank
    ()==15
for r in rr:
 for c in T: assert r.dot_product(c)==0
kernel=M.transpose().right_kernel().basis(); assert len(kernel)==1
lam=kernel[0]
if all(x<0 for x in lam): lam=-lam
assert all(x>0 for x in lam); lam=lam/sum(lam); assert lam*M==0
print('HKO appendix core passed: rows=26 rank=25 symmetry-rank=15')
```

The four groups of predicates in the main text can be located directly: `section` and line 84 check the positive feasible contacts; line 85 checks the two ranks; lines 86–87 check symmetry annihilation; and lines 88–91 compute and check the positive relation. The program's `M` is the derivative matrix called R in the proof, and `T` stores the symmetry columns. It checks the finite linear algebra, including the one-dimensional left kernel. Smooth continuation follows from the displayed inverse-minor construction, and the strict neighborhood and equality statements follow from the geometric lemmas; they are not additional predicates checked by this program.

The listed file is `thesis/appendices/hko_core.py`. From the repository root, run it with Python optimization disabled so that its assertions are executed:

```
PYTHONOPTIMIZE=0 conda run --name sage python thesis/appendices/hko_core.py
```

The recorded run returned

`HKO appendix core passed: rows=26 rank=25 symmetry-rank=15`. These checks establish the finite hypotheses used in the main proof. The geometric chart and volume formula are supplied by Lemmas 7.2 and 7.3; Lemmas 7.5–7.7 then pass from the computed derivatives to local maximality.

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